Health Costs of Firms

Vahid Moghani*

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Abstract

Little is known about the contribution of firms to variation in workers' healthcare expenditure. I use linked Dutch employer–employee data and workers moving across firms to separate persistent worker and firm differences in healthcare expenditure. Moving a worker to a firm with a fixed effect one standard deviation higher would raise expected annual expenditure by 17.8%. The top–bottom quintile gap in firm effects is comparable to the expenditure gradients by income and wealth. Over half of the across-firm dispersion is within industry. Supporting a health-risk interpretation, moving to a firm with higher coworker expenditure increases use of pain management medication, and exposure to high-effect firms predicts disability and mortality. These firms pay higher wage premia, but mainly between rather than within industries. Because healthcare is collectively financed, firms generate a fiscal externality: healthcare expenditure they do not fully bear.

*Erasmus School of Economics, Tinbergen Institute, and Erasmus Center for Health Economics Rotterdam. I thank Owen O'Donnell, Jan van Ours, Sonia Bhalotra, Tom Van Ourti, Hans van Kippersluis, Patrick Kline, Pilar Garcia-Gomez, Peter Hull, Olivier Marie, Bettina Siflinger, Erik Schut, Lieke Beekers, Lisa Voois, Simon Rabaté and Shobhit Kulshreshtha for helpful comments and discussions. I gratefully acknowledge financial support from the Erasmus Initiative Smarter Choices for Better Health, which supported my position. I declare that there are no conflicts of interest.

I Introduction

Healthcare expenditure varies systematically with the environments people live in (Finkelstein et al., 2016, 2021). Workers spend a large portion of their daily time at the workplace and a large literature documents channels through which employers may affect health and healthcare use, including physical demands, psychosocial stress, management practices, and the design of insurance networks (Johnson, 2020; Jones et al., 2019; Ho and Lee, 2019; Tilipman, 2022; Hummels et al., 2025; Ahammer et al., 2026). Yet we know much less about whether and to what extent otherwise similar employees systematically incur different healthcare expenditure depending on their employer, how much of this variation arises across firms within the same industry, and whether these differences are reflected in wages.

I answer these questions using linked employer–employee administrative data from the Netherlands, where basic health insurance is universal, portable across jobs, and not tied to the employer. The data link every full-time male employee aged 25 to 65 to his employer and basic-insurance claims: 17 million worker-years, 1.2 million movers, and 133,000 firms of all sizes connected by worker mobility.[1] Using workers' moves across firms to separate persistent worker differences from persistent firm effects, I provide the first bias-corrected estimates of the dispersion of firm effects on annual healthcare expenditure.[2] The firm effect has a simple interpretation: by how much would the same worker's expected healthcare expenditure change if he worked at a different firm? Because employers do not determine coverage, provider networks, or cost sharing, and primary care is free at the point of use, these differences capture workplace influences operating through channels other than employer-provided access or health insurance.

I document that moving to a firm one standard deviation higher in the firm-effect distribution increases expected annual healthcare expenditure by 17.8%. Moment-matched distributions imply a top–bottom quintile gap in firm effects of

[1] I focus on men because women's job-to-job moves are closely tied to fertility decisions (Figure A7), which can violate the mobility assumptions behind the decomposition. The results for the female sample are reported in Appendix A9 and are qualitatively very similar to those for the male sample.

[2] Formally, I estimate an Abowd–Kramarz–Margolis (AKM) worker–firm two-way fixed effects model of log annual healthcare expenditure (Abowd et al., 1999) and apply the leave-one-out bias correction of Kline et al. (2020) to estimate the variance components.

about 0.25–0.29 log points, comparable to the gradients in healthcare expenditure associated with wealth (0.26) and income (0.29) in the sample. Among firms that can be linked to the business registry, more than 50% of the dispersion in firm effects remains within narrowly defined industries, so the dispersion is not only a matter of what firms produce but, to a large extent, of differences among firms doing the same work.[3] Firms with higher effects also pay higher wage premia, about 1.3% per standard deviation, consistent with compensating differentials, but only between industries: within narrowly defined industries, firms with higher effects pay no higher premia. The firm-specific component of healthcare expenditure is therefore not tied to wage premia.

To assess whether the firm effects reflect health rather than only healthcare use, I study medication, disability, and mortality. Moves to firms with higher coworker expenditure are followed by increased use of pain-management, anti-inflammatory, and muscle-relaxant medications, with little change in the classes tied to screening or diagnosis, a pattern consistent with workplace strain or injury rather than diagnostic intensity. Workers with greater exposure to high-effect firms are more likely to enter disability and face higher mortality. The same gradient appears among workers displaced by firm closures, whose separations are involuntary: reemployment at a destination firm one standard deviation higher in the firm-effect distribution is associated with 0.22 and 0.08 percentage point higher probabilities of disability and death by 2019, about 6% of the respective means. The gradient is also present in GP expenditure, which is free at the point of use, and among high-expenditure workers facing a zero marginal price, so it is not driven by financial incentives to seek care. On average, then, the firm effects carry a health component.

Two features of the estimation matter for interpretation. First, limited worker mobility introduces substantial upward bias in the estimated dispersion of firm effects. I therefore apply the leave-one-out correction of Kline et al. (2020) and show that the estimates are robust to alternative sample-splitting procedures. Then, following Kline (2024), I focus on the magnitude of firm effects rather than their share of total expenditure variation. A one-standard-deviation difference in firm effects corresponds to 17.8 percent higher annual healthcare expenditure,

[3] I use 898 detailed Dutch standard industrial classification (SBI) codes, which are comparable in spirit to SIC or NAICS industry classifications.

yet firm effects account for only about one percent of the variance of annual log expenditure. The contrast reflects the large idiosyncratic variation in annual healthcare expenditure, which can obscure economically meaningful differences across firms when expressed as a variance share (Kline, 2024). Second, for the analyses of disability, mortality, and the relationship between healthcare expenditure effects and wage premia, I use sample splitting: firm effects estimated from one group of workers are assigned to the other. A worker's own healthcare history therefore does not enter the firm-effect measure assigned to him.

The estimates survive several threats to the identification in this design. Workers do not appear to move in anticipation of health changes: event studies show little evidence of pre-trends, and the dispersion among workers aged 45 or younger, for whom health-motivated moves are rarer, is 0.029 against 0.027 overall. The log-additive specification fits: allowing worker-by-firm effects raises the adjusted R^2 from 0.516 to only 0.532, mean residuals are near zero across deciles of worker and firm effects, and movers' expenditure changes are linear in coworker expenditure gaps. The dispersion is unchanged after controlling for pay, residential postcode, and local healthcare access; firm rankings, overall, agree across collar, wage, and baseline-expenditure; firm effects on future expenditure attenuate with the horizon; and a Poisson specification yields firm effects that closely align with those from the log-linear specification, with a slope of 1.06.

This paper adds healthcare expenditure, and the health outcomes it predicts, to the set of worker outcomes shaped by firms. Firms pay different wage premia to otherwise similar workers and account for a substantial share of earnings inequality (Kline, 2024; Card et al., 2013, 2018, 2016), differ in amenities and working conditions (Sockin, 2022; Lamadon et al., 2022), and treat workers differently by gender and origin (Kline et al., 2022); whether non-wage attributes offset or reinforce wage differences remains unsettled (Sockin, 2022; Lamadon et al., 2022; Sorkin, 2018). I estimate the variation in healthcare expenditure that firms induce among otherwise similar workers and show that it has longer-run health implications.

The wage result speaks to the compensating-differentials literature (Rosen, 1974; Marin and Psacharopoulos, 1982; Arnould and Nichols, 1983; Viscusi and Moore, 1987; Viscusi and Aldy, 2003; Mas and Pallais, 2017), which measures job risk at the level of industries or occupations, where fatality and injury rates are

observed. Here risk is measured at the firm, as the healthcare expenditure a firm generates for a given worker, and the compensating relationship appears only between industries: workers are compensated for risks visible at the industry level, not for firm-specific risk.

The findings also connect the firm-heterogeneity literature to the literature on health inequality, which has identified income, education, occupation, and place as important sources of health disparities (Cutler et al., 2006; Lleras-Muney et al., 2025; O'Donnell et al., 2015; Ravesteijn et al., 2018; Finkelstein et al., 2021, 2016). While earlier work estimates the causal effects of specific workplace conditions on average employee health (e.g., Hummels et al., 2025; Lee and Taylor, 2019), this paper quantifies the contribution of the firm environment to disparities in healthcare expenditure. In Hungarian data, Bíró and Elek (2025) show that moving to a higher paying firm raises the detection of chronic conditions through physician visits and cardiovascular medication. Here, I instead quantify how much otherwise similar workers' healthcare expenditure differs depending on their employer. In contemporaneous and independent work, Ahammer et al. (2026) use data from one Austrian state, restricted to larger firms, and a mover event-study design to show that roughly one quarter of the observed expenditure gap between origin and destination firms is transmitted to movers, and that accidents and injuries account for much of this response. They also estimate firm-specific healthcare expenditure and wage components and find that firms with higher effects have higher worker mortality and pay higher wage premia. This paper documents for the first time, the bias-corrected dispersion in healthcare expenditure induced by firms among otherwise similar workers, across the universe of firms of all sizes in the Netherlands. It further shows that most of this variation arises within narrowly defined industries, where firms with higher healthcare expenditure effects do not systematically pay higher wage premia.[4]

The findings of this paper point to a fiscal externality in collectively financed

[4]As Moghani (2026) shows, the mover-design event study coefficient of Finkelstein et al. (2016) is a transition-weighted statistic and therefore depends on which moves occur. It admits a cross-sectional interpretation only under restrictions on mobility patterns and, even then, concerns variation in firm averages, a different object from the contribution of firm effects to individual-level expenditure variation. Large variation in healthcare expenditure among workers within the same firm can therefore coexist with a sizable mover coefficient and a small firm-effect variance share, as documented here.

healthcare systems such as the Dutch one. Because health insurance contributions are not tied to realised healthcare expenditure at the firm level, firms do not bear the marginal cost of the healthcare expenditure generated by their workforce, which weakens incentives to reduce the healthcare expenditure their workplaces generate even under stringent safety regulation and experience rating in sickness and disability insurance. The fact that more than half of the dispersion remains within narrowly defined industrial codes suggests that this problem is not only about what firms produce but also about differences among firms doing the same work. While wages may partly compensate workers for the disutility of health risks—and within industries appear not to—they do not reimburse the public financing system for the resulting medical costs. This creates an unpriced fiscal externality and points to the value of policies that better align firms' incentives with the healthcare expenditure they generate.

Whether such policies can be targeted is a separate question, because the firm effects available to a regulator are estimated with error. Rather than recovering or classifying individual firms' latent effects (Kline et al., 2022, 2024), I ask whether the noisy estimates are informative enough to assemble a portfolio of high-expenditure employers: I rank firms on effects estimated from one half of the workforce and evaluate the portfolio on the other half in the spirit of Kline (2025), so that firms selected on estimation noise do not appear high-cost when evaluated. For the top decile of firms, moving their effects to the overall mean would reduce the portfolio's healthcare expenditure by 6.1%, and to their own industry means by 4.0%. The rule falls short of an infeasible oracle that ranks on latent effects at the most selective cutoffs, where ranking noise matters most, and closes the gap as targeting widens. Noisy administrative firm effects are thus a feasible targeting signal even where they cannot classify individual firms.

II Setting

Two features of the Dutch healthcare system matter for interpreting the estimates; Appendix A1 provides further institutional details. First, basic health insurance is universal, individual, and portable across jobs. Every resident is required to hold statutory basic insurance, which covers a common set of services across insurers and remains unchanged when a worker changes employer or becomes

unemployed. Cost sharing is governed by a statutory annual deductible, from which general practitioner (GP) care is exempt. GPs also serve as gatekeepers for non-urgent specialist and hospital care. Employers do not determine basic insurance coverage or cost sharing, although they may facilitate collective insurance arrangements, including supplementary coverage. Because my data cover claims under the statutory basic package, differences in expenditure across firms cannot be attributed to differences in employer-provided insurance coverage.

Second, employers nonetheless carry substantial responsibility for their workers' health. Under the Working Conditions Act (*Arbowet*) firms must monitor workplace risks, contract occupational-health services, provide access to a company doctor, and continue wage payments for up to two years during illness, and these duties are reinforced by experience rating in disability insurance. Day-to-day medical care, however, runs through the GP-based system, and occupational physicians provide work-related advice and reintegration support rather than treatment. The setting therefore combines employer responsibility for prevention and sickness management with no employer control over the content or financing of ordinary care. The expenditure that workplaces generate is paid by the collective insurance system, which is the fiscal externality the paper returns to in Sections VI.III and VII.

III Framework

This section defines the object of interest and the assumptions required for its interpretation. Appendix A4 provides a health-capital foundation for the framework.

The object. Healthcare expenditure is the price of care multiplied by the quantity received, which reflects both a worker's healthcare needs and the extent to which those needs translate into healthcare use. In the Dutch setting, employers do not set the prices of covered care, but both need and care-seeking may have persistent worker and firm components. Writing these components multiplicatively, and allowing for the transformation used in the empirical analysis, expected expenditure can be represented as

$$\mathbb{E}[c_{it} \mid i, j, t] = \exp\Big(\alpha_i^L + \psi_{J(i,t)}^L + x_{it}'\beta^L\Big), \tag{1}$$

where α_i^L captures persistent worker differences, ψ_j^L captures firm effects, and $x_{it}'\beta^L$ captures observed time-varying determinants, including age profiles of healthcare use that vary by education. In the empirical analysis, I estimate the additive model for transformed expenditure,

$$\log(c_{it} + 1) = \alpha_i + \psi_{J(i,t)} + x_{it}'\beta + r_{it}, \tag{2}$$

which is the two-way fixed-effects model estimated in Section V. This specification identifies firm effects on log expenditure under the AKM assumptions.[5]

Moving the same worker from firm A to firm B, holding his characteristics fixed, changes expected expenditure plus one by the factor $\exp(\psi_B - \psi_A)$, approximating the proportional change in expected expenditure when expenditure is sufficiently large. Importantly, the multiplicative structure allows firm effects to interact with worker characteristics in levels: the same firm difference produces a larger euro effect for workers with higher baseline expenditure, even though the proportional effect is common across workers.

The object of interest is the employment-weighted dispersion of ψ_j across firms. I report the percentage difference associated with a one-standard-deviation gap in firm effects, $\exp(\sigma_\psi) - 1$, and the gap between the top and bottom quintiles of the firm-effect distribution, and compare it to gradients commonly reported by income, wealth, and education. I focus on these magnitudes rather than the share of annual log expenditure variation explained by firms. That share can be small even when differences across firms are economically substantial, because the denominator is dominated by idiosyncratic variation in annual healthcare expenditure. It also depends on the volatility of the outcome, rather than solely on the dispersion of workplace effects (Kline, 2024).

The mover design event-study in Section IV.I serves a different purpose. By relating expenditure changes around job moves to the gap in average expenditure between destination and origin firms, it provides evidence on the timing of

[5]Suppose $c_{it} + 1 = \exp(\alpha_i^L + \psi_j^L + x_{it}'\beta^L)u_{it}$, where $\mathbb{E}[u_{it} \mid i, j, t] = 1$. A sufficient condition for the two specifications to have the same firm effects is that $\mathbb{E}[\log u_{it} \mid i, j, t]$ is constant, so the difference between the log of expected expenditure and expected log expenditure can be absorbed into the intercept. More generally, this difference must not introduce additional firm components. This restriction is not required to estimate the AKM model for log expenditure, but is needed for its interpretation in terms of proportional differences in expected expenditure.

workplace effects and possible pre-trends. I do not use its coefficient to quantify their magnitude as the slope depends on which firms movers connect and the composition of their average outcomes, and its interpretation requires additional restrictions on mobility patterns (Moghani, 2026).

What has to hold. Four conditions carry the interpretation of ψ_j as the effect of the firm on a given worker, each examined in the paper. Workers must not move in anticipation of changes in their own health, and must not follow health trajectories that differ by the type of firm they join (exogenous mobility): Sections IV.I, V, and VI. Worker and firm components must enter additively in logs, so that a firm shifts all its workers' expenditure by the same proportion: Sections V and VI. Firm effects must be constant over the 2009–2016 window; the short panel and the limited dynamics after a move in Section IV.I support this. And because each ψ_j is estimated from the moves of a finite number of workers, the plug-in dispersion of the estimates overstates the dispersion of the true effects, and any regression with an estimated firm effect on the right-hand side is attenuated; the leave-out correction of Kline et al. (2020) and the sample-split designs of Section V and Appendix A6 address this.

Expenditure and health. Equation (1) makes ψ_j a net effect: a firm can generate expenditure through the health consequences of its workplace—injury, strain, stress, exposure—or through care use conditional on health—time off for appointments, screening, occupational health services, norms about seeking treatment. The Dutch setting removes the channels that run through the insurance contract, since coverage, networks, and cost sharing are the same across firms, GP care is free at the point of use, and income effects on care are small (Miller et al., 2024; Cesarini et al., 2016). It does not remove care-seeking differences that arise inside the workplace. The paper therefore tests the (average) health interpretation rather than assuming it: whether moves toward higher-effect firms raise medication use in the classes that treat injury and strain rather than those that follow screening; whether exposure to higher-effect firms predicts disability and mortality, which care-seeking does not produce; whether the gradient appears in GP care, which is free, and among workers who have exhausted their deductible; and whether it

survives controlling for pay, residential location, and local healthcare access.[6]

IV Dataset Description, Sample Selection, and Descriptive Statistics

I use Dutch administrative employer–employee data covering all male employees living in the Netherlands.[7] The data contain information on employer–employee matches, demographics, income, contract start and end dates, and firms' Dutch standard industrial classification (SBI) codes.[8] I also observe workplace postal codes, individuals' residential locations over time, highest completed education, and annual total and component-level basic healthcare expenditure (general practitioner, medication, specialized mental healthcare, and hospital care).[9]

Employment information is available annually from 2009 to 2016, while healthcare expenditure is observed annually from 2009 to 2019. I focus on male employees aged 25–65 who, in each year, work at least 11 months in jobs with a contract of at least 0.9 full-time equivalent.[10,11] When an individual is employed by multiple firms within the same year, I assign the worker to the firm where he spent the most

[6] These tests speak to the average firm effect. They do not identify, firm by firm, how much of an effect is health and how much is care use; a firm with a low estimated effect may employ workers whose health is good, who seek less care, or whose care falls outside the basic package, and the lower tail of the distribution is read in that light in Sections VI and VII.

[7] The corresponding results for female employees are reported in Appendix A9. For women, the patterns in Figure A7 indicate that firm changes are endogenous to childbearing decisions. Given the direct and indirect healthcare expenditure of childbearing (Ahammer et al., 2023; García-Gómez et al., 2026), estimates based on firm moves should therefore be interpreted with caution for female workers.

[8] Standaard Bedrijfsindeling (SBI) is the Dutch standard industrial classification used by Statistics Netherlands and the Chamber of Commerce to assign each firm an industrial code. The hierarchy spans five levels, with up to five digits identifying increasingly specific industries (e.g. code 01131 "Growing of vegetables in open fields" versus code 01132 "Greenhouse growing of vegetables"). The analysis data contain 898 unique SBI codes.

[9] The baseline data do not contain occupation information. In Section VI, I examine heterogeneity by education-proxied collar type and show that the estimated firm effects are largely shared across groups. Moreover, I document substantial firm-effect dispersion within narrowly defined SBI industrial codes, suggesting that the results are not driven solely by broad industry or occupation groups.

[10] Full-time employment in the Netherlands ranges from 38 to 40 hours per week.

[11] Requiring at least 0.9 FTE excludes part-time workers, who constitute less than 20% of employed men between ages 25 and 65 ((CBS), 2025), and removes years in which employees shift into part-time work or leave employment. Sick employees continue to receive employer-paid wages for up to two years before entering sickness or disability schemes, so these periods remain recorded as employment, which is important for observing healthcare expenditure during sickness.

months, since healthcare expenditure is observed annually.[12] These restrictions reduce reliance on functional-form assumptions relating work intensity to healthcare expenditure and simplify the assignment of workers to firms. A consequence is that workers who move late in the year remain assigned to their origin firm for that year.

For these full-time workers, I observe background characteristics such as birth year, migration background, and—for most individuals—education.[13] The data further include annual salary, employer identifier, employer sector, and firm size.

For each worker, I observe total and category-specific basic healthcare expenditure from 2009 to 2019. I observe all claims billed to the basic insurance package.[14] Total expenditure combines insurer reimbursements and individuals' deductible payments, which are not separately identified in the data. The categories are: (1) primary care (general practitioner services); (2) medication; (3) specialized mental healthcare outside primary care (psychology and psychiatry); and (4) hospital care, covering specialist treatments and inpatient admissions.[15] Employees in the defence sector (SBI code 8422) obtain healthcare through the Ministry of Defence, which operates a parallel system not captured in these data. I therefore exclude defence-sector firms and their employees, who account for about 1.5% of the sample.

The sample is restricted to firms connected through worker mobility, which is required for estimating the AKM model because it relies on an employer network linked by movers (Abowd et al., 1999). To address limited-mobility bias, I apply the leave-one-out correction proposed by Kline et al. (2020), which removes each employer–employee match iteratively during estimation. This procedure requires that the network remains connected after excluding any given match. As a result, the sample is further restricted to ensure connectivity under leave-one-out removal. Summary statistics for the main sample, the connected sample, and the movers

[12] The 0.9 FTE requirement effectively rules out simultaneous substantial contracts.

[13] Education information is complete for cohorts born after 1990. For older cohorts, education is observed only for those who interacted with administrative processes requiring educational documentation. Individuals with missing education are treated as a separate category. Results are robust to excluding these individuals.

[14] Care outside the basic package is therefore not observed. This is most relevant for short-term physiotherapy, which is often financed out of pocket or through supplementary insurance.

[15] Together, these categories represent about 90% of total basic healthcare expenditure; remaining expenditure includes categories such as physiotherapy and patient transport.

sample appear in Table 1.

Table 1 shows that restricting the data to the connected set leaves the composition of the sample essentially unchanged. Relative to the initial full-time sample, workers in connected firms have very similar age, migration background, education, birth rates, and healthcare expenditure, with only small increases in average age (44.2 vs. 44.0), wages, firm size, and total healthcare expenditure. By contrast, within the connected set, movers differ systematically from stayers. Movers are on average about three years younger (42.6 vs. 45.6), somewhat better educated (higher shares with upper secondary, bachelor, and master degrees and fewer missing-education records), and earn slightly higher hourly wages. They are also more likely to experience a birth in a given year (5% vs. 3%) and have marginally fewer children on average, consistent with mobility being concentrated among younger workers. Movers tend to work in slightly smaller firms. Finally, movers have lower total healthcare expenditure, hospital expenditure, and specialized mental-health expenditure than stayers, consistent with their younger age and suggesting that movers are somewhat positively selected on health. The AKM framework assumes that firm effects are common across subgroups, so that effects identified from movers also apply to stayers and to workers with different observable characteristics; this restriction is not directly testable for stayers, who never switch firms. In Section VI, I examine robustness of findings across subsamples defined by age, income, and baseline healthcare expenditure and show that the estimated firm effects and their dispersion are similar. In Section V, I also provide additional empirical support for the common-firm-effect assumption.

For the main analysis, I model annual healthcare expenditure in logs. Under a multiplicative structure of health investments, such as Cobb–Douglas, taking logs yields additive individual and firm components.[16] This property is important for interpreting firm effects as "shared" across subgroups—such as blue- and white-collar workers, high- and low-income workers, or high- and low-expenditure individuals—because the same firm component on the log scale implies proportionally similar effects while preserving larger level effects for workers with higher

[16] Appendix A4 shows, in a Grossman-style framework and up to standard approximations, that multiplicative individual and firm components in health depreciation and care-seeking costs deliver a log-additive decomposition into individual, firm, and age terms.

	Initial Sample *(Full-time Employees)* N = 18,705,218	**Connected Set** *(Connected Firms)* N = 16,976,071	**Stayers** *(Connected Firms)* N = 9,236,759	**Movers** *(Connected Firms)* N = 7,739,312
Variable	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
Age	43.99 (10.48)	44.24 (10.39)	45.60 (10.51)	42.62 (10.01)
Dutch background (dummy)	0.83 (0.38)	0.83 (0.37)	0.83 (0.37)	0.83 (0.37)
First-generation migrant (dummy)	0.10 (0.30)	0.10 (0.30)	0.10 (0.31)	0.09 (0.29)
Second-generation migrant (dummy)	0.07 (0.25)	0.07 (0.25)	0.06 (0.24)	0.07 (0.26)
Education: Elementary (dummy)	0.02 (0.16)	0.02 (0.15)	0.02 (0.15)	0.02 (0.15)
Education: Lower sec./MBO1 (dummy)	0.06 (0.24)	0.06 (0.24)	0.06 (0.24)	0.06 (0.24)
Education: Upper sec./MBO (dummy)	0.23 (0.42)	0.23 (0.42)	0.20 (0.40)	0.26 (0.44)
Education: Bachelor (HBO/WO) (dummy)	0.17 (0.37)	0.17 (0.38)	0.15 (0.36)	0.20 (0.40)
Education: Master/Doctorate (dummy)	0.11 (0.31)	0.11 (0.31)	0.10 (0.29)	0.12 (0.33)
Education: Missing (dummy)	0.41 (0.49)	0.41 (0.49)	0.46 (0.50)	0.34 (0.47)
Total # children	1.38 (1.23)	1.40 (1.23)	1.44 (1.24)	1.35 (1.21)
New child (dummy)	0.04 (0.20)	0.04 (0.20)	0.03 (0.18)	0.05 (0.22)
Gross payment of contract (€)	49,616 (46,120)	50,545 (46,821)	50,510 (47,553)	50,586 (45,932)
Mover (dummy)	0.44 (0.50)	0.46 (0.50)	0.00 (0.00)	1.00 (0.00)
Full-time factor (FTE)	0.99 (0.02)	0.99 (0.02)	1.00 (0.02)	0.99 (0.02)
Estimated hourly wage (€)[a]	24.32 (22.55)	24.76 (22.91)	24.48 (23.00)	25.09 (22.79)
Business size (employees)	692.46 (800.51)	741.76 (806.30)	763.30 (804.44)	715.98 (807.77)
Total healthcare expenditure (€)	1,143 (4,800)	1,149 (4,780)	1,260 (5,263)	1,015 (4,127)
Mental health (specialist) (€)	84.46 (1,434)	83.22 (1,389)	90.03 (1,573)	75.10 (1,130)
Hospital expenditure (€)	687.31 (4,040)	691.63 (4,036)	767.33 (4,451)	601.28 (3,474)
GP expenditure (€)	109.70 (69.00)	110.09 (68.58)	111.42 (70.71)	108.51 (65.92)
Number of Observations	18,705,218	16,976,071	9,236,759	7,739,312

[a] Hourly wage computed from gross contract payment assuming a full FTE works 52 weeks × 40 hours per week.

Notes: Universe of male employees aged 25–65 observed between 2009 and 2016. The sample is restricted to full-time jobs with FTE ≥ 0.9 and at least 11 months employed in a given year; if an individual is employed by multiple firms within the same year, the employer with the largest number of months is used. "Connected set" refers to employees in firms that belong to the leave-one-out connected component of the employer network, i.e. the set of firms that remain connected through worker moves when any single worker–firm match is removed. "Stayers" are worker–year observations for individuals who remain with the same employer throughout the 2009–2016 period; "Movers" are worker–year observations for individuals who change employer at least once over this period. Sectors are defined on the basis of the first two digits of the Dutch Standaard Bedrijfsindeling (SBI) classification, grouped into 17 broad sectors. Business size is measured as the midpoint of the categorical firm–size intervals reported in the data (interpreted as the approximate number of employees at the firm in the given year). Healthcare variables refer to annual expenditure under the basic health insurance package (insurer payments plus deductible). "Total healthcare expenditure" is the sum of all basic-package claims; "Mental health (specialist)" includes specialized mental healthcare outside primary care; "Hospital expenditure" cover specialist and inpatient hospital care; and "GP expenditure" refer to primary care delivered by general practitioners. Employees in the defence sector (SBI 8422), who obtain healthcare through a separate system, are excluded from all samples.

Table 1: Summary Statistics

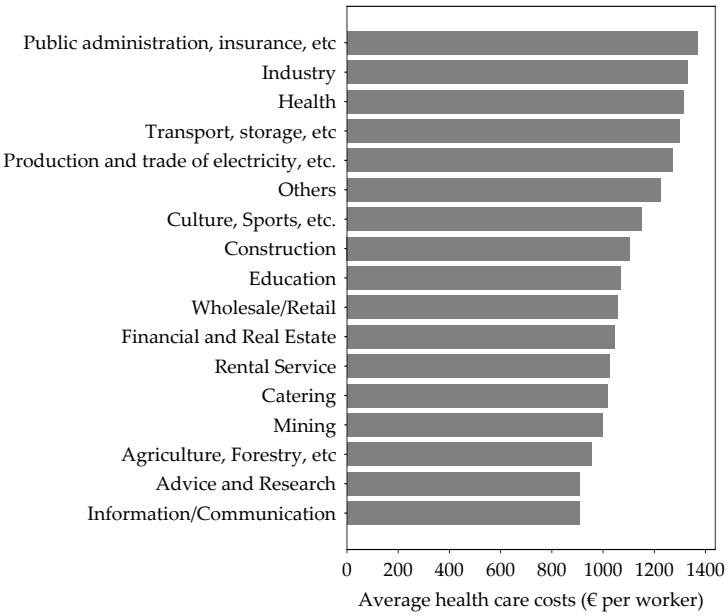
individual expenditure.[17][18]

A common concern with log transformations is the treatment of zero outcomes, since estimates can then depend on the measurement unit and on the mix of extensive- and intensive-margin responses (Chen and Roth, 2024; Mullahy and Norton, 2024). In my data, however, the share of observations with zero total healthcare expenditure is very small (Figure A1). As a functional-form check, I also estimate firm effects using a Poisson specification; regressing these estimates on the baseline log-linear firm effects yields a slope of 1.056 (Appendix Figure A13 and Table A17). Using contemporaneous annual healthcare expenditure is also appropriate in this setting. Dutch employers must continue wage payments for up to two years during sickness, so periods of poor health remain attached to the firm where they arise. Moreover, Table A16 shows that firm effects on future healthcare expenditure are noticeably smaller than firm effects on contemporaneous expenditure. This supports using contemporaneous healthcare expenditure as the baseline outcome and interpreting the estimated firm effects as contemporaneous workplace-related contributions to healthcare expenditure.

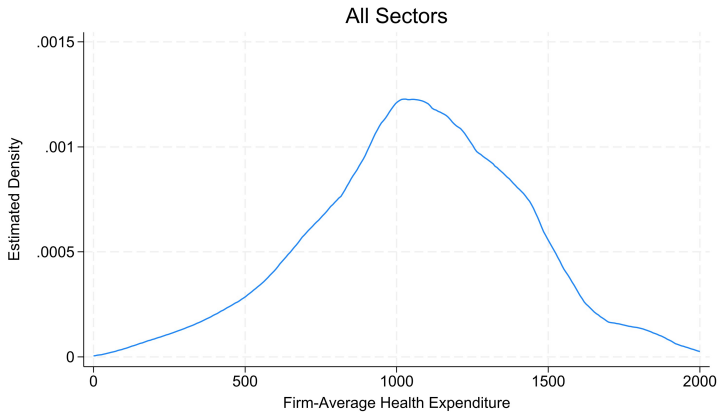
Firms differ in the average basic healthcare expenditure of their workers, both within and across sectors (Figure 1). These raw differences, however, cannot be interpreted as firm effects. Workers and firms sort: lower-skilled and otherwise disadvantaged workers are more likely to be employed in low-wage, low-amenity firms (Card et al., 2013; Sockin, 2022), and health is strongly graded by socioeconomic status (O'Donnell et al., 2015; Danesh et al., 2024). As a result, firms employing workers with poorer underlying health or higher healthcare needs will mechanically display higher average healthcare expenditure even if the workplace itself has no effect on health. I therefore use within-worker changes in healthcare expenditure around firm moves to separate persistent worker components from firm components.

[17] Section VI shows that firm effects and their variance are broadly similar across these subgroups, consistent with such a shared firm component.

[18] Additionally, the KSS correction is derived for OLS estimators in linear models; this motivates the use of a linear AKM specification in logs rather than nonlinear alternatives.



(a) Mean basic healthcare expenditure by industrial code (€, per worker-year)



(b) Distribution of firm-level mean basic healthcare expenditure (€, per worker-year)

Figure 1: Raw sectoral and firm-level variation in mean annual basic healthcare expenditure

Note: Sample of male employees aged 25–65 in the connected set of firms used for the AKM estimation, observed between 2009 and 2016. Workers are restricted to full-time jobs with FTE ≥ 0.9 and at least 11 months in a given year; if an individual is employed by multiple firms within a year, the firm with the largest number of months is used. Healthcare expenditure refers to annual expenditure under the statutory basic health insurance package (insurer payments plus deductible) and exclude care covered only by supplementary insurance or in parallel systems (such as the defence sector, which is excluded). Sectors in panel (a) are defined from two-digit industrial codes grouped into 17 broad sectors. Panel (b) shows the cross-sectional distribution of firms' mean annual basic healthcare expenditure per worker-year, where each firm's mean is computed over all full-time male worker-year observations in the connected sample.

IV.I Mover evidence on timing and pass-through

A natural next step is to examine healthcare expenditure among workers who move between firms with different average expenditure. The event study provides two pieces of evidence. Pre-move outcomes show whether workers moving toward firms with higher average coworker expenditure were already on different trajectories; post-move outcomes show whether and when their own expenditure adjusts after arrival.

Mover designs measure pass-through along observed moves: the coefficient relates a worker's change in expenditure to the origin–destination gap in coworker expenditure. This transition-based object is shaped by the routes movers take through the firm distribution. Even under conditions that give it a cross-sectional interpretation, that interpretation concerns variation across firm averages, not firms' contribution to individual-level healthcare-expenditure variation (Moghani, 2026). Appendix A3 develops this distinction.

The event study and the main firm-effect estimates consequently have distinct roles. I use the event study to assess pre-trends and trace the timing and pass-through of healthcare expenditure following a change of employer. I estimate firm-effect dispersion and its contribution to individual-level expenditure variation directly with the bias-corrected two-way fixed-effects decomposition in Section VI.

Here, I focus on individuals who switch employer exactly once during 2009–2016 and work with age- and year-residualised log healthcare expenditure. Let $\hat{h}_{it}$ denote the residual from a regression of log-healthcare expenditure ($h_{it} = \log(c_{it}+1)$) on age and calendar-year dummies. For each mover i, define

$$D_i \equiv \bar{\hat{h}}^{(-i)}_{d(i)} - \bar{\hat{h}}^{(-i)}_{o(i)},$$

where $\bar{\hat{h}}^{(-i)}_{o(i)}$ and $\bar{\hat{h}}^{(-i)}_{d(i)}$ are the leave-one-out mean residualised expenditure at the origin and destination firms. Thus, D_i measures how much higher or lower average residualised expenditure is at the destination firm relative to the origin.

Figure 2a relates the change in a mover's own residualised expenditure between the pre- and post-move years to D_i nonparametrically.[19] Two features stand out.

[19] Appendix Figure A6 reports a binscatter with the linear approximation and the distribution

First, the relationship is close to linear: workers who move to firms with higher average residualised coworker expenditure experience higher own expenditure after the move. Second, the fitted relationship has a positive intercept at $D_i = 0$: even movers between firms with similar average expenditure exhibit a modest increase in healthcare expenditure around the job change. This pattern is consistent with a common cost of moving—for example, due to job loss, or displacement—that shifts expenditure for all movers, while the slope with respect to D_i measures pass-through associated with the origin–destination gap in average coworker expenditure. In turn, the approximate linearity of the relationship is consistent with the log-additive specification motivated above.[20]

To study timing more systematically, I next estimate an event-study specification that follows one-time movers from four years before to four years after the switch. Using event times $\mathcal{T} = \{-5, \ldots, 5\}$ and taking $t = -2$ as the reference year, I estimate

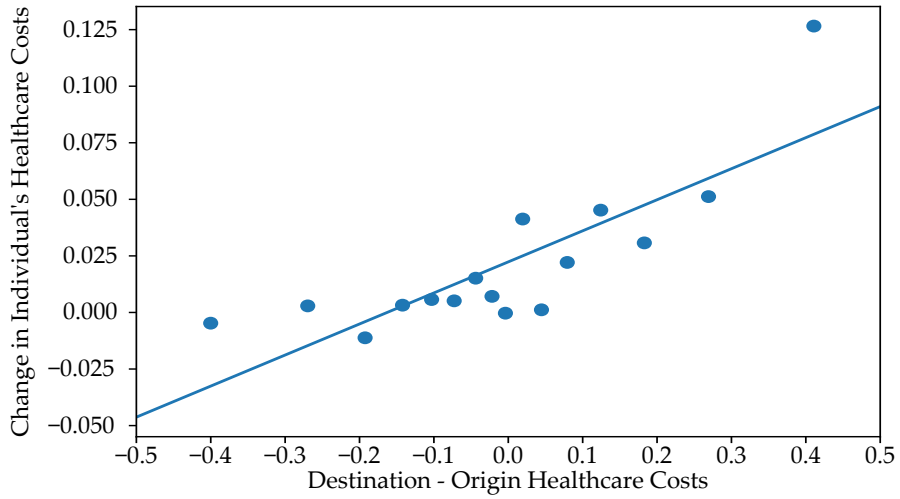
$$\hat{h}_{it} = \eta_i + \sum_{\tau \in \mathcal{T},\ \tau \neq -2} \lambda_\tau \mathbf{1}\{t = \tau\} + \sum_{\tau \in \mathcal{T},\ \tau \neq -2} \theta_\tau \mathbf{1}\{t = \tau\} D_i + u_{it}, \tag{3}$$

where η_i are individual fixed effects and $\lambda_{-2} = \theta_{-2} = 0$, so all coefficients are relative to event time $t = -2$.

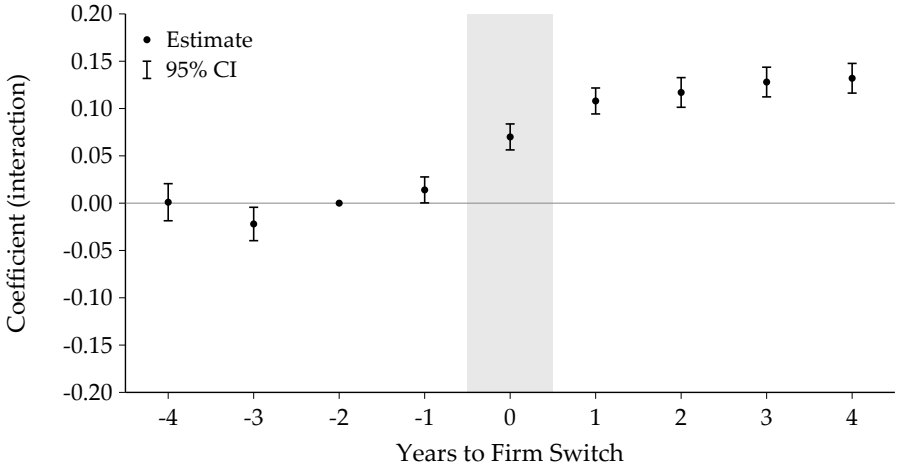
Figure 2b plots the coefficients θ_τ, which trace how pass-through evolves around the move. The event study delivers three main messages. First, the interaction coefficients, θ_τ, are small and close to zero in the years before the move, in line with no systematic pre-trends correlated with the origin–destination gap in average coworker expenditure. This argues against selection into destination firms based on differential pre-move health shocks. Second, θ_τ jumps sharply at the time of the move, showing that workers' expenditure responds promptly. Third, only modest further adjustment follows, and the response stabilizes within the first few years. This limited adjustment shows that pass-through in the mover design occurs

of D_i across movers. The figure focuses on the central range of D_i, where almost all moves lie; estimates are naturally less precise in the tails.

[20] By contrast, the origin–destination cell approach of Card et al. (2018) effectively imposes symmetry: the estimated effect of moving from firm a to firm b is constrained to be the mirror image of the effect of moving from b to a. In the setting of healthcare expenditure—where job loss and search themselves have health consequences—working at the individual level instead allows these common moving costs to show up as an intercept shift, while the slope with respect to D_i measures pass-through associated with the origin–destination gap in average coworker expenditure and provides a direct functional-form check. The four group split results are shown in Figure A8.



(a) Average change vs. exposure gap



(b) Event-study of mover responses

Figure 2: Workers' healthcare expenditure around moves between firms

Note: Sample of full-time male employees aged 25–65 in connected firms who switch employer exactly once during 2009–2016. Total basic-package healthcare expenditure is transformed as $\log(c_{it}+1)$ and residualised on age and calendar-year dummies. The exposure gap D_i is destination minus origin leave-i-out mean residualised log expenditure. Panel (a) shows the movers' changes in residualised expenditure from before to after the move against D_i. Panel (b) reports pass-through coefficients from the event-time interactions with D_i in equation (3), relative to $t = -2$.

primarily at the move and stabilizes thereafter; it is also consistent with a time-invariant firm component in annual expenditure. Along with the approximately linear relationship in panel (a), these patterns support the log-additive two-way fixed-effects specification used in the next section. Figure 4 provides additional evidence on interpretation: moves toward firms with higher average coworker expenditure are followed by increases in medication categories associated with pain, inflammation, or related physical strain.

V Method

To quantify how much firms contribute to variation in healthcare expenditure, I estimate a two-way fixed-effects model of log annual healthcare expenditure in the spirit of Abowd et al. (1999):

$$h_{it} = \alpha_i + \psi_{J(i,t)} + x'_{it}\beta + r_{it}, \tag{4}$$

where h_{it} is log basic-package healthcare expenditure (total paid by the insurer plus deductible paid by the individual) of worker i in year t, $J(i,t)$ indexes the firm employing i in t, and x_{it} collects time-varying observables (a 10-year split quadratic age profile by education group and calendar-year dummies). The term α_i is a worker fixed effect capturing time-invariant determinants of expenditure such as genetic endowments, early-life health, persistent preferences for care, and long-run earnings capacity. The term $\psi_{J(i,t)}$ is a firm fixed effect capturing each firm's persistent contribution to workers' healthcare expenditure after removing time-invariant worker heterogeneity and life-cycle and calendar-time factors, and is interpreted below as a net workplace-related expenditure effect.

Annual basic-package healthcare expenditure is the main outcome because it is measured for every worker in every year and varies continuously rather than as a rare event. These features make it preferable, for the AKM analysis, to "hard" health outcomes such as mortality or disability, which are low-frequency tail events, observed at most once over the sample window and recorded as binary indicators. The outcome also maps naturally into a Grossman-style interpretation in which firms may affect expenditure through the workplace-specific health-depreciation rate as well as through other determinants of care use. Appendix A4 motivates

the log-linear specification in a simple Grossman-style health-capital framework. Under a multiplicative structure of health investments and care-use determinants, a log specification yields additive worker and firm components, so that a given difference in firm effects implies the same percentage difference in expenditure for workers with different baseline expenditure levels, while allowing larger level effects (in euros) for higher-expenditure workers.

V.I Identification and assumptions

Equation (4) relies on several assumptions.

Individual health trajectories. The residual r_{it} captures, among other things, individual-specific health trajectories over the life cycle that are not fully absorbed by the flexible age and year controls in x_{it}. Following Card et al. (2013), I assume that any remaining worker-specific trend by age has zero mean and is not systematically related to the firm effect of the worker's employer, conditional on the included age–by–education profiles and calendar-year dummies. This rules out unobserved individual characteristics that generate differential age patterns which are themselves correlated with the firm effect at the worker's destination. The rich age controls and the absence of pre-trends in the mover event studies (Figure 2) support this assumption: there is no evidence that workers who move toward firms with higher versus lower average coworker expenditure already display diverging healthcare trajectories in the years before the move.

Additive separability in logs. The model assumes that worker and firm effects enter additively in logs, i.e. there are no systematic worker–firm interaction terms in r_{it}. To gauge the importance of such interactions, I estimate a saturated model with contract fixed effects, replacing $\alpha_i + \psi_{J(i,t)}$ by a full set of worker×firm dummies, and compare the fit to the two-way fixed-effects model. As shown in Table 2, the R^2 increases from 0.599 to 0.657 when moving from the AKM to the saturated specification, but the adjusted R^2 rises only modestly, from 0.516 to 0.532, despite an increase of 56% in the number of parameters.

Figure 3 plots the mean residuals $\hat{r}_{it}$ from the AKM model by deciles of the estimated worker and firm effects. While the the variance of $\hat{r}_{it}$ in the sample

is close to one, the averages are close to zero across almost all cells, with small deviations only in the cell combining the lowest worker and firm deciles, where truncation at zero and extensive-margin responses are most relevant. This pattern suggests that firm effects may differ somewhat for individuals at the very bottom of the expenditure distribution, but that additive separability in logs is a good approximation for the large majority of workers. Additionally, in Section VI, I re-estimate firm effects separately for subsamples defined by education (blue- vs. white-collar), income, and baseline expenditure and show that the dispersion of firm effects and their correlation across subsamples are high, supporting the interpretation of a common firm component.

A further functional-form check comes from the mover design in Figure 2a. The non-parametric relationship between the change in an individual's residualised log expenditure and the exposure gap D_i is approximately linear, with a common positive intercept at $D_i = 0$. The approximate linearity with a constant intercept is therefore consistent with the log-linear, additively separable structure in Equation (4).

	Two-Way Fixed Effects		**Saturated with Job ID**	
Model	R^2	Adjusted R^2	R^2	Adjusted R^2
	0.599	0.516	0.657	0.532
No. of Parameters	2,901,488		4,527,516	
# Person–Year Observations: 16,976,071				

Note: Sample of full-time male employees aged 25–65 in the connected set of firms, observed between 2009 and 2016 (# person–year observations = 16,976,071). The dependent variable is log annual basic-package healthcare expenditure (insurer payments plus deductible), $h_{it} = \log(c_{it} + 1)$. The two-way fixed-effects model (Eq. 4) includes worker and firm fixed effects and controls for age–by–education profiles and calendar-year dummies. The saturated model replaces worker and firm fixed effects with job-ID (worker×firm) fixed effects while keeping the same controls. The table reports R^2, adjusted R^2, and the total number of estimated parameters in each specification.

Table 2: Two-Way Fixed Effects vs. Saturated (Job-ID) Model

Time-invariant worker and firm effects. I treat α_i and ψ_j as constant over the 2009–2016 period. Any transitory firm shocks or worker-specific changes in long-run health are absorbed into r_{it}. Given the relatively short panel and the limited

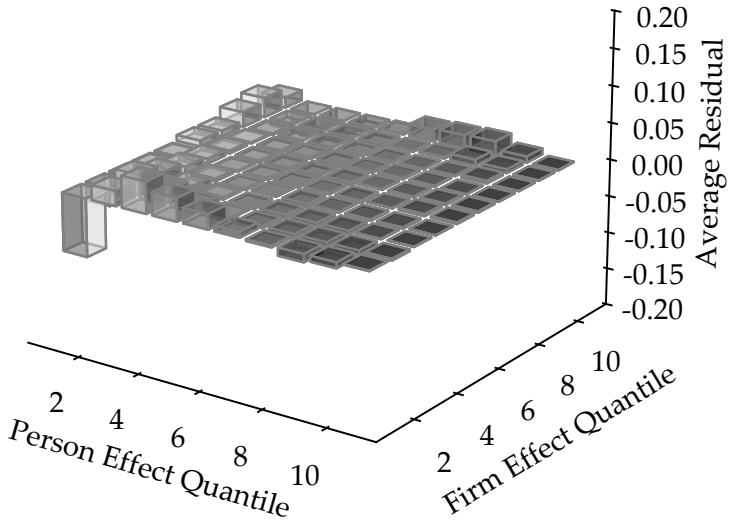


Figure 3: Mean residuals of the two-way fixed-effects model (Eq. 4) by decile of firm and individual fixed effects.

Note: Sample of full-time male employees aged 25–65 in the connected set of firms, observed between 2009 and 2016. The figure plots mean residuals from the two-way fixed-effects regression in Eq. 4, where the dependent variable is log annual basic-package healthcare expenditure, $h_{it} = \log(c_{it} + 1)$, and the controls are worker and firm fixed effects, age–by–education profiles, and calendar-year dummies. Residuals are averaged within 10×10 cells defined by deciles of the estimated worker and firm fixed effects (decile 1 = lowest, decile 10 = highest). The variance of the residuals is approximately 1, so the cell means shown in the figure represent small deviations relative to the overall idiosyncratic dispersion.

dynamic adjustment after job moves shown in Figure 2b, this appears to be a reasonable approximation.[21]

Selective mobility and pre-trends. A central concern is that workers may move to firms with different health environments in anticipation of future health shocks, violating the implicit exogeneity condition that, conditional on α_i, and age-education patterns and time dummies, the firm effect is unrelated to unobserved changes in health. I provide two pieces of evidence against strong anticipatory mobility. First, the mover event study in Figure 2b, which interacts event-time indicators with the exposure gap D_i defined in Section IV, shows that the coefficients θ_τ are flat and close to zero in the years before the move, while expenditure differentials open up sharply at the time of the move and then remain roughly constant. Second, Table A7 shows that excluding workers above age 45, for whom future health declines are more common and predictable, leaves the estimated dispersion of firm effects robust. This age split is consistent with life-cycle health-capital models in which individuals are close to full health throughout most of their prime working years and more rapid health deterioration sets in mainly at older ages (Galama and Van Kippersluis, 2019). Moreover, existing evidence for the Netherlands documents substantial worker and job dynamics, with job-to-job flows an important component of labour reallocation (e.g. Adema et al., 2025), making it plausible that many employer switches reflect career progress or job matching rather than responses to anticipated health problems. Taken together, these patterns suggest that selective mobility based on unobserved, destination-specific health trends is limited.

Idiosyncratic shocks. The remaining component of r_{it} reflects idiosyncratic, time-varying shocks to healthcare use that are not captured by the fixed effects or observed covariates. I assume that these shocks have zero mean and are not systematically related to firm assignment or the decision to move, conditional on α_i, ψ_j, and x_{it}. The large variance of r_{it} and the small average residuals across worker–firm cells in Figure 3 support this view that most of the residual variation reflects noise rather than systematic departures from the model structure.

[21] Allowing fully time-varying firm-specific health effects would require a longer panel and richer mobility than the data permit.

V.II Object of interest and inequality interpretation

The main object of interest is the dispersion of firm effects,

$$\sigma_\psi^2 \equiv Var(\psi_j), \qquad \sigma_\psi = \text{sd}(\psi_j).$$

A move from firm A to firm B changes a worker's expected log expenditure by $\Delta\psi = \psi_B - \psi_A$, holding α_i and x_{it} fixed, so that

$$\frac{\mathbb{E}[c_{it}^{(B)}]}{\mathbb{E}[c_{it}^{(A)}]} = \exp(\Delta\psi).$$

In particular, expected expenditure for the same worker differs by a factor $\exp(\sigma_\psi)$ between firms whose effects are one standard deviation apart. I therefore report σ_ψ and $\exp(\sigma_\psi) - 1$ as the primary measures of how much firms can shift individual healthcare expenditure.

I use this absolute measure because it answers the substantive question directly: how much expected healthcare expenditure differs across firm environments for the same worker. A variance share instead scales firm effects by total individual-level variation, much of which reflects short-lived and idiosyncratic healthcare shocks, and can therefore obscure economically meaningful workplace differences (Kline, 2024). The mover analysis answers a complementary question—how much of the observed origin–destination gap passes through along the transitions seen in the data—rather than directly estimating the dispersion of firm effects across the workforce or their contribution to individual-level expenditure variation (Moghani, 2026). I therefore estimate σ_ψ directly and translate it into changes in workers' expected expenditure.

This choice aligns closely with how health disparities are typically measured in the literature. For wages or incomes, univariate inequality indices (such as the variance or the Gini coefficient) have a natural welfare interpretation because the variable is monetary, cardinal, and directly comparable across individuals. For health expenditure, by contrast, such univariate indices are used much less frequently, in part because a large share of the dispersion reflects idiosyncratic biological risk and random health shocks[22], and there is no clear normative benchmark

[22] In classic earnings data, regressions of log earnings on basic human-capital and labor-supply

for what constitutes "acceptable" or unavoidable dispersion.[23] Instead, empirical work typically studies socioeconomic-related health disparities using bivariate measures based on the joint distribution of health and another variable such as income, education, or socio-economic status—for instance concentration indices, slope indices of disparities, or health gaps across income, education, or regional groups (e.g. Kakwani et al., 1997; Wagstaff, 2002; Wagstaff and Doorslaer, 2004; O'Donnell et al., 2015; Finkelstein et al., 2016). These measures answer questions of the form "by how much is health worse for individuals in disadvantaged positions?" rather than "what fraction of the total variance is explained by a given factor?". In this sense, σ_ψ provides a firm-level analogue to these bivariate gradient measures, with the firm effect replacing income or education and healthcare expenditure replacing the health outcome.

V.III Limited mobility and leave-out variance correction

Equation (4) can be estimated by OLS. However, variance components computed from the plug-in fixed-effect estimates, $\{\hat{\alpha}_i, \hat{\psi}_j\}$, are biased when worker mobility is limited. In employer networks where only few workers move between subsets of firms, each fixed effect is estimated with substantial noise even if it is unbiased individually. When these noisy estimates enter quadratically, as in $\widehat{\sigma^2_\psi}$, the variance of the estimation error inflates the quadratic form and leads to systematic bias (Andrews et al., 2008; Jochmans and Weidner, 2019; Kline et al., 2020; Bonhomme et al., 2023). Intuitively, the plug-in variance of firm effects is biased upward because it adds true dispersion and noise dispersion. This problem is often referred to as *limited mobility bias*.[24]

To address this, I use the leave-out variance estimator proposed by Kline et al. (2020) (KSS). Their approach provides unbiased estimators for variance and covari-

variables (schooling, experience) explain about 30% of the cross-sectional variance ($R^2 \approx 0.3$; see Mincer (1974)), whereas analogous models for annual healthcare expenditure using only age, gender, region, and source of income explain on the order of 6% of the variance ($R^2 \approx 0.06$; see Van Kleef et al. (2013)).

[23] See, for example, Wagstaff et al. (1991), Kakwani et al. (1997), Wagstaff and Doorslaer (2004), Erreygers and Van Ourti (2011), and Schlotheuber and Hosseinpoor (2022), who distinguish overall from socioeconomic-related health disparities and show that empirical work and monitoring practice typically focus on the latter using concentration-type and related bivariate indices.

[24] The same logic applies to other quadratic terms such as $Cov(\alpha_i, \psi_j)$; these are corrected using the same leave-out procedure.

ance components of interest in linear models without imposing homoskedasticity or additional structural restrictions. Formally, any variance or covariance term can be written as a quadratic form $\theta = \nu' A \nu$, where ν stacks all model parameters and A is a known symmetric matrix. The naive plug-in estimator $\hat{\theta}_{PI} = \hat{\nu}' A \hat{\nu}$ has a bias that can be expressed as a weighted sum of observation-specific error variances. KSS shows how to estimate these error variances in a leave-one-out fashion and subtract the implied bias, yielding a bias-corrected estimator $\hat{\theta}_{KSS}$.

In practice, I implement the KSS correction at the worker–firm match level. Following Kline et al. (2020), I first partial out the observed covariates x_{it} by estimating Equation (4) and working with the residualised outcome $h_{it} - x_{it}'\hat{\beta}$. I then compute bias-corrected estimates of σ^2_ψ using the leave-one-out formulas of Kline et al. (2020), treating an entire worker–firm match as the leave-out unit to accommodate serial correlation within matches. Because recomputing the fixed effects separately for each leave-out observation is infeasible in a large panel, I follow KSS and use their analytic leave-out formulas and random-projection approximation based on Achlioptas (2003) to obtain computationally tractable estimates.[25] Appendix A5 provides the quadratic-form representation of the variance components and details of the implementation; see Kline et al. (2020) for a full derivation of the leave-out estimator. As a robustness check, I also re-estimate the variance components on a more strongly connected subset of firms; as expected, the plug-in bias is smaller but not negligible, and the KSS correction still reduces the estimated dispersion of firm effects (Table A15). These strongly connected firms are larger and more selected than the average employer and are therefore not representative of the full economy.

VI Results

Firms differ substantially in the healthcare expenditure they induce among otherwise similar workers. Table 3 reports KSS-corrected variance and covariance components for log annual healthcare expenditure h_{it} in Equation (4). The variance of firm effects is $\text{var}(\psi_j) = 0.027$, implying $\sigma_\psi \approx 0.164$ log points. A one-standard-

[25] All AKM and leave-out computations are implemented in R using the package `LeaveOutKSS` developed by the author, which adapts the original Matlab routines of Kline et al. (2020) available at GitHub: LeaveOutKSS-R.

deviation difference in firm effects corresponds to a difference of $\exp(\sigma_\psi)-1 \approx 17.8\%$ in expected annual basic healthcare expenditure for the same worker. The worker–firm correlation is close to zero ($\rho(\alpha_i, \psi_j) = -0.018$), showing limited sorting on time-invariant expenditure.

To benchmark magnitudes, I compare the workplace gradient to the well-known gradients by wealth, income, and education that are standard in the health-disparities literature.[26] Within the sample, the age- and year-adjusted log-linear estimates imply about 23% lower healthcare expenditure in the top than in the bottom wealth quintile and 25% lower expenditure in the corresponding income comparison; workers with a master's or doctorate have about 33% lower expenditure than those with elementary or lower education (Table A2). The corresponding gap between the top and bottom quintiles of firm effects is 0.25–0.29 log points, a 22–25% difference in expenditure, under two-normal mixtures matched to the variance and the third and fourth cross-split moments of the firm-effect distribution (Appendix A6).[27] The dispersion across firm environments is therefore of the same order as the expenditure gradients by wealth and income in the sample and somewhat below the education gradient.[28]

For completeness, the components in Table 3 imply that firm effects account for only about 1% of the total variance in h_{it} at the individual-year level. As discussed in Section V, this variance share is not the most informative summary statistic, because the denominator is dominated by large idiosyncratic individual-level shocks—such as cancers, accidents, and chronic conditions—that no employer can plausibly control and that are at least partly inherently random.

[26] Highest completed education is used to define education; household wealth and total personal income are measured in 2009, before the main analysis period, to reduce concerns about reverse causality. These SES gradients are descriptive correlations rather than causal effects, and I report them only as a familiar benchmark for the magnitude of the firm-induced expenditure gradient.

[27] The quintile gap is the difference between mean firm effects in the top and bottom quintiles of the worker-year-weighted distribution. The moments identify the gap only up to the shape of the distribution; the mixtures are the reference fits, and Appendix A6 reports a symmetric normal benchmark (0.46) and a generalised-lambda fit that compresses the centre (0.09) as the outer range.

[28] The sample is positively selected in terms of socioeconomic characteristics: it consists of full-time male workers with relatively high income and wealth by construction (Figure A9), which likely underestimates the SES gradients in the Netherlands. In the same institutional context, Loef et al. (2021) show that healthcare expenditure is between 50% and 150% higher among individuals in the lowest education or income group than among those in the highest group across age bands. This suggests that, in the full population, SES-related gradients are larger than the 29–49% higher expenditure among the lower-SES groups in my sample.

Statistic	**KSS Estimates**
Log Health Exp. Var.: var(h_i)	2.617
Firm FE Var.: var(ψ_j)	0.027
Firm FE SD: $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$	$\simeq 0.164$
One-SD expenditure change: $100[e^{0.164} - 1]$	17.8%
Employee FE Var.: var(α_i)	1.516
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.018
Explained share: workers & firms	0.587
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Health Exp. : mean(h_i)	5.608

Notes: Full-time male employees aged 25–65 in the leave-one-out connected set, 2009–2016. The outcome is $h_{it} = \log(c_{it} + 1)$, where c_{it} is annual basic-package healthcare expenditure. Components use the leave-one-match-out Kline–Saggio–Sølvsten (KSS) correction (Kline et al., 2020) with 50 random-projection simulations. In the log-linear model, a one-SD increase in the firm effect multiplies $c_{it} + 1$ by e^{σ_ψ}. Using the reported SD of 0.164 gives $100[e^{0.164} - 1] \simeq 17.8\%$. Firm effects account for about 1% of the worker–year variance of h_{it}. Because variance shares depend on the noise in the outcome—here dominated by idiosyncratic health shocks and persistent individual heterogeneity—they are a poor summary of the firm margin; following Kline (2024), magnitudes are therefore reported as the firm-effect standard deviation in log points, interpreted as the change in a given worker's expected expenditure from reassignment across firms. Firm-effect variance is similar using cross-split covariances from two (0.0261) or four (0.0265) random worker splits, each evaluated on its common firm support (Tables A21 and A18). The four-split estimate averages six pairwise covariances, which range from 0.0257 to 0.0271 with a standard deviation of 0.00054.

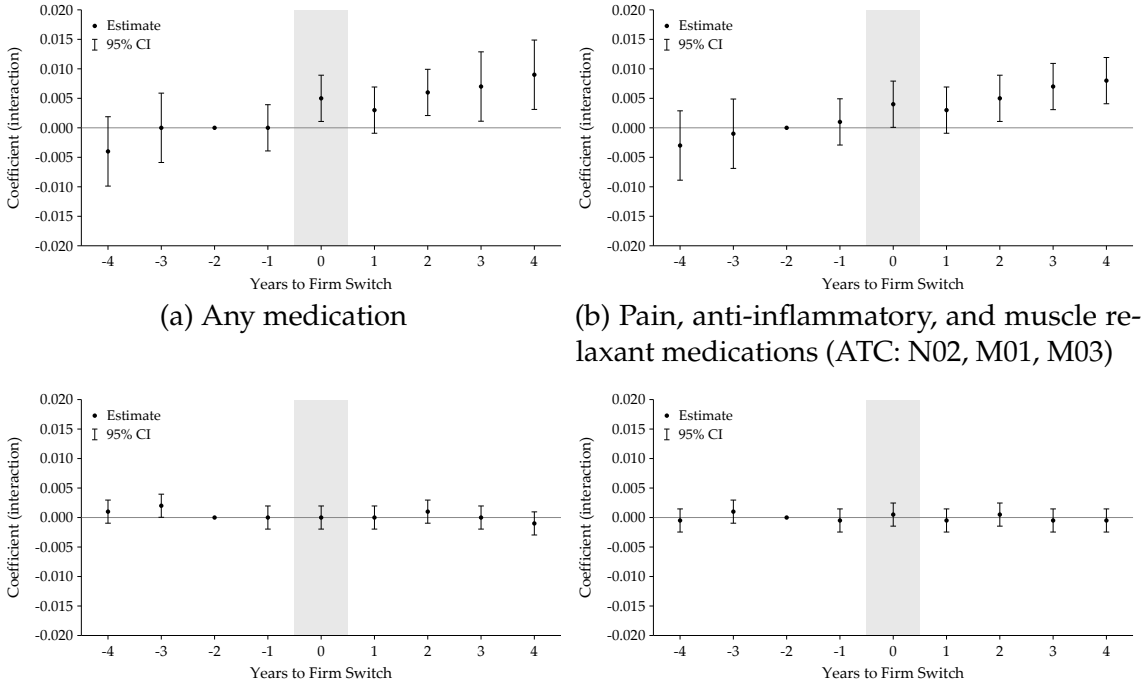
Table 3: Variance Decomposition of Log Healthcare Expenditure

VI.I Firm Fixed Effects and Health

Several results suggest that the estimated firm effects are more consistent with workplace-linked differences in health risk than with insurance design, income-related care use, or short-run utilisation incentives. Adding annual gross contract earnings leaves both the dispersion and the ranking of firm effects essentially unchanged (Table A4), suggesting that the estimates do not simply proxy for pay-related differences in healthcare use. Dispersion remains sizeable among workers with high baseline expenditure, who typically exceed the deductible and face a zero marginal price for additional basic care, and firm rankings are aligned across high- and low-baseline expenditure groups (cross-sample slope 0.60) (Table A5). The same alignment holds when splitting by hourly wages (Table A6). Category-specific estimates point in the same direction. GP expenditure, which is fully covered and deductible-exempt, exhibits a firm-effect standard deviation close to that for total expenditure (Table A9). These patterns are difficult to reconcile with employer-specific insurance design, prices, or income-related care use as the main drivers of the firm effects.

To provide complementary evidence on whether the expenditure changes observed around moves reflect health-risk channels rather than broad differences in utilisation or diagnosis intensity, I examine changes in medication use around firm moves using Equation (3). I consider both an indicator for any medication use and indicators for three subgroups: pain, anti-inflammatory, and muscle relaxant medications; cardiometabolic medications; and cancer-related medications. Consistent with the rise in healthcare expenditure among workers moving toward firms with higher average coworker expenditure (Figure 2), Figure 4 shows an increase in overall medication use following such moves. This increase is not accompanied by changes in cardiometabolic or cancer-related medications, suggesting that differential screening or referral for previously undiagnosed conditions is unlikely to explain the expenditure pass-through documented in the mover design. Instead, the increase is concentrated in pain, anti-inflammatory, and muscle relaxant medications, a pattern more consistent with workplace strain, injury, or work-related stress than with broader changes in diagnosis intensity.

To further study the link between the estimated firm effects to health outcomes, I study two long-run measures observed in 2019. The first is an indicator for death



(a) Any medication

(b) Pain, anti-inflammatory, and muscle relaxant medications (ATC: N02, M01, M03)

(c) Cardiometabolic (ATC: C03, C07, C08, C09, C10, A10, B01)

(d) Cancer-related (ATC: L01, L02, L03)

Figure 4: Medication use around moves between firms

Note: Sample of full-time male employees aged 25–65 in connected firms who switch employer exactly once during 2009–2016. Panels report mover event-study coefficients from equation (3) for annual medication-use indicators (any reimbursed dispensing), residualised on age and calendar-year dummies. The exposure gap D_i is destination minus origin leave-i-out mean residualised log total healthcare expenditure. Event-time interactions with D_i are relative to $t = -2$. Coefficients are probability changes per log-point exposure gap. Markers show point estimates and vertical bars show 95% confidence intervals. Medication groups are defined by ATC prefixes: pain, anti-inflammatory, and muscle relaxant medications (N02 analgesics, M01 anti-inflammatory/antirheumatic products, M03 muscle relaxants; consistent with musculoskeletal strain and acute injury-related treatment), cardiometabolic medications (C03 diuretics, C07 beta blockers, C08 calcium channel blockers, C09 RAAS agents, C10 lipid modifiers, A10 diabetes drugs, B01 antithrombotics), and cancer-related medications (L01 antineoplastic agents, L02 endocrine therapy, L03 immunostimulants).

by 2019, which captures cumulative severe health deterioration. The second is an indicator for disability receipt in 2019,[29] which captures persistent work-limiting health conditions. I relate these outcomes to firm effects on healthcare expenditure, conditioning on baseline demographics, pre-period expenditure, and origin-firm characteristics.

	Mortality in 2019		**Disability in 2019**	
	(1)	(2)	(3)	(4)
Average sample-split firm-effect exposure	0.00111***	0.00212***	0.00377***	0.00779***
	(0.00020)	(0.00020)	(0.00036)	(0.00035)
Worker FE control	No	Yes	No	Yes
Migration background controls	Yes	Yes	Yes	Yes
YOB × education FEs	Yes	Yes	Yes	Yes
Postal code FEs	Yes	Yes	Yes	Yes
N	2,485,666	2,485,666	2,432,508	2,432,508
R^2	0.0417	0.0483	0.0445	0.0754

Notes: Each column reports a worker-level linear probability model. The regressor is the worker's average exposure, over observed years in 2009–2016, to firm effects on log healthcare expenditure estimated in the other worker split; coefficients are probability changes per log point of average exposure. All specifications include migration background indicators, postal code fixed effects, and year-of-birth × education-category fixed effects. Columns (2) and (4) additionally control for the worker fixed effect estimate. Robust standard errors in parentheses; they may understate uncertainty because the exposure measure is constructed from estimated firm effects. *** $p < 0.01$.

Table 4: Average Estimated Firm-Effect Exposure and Subsequent Mortality and Disability (2019)

To prevent the same workers' healthcare histories from determining both the firm-effect measure and their downstream outcomes, I follow Kline (2025) and randomly split workers into two subsamples. I estimate the two-way fixed-effects model separately in each,[30] keep firms observed in both, and assign each worker the demeaned firm effect estimated in the other subsample. Table 4 shows that workers with greater exposure to firms with higher estimated healthcare-expenditure effects are more likely to be on disability and more likely to have died by 2019, conditional on migration background, postal-code fixed effects, and year-of-birth × education

[29] Disability status is a truncated outcome: by construction, only individuals who are alive and not on other competing benefits or social welfare programmes can be on disability.

[30] See Table A21 for the KSS variance estimate within each subsample.

fixed effects. The association becomes larger after controlling for the worker fixed-effect estimate, reinforcing that it is not explained by time-invariant differences in baseline health.

I next use firm closures as an additional source of exogenous mobility. When a firm exits, all workers separate in the same year and move to destination firms with different estimated firm effects. I focus on workers observed from the start of the panel and followed continuously, allowing me to trace healthcare trajectories around displacement. Following the displacement literature (e.g., Halla et al. (2020)), I define closure-induced job loss as a transition in which the employer identifier disappears from the panel and take the closure year as the last year the firm is observed. Figure A11 shows that, after reemployment, workers' expenditure shifts in the direction of the destination firm's average coworker expenditure.

Table 5 reports regressions of disability and mortality in 2019 on the sample-split estimate of the destination firm's effect, controlling for baseline demographics, pre-closure average healthcare expenditure, and the corresponding sample-split estimate of the origin firm's effect.[31,32] The magnitudes are economically meaningful. Using the baseline AKM dispersion ($\sigma_\psi = 0.164$), reemployment at a destination firm one standard deviation higher in the firm-effect distribution is associated with a 0.22 percentage point higher probability of disability in 2019 and a 0.08 percentage point higher probability of death, roughly 6% of the respective sample means.[33] Figure 5 provides a complementary visualization: within broad origin bins, (residualized) mortality and disability are higher for workers reemployed at destination firms with high estimated effects than for those reemployed at destination firms with low estimated effects.[34,35]

[31] See Table A10 for summary statistics of the closure sample.

[32] To reduce noise in the destination measure, I restrict the sample to destination firms above the 25th percentile of the firm-size distribution (at least 75 FTE workers). Origin and destination firm effects are highly correlated ($\rho \approx 0.64$), so specifications with origin-firm fixed effects rely on limited within-origin dispersion and are correspondingly less precise and less robust. Results with origin fixed effects are similar but less precisely estimated.

[33] These effect sizes use the preferred estimates in column (2) of Table 5 and the dependent-variable means reported in the table.

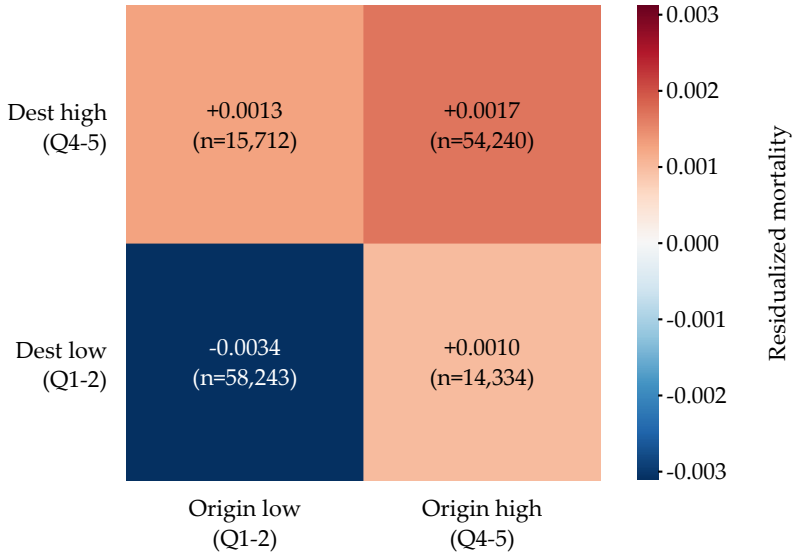
[34] Figure A12 shows a similar relationship using a finer 5×5 (quintile-by-quintile) grouping.

[35] As a placebo, I re-estimate the preferred specification using average pre-move healthcare expenditure as the outcome and the same covariate set as in Table 5. The destination-firm coefficient is not positive; if anything, it turns negative once I condition on the origin firm's effect. Because origin and destination firm effects are highly correlated in the closure sample, placebo estimates that condition on origin-firm fixed effects should be interpreted cautiously. In any case, the lack

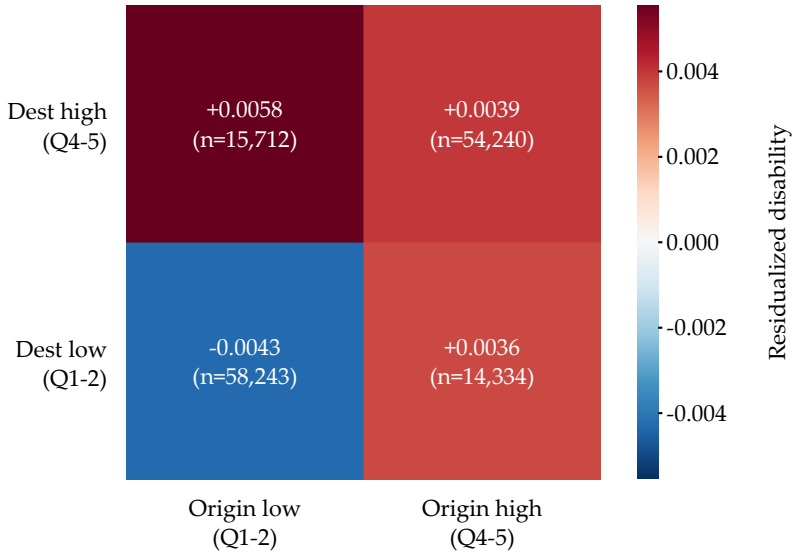
	(1)	(2)	(3)
	Outcome measured in 2019		
Panel A: Disability in 2019			
Destination firm effect (sample-split)	0.0078***	0.0137***	0.0086***
	(0.00133)	(0.00192)	(0.00292)
Mean of dep. var.		0.03929	
N	206,305	206,305	204,012
R^2	0.066	0.066	0.116
Panel B: Mortality in 2019			
Destination firm effect (sample-split)	0.0033***	0.0047***	0.0034*
	(0.00078)	(0.00113)	(0.00173)
Mean of dep. var.		0.01313	
N	209,235	209,235	206,945
R^2	0.048	0.048	0.083
Baseline controls & pre-move postal-code FEs	Yes	Yes	Yes
Origin firm-effect level (linear control)	No	Yes	No
Origin-firm fixed effects	No	No	Yes

Notes: Sample of male workers displaced by firm closures whose destination firms are above the 25th firm-size percentile (at least 75 employees). The regressor is the destination firm's effect on log healthcare expenditure, estimated in the other worker split; coefficients are probability changes per log point. Column (2) also controls for the origin firm effect from the other worker split; column (3) includes origin-firm fixed effects. Baseline controls include pre-move average healthcare expenditure, demographics (as in the text), and pre-move postal-code fixed effects. Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Firm Closures: Destination Firm Effects and Long-Run Outcomes (Continuous Measure)



(a) Mortality in 2019



(b) Disability in 2019

Figure 5: Long-run outcomes by origin and destination firm-effect bins after firm closures

Note: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). Each panel plots binned means of residualized outcomes, where residuals are obtained from the same control set used in Table 5 (baseline demographics, pre-move average healthcare expenditure, and pre-move postal-code fixed effects). The horizontal and vertical axes group workers by the sample-split origin and destination firm effects, respectively, into coarse bins: the bottom two versus the top two quintiles, so the middle quintile is omitted. Cell annotations report the number of observations used to compute each binned mean. Appendix Figure A12 shows the same relationships using the full 5×5 quintile-by-quintile binning.

To characterise what the firm effects capture within healthcare expenditure and related health outcomes, I decompose expenditure by care category and relate workers' exposure to estimated firm effects on healthcare expenditure to cause-specific mortality. Dispersion in firm effects is present across care domains, and it is largest for hospital and specialist somatic care, smaller for GP services and medication—suggesting that firm effects are present for both severe and more routine care categories—and modest for specialised mental healthcare (Table A9). Higher destination-firm effects are associated with higher cancer mortality in both the all-mover and closure samples. Among all movers, I also find evidence of higher mortality from accidents and injuries when controlling for the origin firm's estimated effect, although the evidence is weaker with origin-firm fixed effects and among closure movers (Table A12). These patterns, along with the increase in the use of pain, anti-inflammatory, and muscle relaxant medications (Figure 4), suggest that firms' contributions to healthcare expenditure are multidimensional and appear most closely linked to somatic and longer-run physical risks.

VI.II Heterogeneity Across Sectors, Firms, and Workers

Because production technologies differ sharply across sectors, gaps between, for example, construction and finance in terms of induced health expenditure, mostly reflect what firms produce and leave limited scope for policy short of changing the activity mix. The more relevant question is how much firm-effect dispersion remains among firms that ostensibly do the same thing. Figure 6 plots observation-weighted sector means of the workplace fixed effects. As expected, traditionally high-risk sectors such as mining, construction, and manufacturing are among the worst performers, and the financial sector—where work is less physically demanding but characterised by high pressure—also stands out with large adverse effects on employee healthcare expenditure, consistent with both physical and psychosocial channels. Using 17 broad sector classifications, sector differences account for only about one sixth of the dispersion in firm health effects. When I replace these broad groups with detailed, highly granular industrial codes,

of a positive relationship between destination fixed-effect and pre-move healthcare expenditures reduces concerns that the documented relationship reflects pre-existing differences in health (Table A11).

industrial-code differences explain just under half of the dispersion,[36] so more than half of the dispersion remains within narrowly defined industrial-code cells. Thus, while some activities are clearly more hazardous than others, a large share of firm-effect dispersion remains among firms that do very similar work, pointing to firm-specific practices—and hence to scope for firm-level policy interventions—as a central driver.

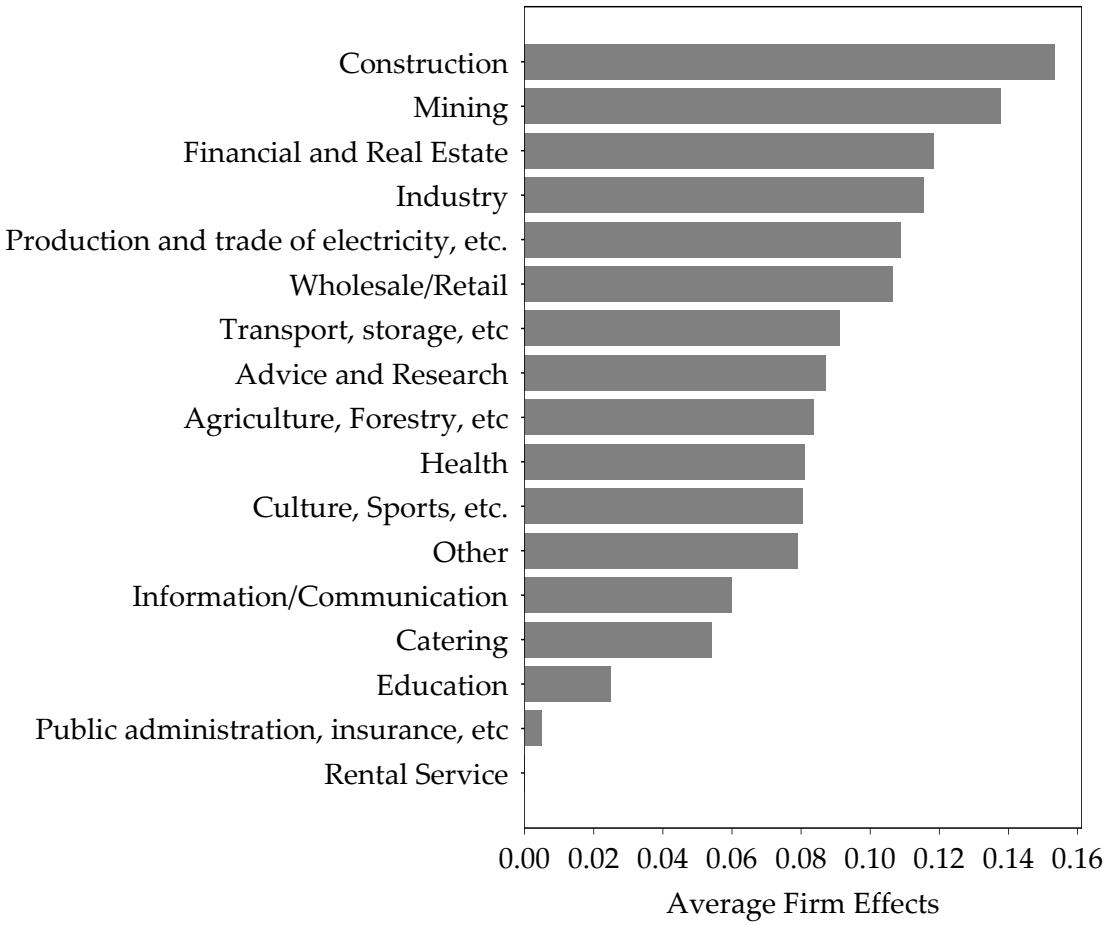
Firm characteristics and firm effects. To characterise which firms drive the estimated firm effects, I link the AKM firm effects to observable firm characteristics from the business registry. Figure 7 reports these relationships as simple bivariate regressions of the firm health effect on each characteristic (one at a time). Firms with larger workforces and higher female and migrant shares have lower firm effects, while firms with more overtime have higher firm effects. The coefficient on worker age is small and positive. Firms that employ an in-house company doctor are also associated with lower firm effects.

Wage premia between and within industries. Firms with higher estimated healthcare-expenditure effects tend to have higher wage premia, consistent with compensating differentials.[37,38] Table 6 compares the pooled wage–health association with the corresponding within-industry association using split-sample instrumental variables. The pooled slope is 0.078 across firms in the split-sample comparison (column 1) and 0.116 among firms with an observed industry code

[36] The sector and industrial-code shares are computed in the subsample of firms that can be linked to the business registry, which removes about 12% of firms (around 1.4% of person–year observations). In this subsample, the overall variance of firm effects is smaller ($\sigma_\psi \approx 0.11$ instead of 0.16; see Table A13), so the same sector differences translate into somewhat larger variance shares than in the full connected sample. Because some SBI industrial-code cells contain relatively few firms and workers, the between-code component may also be slightly upward biased; the reported industrial-code shares should therefore be interpreted as upper bounds, and the within-code component as a conservative estimate of within-industry firm-effect dispersion. Cross-split second moments imply that approximately 75% of firm-effect variance remains within SBI codes in the four-way common-support sample of 19,650 firms with more than 100 worker–year observations (Table A19).

[37] This follows the compensating-differentials tradition (e.g. Rosen, 1986), and connects to recent firm-level evidence on wage premia and amenities (e.g. Sorkin, 2018; Sockin, 2022; Lamadon et al., 2022).

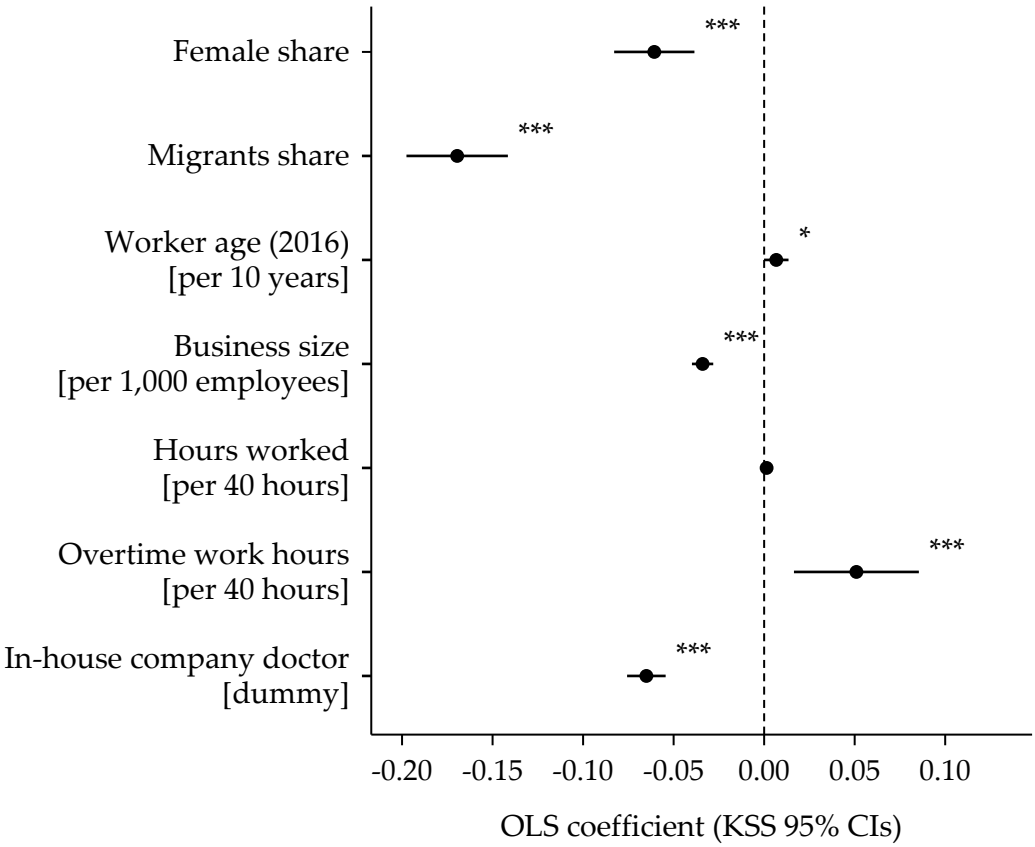
[38] Firm wage premia are estimated using the same specification as in Equation (4), replacing log healthcare expenditure with log annual earnings and restricting to full-time workers; see Table A3. Hourly wages are approximated by dividing annual earnings by a full-time schedule.



Notes: The figure plots observation-weighted sector means of estimated firm effects ψ_j in log healthcare expenditure from Equation 4, for firms with observed sector codes. Sectors are defined using firms' two-digit SBI industrial codes.[a] I normalise the average effect in the Rental Services sector to zero; bars are log-point differences relative to this baseline.

[a] Overzicht Standaard Bedrijfsindeling (SBI) codes are Dutch industrial codes used to classify firms by their main industry.

Figure 6: Average Sectoral Fixed Effects on Total Healthcare Expenditure



Notes: The figure reports OLS coefficients from separate bivariate linear regressions of estimated firm fixed effects in log healthcare expenditure on the indicated firm characteristic. Points show coefficient estimates; horizontal bars show 95% confidence intervals constructed using KSS standard errors for linear combinations of the estimated effects (Kline et al., 2020). The first six estimates use the male-worker sample with complete firm-characteristic data; the company-doctor estimate uses the male-worker sample excluding occupational-health-service firms (SBI 86922). Units are indicated in the variable labels. Significance codes: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 7: Correlates of Firm Fixed Effects in Healthcare Expenditure

(column 2). On the same SBI-coded sample, adding detailed industry fixed effects brings the slope close to zero and reverses its sign (0.027 against the pooled 0.078)[39], so the positive association disappears (column 3). This contrast suggests that the positive wage–health association arises mainly between industries, rather than among firms in the same industry.

These estimates do not measure how fully workers are compensated for health risk: wage premia also reflect rent sharing, work intensity, and other firm attributes. The split-sample instrument is intended to address measurement error under independence of estimation errors across worker splits.

Differences across worker groups. Subsample analyses show that firm effects are largely common across workers, but with meaningful heterogeneity in magnitude. Re-estimating the AKM model separately by age, income, baseline expenditure, and collar type, the estimated firm effects line up closely across groups: firms with higher estimated healthcare-expenditure effects for one group tend also to have higher effects for others, as reflected in the cross-sample slope regressions of firm effects reported in Appendix Tables A5–A8. At the same time, the dispersion of firm effects is systematically larger for younger, lower-income, low-expenditure, and blue-collar workers, consistent with these groups being more exposed to channels such as physical hazards, work intensity, or tighter budget and time constraints.[40]

VI.III Identifying High-Expenditure Firm Environments

The results so far establish that firm effects on healthcare expenditure are dispersed, that most of the dispersion lies within industries, and that the costs are borne by the public insurer. A regulator who wanted to act on this—through inspection, experience rating, or prevention support—would have to decide which employers

[39]The standard errors overstate precision for this generated-regressor comparison due to dependence of the dependent variables and individual-year level regression; I read the evidence as an absence of within-industry compensation rather than as a negative premium.

[40]Blue- and white-collar status is proxied from fields of study using a text-based classification. See Appendix A8 for details. The mapping from majors to collar types uses a simple text-based classification implemented with a large language model, in line with recent evidence that such models match or exceed human accuracy in straightforward coding tasks (e.g. Gilardi et al., 2023; Chang et al., 2024).

	(1) Pooled All comparison firms	(2) Pooled SBI-coded firms	(3) Within industry SBI-coded firms
Panel A. Second stage			
Dependent variable: own-split firm wage effect			
Own-split firm effect	0.0778**	0.116**	−0.0265
	(0.0366)	(0.0497)	(0.0337)
Panel B. First stage			
Dependent variable: own-split firm effect			
Other-split firm effect	0.199***	0.154***	0.110**
	(0.0579)	(0.0466)	(0.0480)
First-stage F (cluster-robust)	11.8	10.9	5.3
Detailed SBI industry fixed effects	No	No	Yes
Observations (worker–years)	15,447,395	13,491,466	13,491,466

Notes: Panel A reports worker–year two-stage least-squares regressions of firm wage effects on firm healthcare-expenditure effects estimated in the same worker split, using the firm effect estimated in the other split as an instrument. Panel B reports the corresponding first stages; the first-stage F is the cluster-robust statistic on the excluded instrument. Anderson–Rubin tests of the second-stage coefficient, which remain valid when the instrument is weak and are equivalent to the cluster-robust test of the excluded instrument in the reduced form with the same controls, yield $p < 0.001$ in columns 1 and 2 and $p = 0.55$ in column 3. Columns 2 and 3 restrict the sample to firms with an observed SBI industry code; column 3 includes detailed SBI industry fixed effects in both stages. All regressions include an intercept, which is not reported. Standard errors, clustered at the firm level, in parentheses. They may still understate uncertainty because the dependent variable and the regressor are themselves estimated firm fixed effects, so estimation error can remain dependent across firms. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Firm Wage Premia and Firm Effects on Healthcare Expenditure: Pooled and Within-Industry IV Estimates

to act on, and the only firm-level measure available would be an estimate of the kind constructed here, with sampling error that varies across firms. This subsection asks whether such estimates are informative enough to be useful: can a policymaker rank firms on estimated effects, select the top of the ranking, and find that the selected firms carry excess healthcare expenditure among workers whose data played no part in the ranking?

The question differs from the one addressed by the empirical-Bayes literature on noisy firm-level estimates (Kline et al., 2022, 2024), which asks how to recover or classify individual firms' latent effects. Here no attempt is made to shrink, rank correctly, or classify any firm. The rule is the naive one a regulator could implement with the published estimates, and the question is only whether the portfolio it selects is high-cost out of sample. Noise then enters as a feature of the policy problem rather than as a bias to be removed: false positives and false negatives lower what the rule captures, and the exercise measures how much.

The exercise is confined to firms with more than 100 worker-year observations, 20,926 firms that employ about four fifths of the worker-years in the sample. The restriction follows the policy problem rather than the estimation. Small firms contribute little to aggregate expenditure, so the return to inspecting one is low whatever its effect; firm-specific instruments typically exempt them in practice, as experience rating of disability contributions and the full risk-inventory requirement do for small Dutch employers; and their effects are estimated from few workers, so a ranking that admitted them would fill its top ranks with noise.[41] Firms are ranked on effects estimated in one worker split and the selected portfolio is evaluated on the other split, so that a firm selected because its estimate was high by chance does not appear high-cost when it is assessed (Appendix A7). Let $\bar{\psi}(q)$ denote the held-out, size-weighted mean firm effect, relative to a benchmark, among the top q share of ranked firms. Under the log-linear specification, moving the selected firms' effects to the benchmark would change their workers' expected expenditure by the factor $\exp[-\bar{\psi}(q)]$, a reduction of $100\{1 - \exp[-\bar{\psi}(q)]\}$ percent of the current level. I consider two benchmarks: the overall mean firm effect, and each firm's own SBI industry mean, in which case firms are ranked on their excess effect above

[41] The restriction also keeps the feasible exercise comparable with the oracle benchmarks below. The moment-matched distributions are worker-weighted, so small firms carry little weight in the shape the oracle ranks on, whereas a feasible ranking by firm count would be dominated by them.

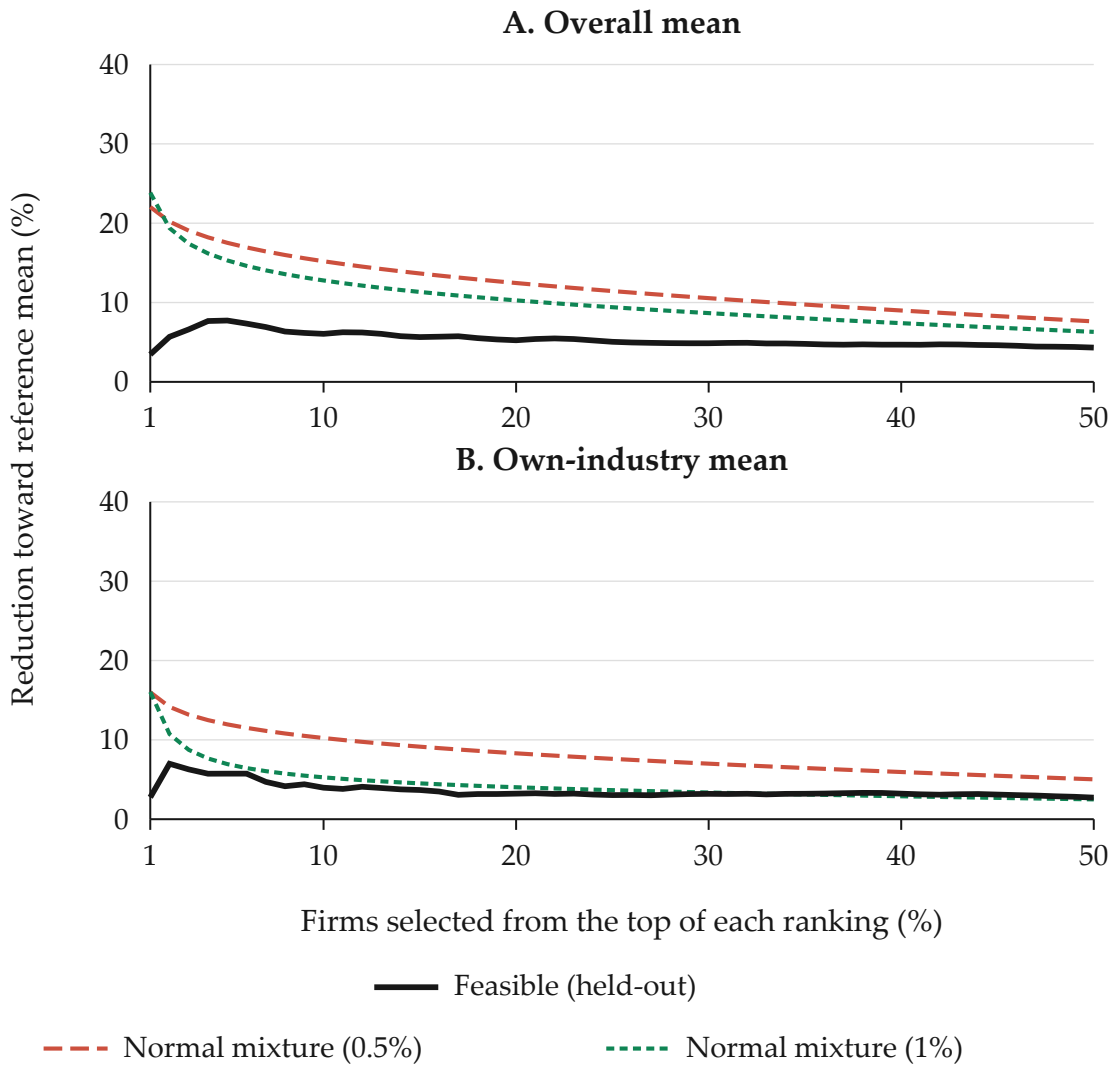
their industry. The second benchmark asks the sharper question, whether the rule can locate firms that generate more expenditure than others doing the same work.

Figure 8 plots the held-out reduction against the share of firms selected. For the top decile of firms, moving effects to the overall mean would reduce the portfolio's healthcare expenditure by 6.1%; moving them to their own industry means would reduce it by 4.0%. The curves are nearly flat beyond the top few percent, which reflects a property of the data rather than of the rule: the most highly ranked firms are small and their estimates noisy, so the top of the ranking mixes genuinely high-effect firms with firms selected by chance. These are counterfactual changes in firm effects, not the effect of an intervention; they measure what a rule could locate, not what a policy would save.

To see how much is lost to noise, I compare the feasible curves with the reduction that ranking on latent effects would deliver under the moment-matched distributions of Appendix A6, using the SBI-adjusted moments of Table A19 for the industry benchmark. Under the two mixture fits, this infeasible oracle would reduce expenditure in the top decile by 13–15% toward the overall mean and by 6–10% toward industry means. The feasible rule therefore recovers roughly two fifths to one half of the oracle reduction at the top decile and a larger share as the portfolio widens, where ranking noise matters less. The comparison assumes that expected firm size does not vary with the latent effect and uses different estimation populations, so the oracle is a benchmark rather than a bound. Its message is that a regulator ranking employers on noisy administrative estimates would locate a substantial part of the excess burden that perfect information would, without any attempt to classify individual firms.

VI.IV Robustness Checks

A key identifying assumption is that workers do not change employers in anticipation of unobserved health deterioration. The mover event studies in Figures 2 and A8 show no systematic pre-trends associated with the origin–destination gap in average coworker expenditure. To further reduce scope for health-driven mobility, Table A7 re-estimates the AKM model and corresponding variance decomposition for workers aged ≤ 45, a life-cycle phase in which health and healthcare use are typically stable (Galama and Van Kippersluis, 2019). The KSS-corrected dispersion



Notes: The horizontal axis is the cumulative share q of firms selected from the top of each ranking. The feasible curves rank firms in one worker split and evaluate their size-weighted mean effect in the other. Panel A uses raw effects; Panel B reranks effects relative to each firm's SBI industry mean. The vertical axis is $100[1 - \exp(-\bar{\psi}(q))]$, where $\bar{\psi}(q)$ is the selected size-weighted mean effect relative to the reference. Dashed curves rank latent effects in the fitted two-normal mixtures of Appendix A6; mixture percentages denote the lower-mean component's weight. The feasible exercise covers 20,926 eligible firms with an SBI code and more than 100 worker-year observations; the overall fits use 47,144 four-way common firms, and the SBI fits use 19,650 firms. The benchmarks assume $\mathbb{E}[N_j \mid x_j] = \mathbb{E}[N_j]$ for the relevant latent effect x_j, where N_j is firm size. The different populations make the comparison a benchmark rather than a bound.

Figure 8: Feasible Targeting and Calibrated Oracle Benchmarks

of firm effects is close to the baseline estimate in Table 3, suggesting that the main results are not driven by late-career, health-motivated job changes.

A separate concern is that estimated firm effects proxy geography or access rather than workplace conditions. In the Netherlands, job changes rarely coincide with home moves (<18%; Kronenberg and Carree, 2010), but I nonetheless add residential four-digit postcode fixed effects and, in a further specification, control for workplace-area healthcare access at PC4 (GP density, distance to pharmacies, and nearby hospitals). Table A14 shows that the estimated firm-effect dispersion changes little (0.025 versus 0.027), indicating that residential sorting and local access do not drive the findings.

Because variance components from the AKM decomposition are sensitive to limited-mobility bias, the baseline estimates use the leave-one-match-out correction of Kline et al. (2020). Table A15 shows that restricting to a highly connected subset of firms (at least 50 worker moves) reduces, but does not eliminate, plug-in bias and comes at substantial sample loss, while the KSS-corrected dispersion remains stable. Table A21 provides an additional validation by estimating firm effects in two random worker splits, and reporting a cross-split covariance (Panel B) that is very close to the full-sample KSS estimate of $\text{Var}(\psi_j)$ in Table 3, consistent with the baseline dispersion reflecting signal rather than estimation noise.

Using a Poisson specification that accommodates zero expenditure (Chen and Roth, 2024; Mullahy and Norton, 2024), I find an approximately linear relationship between its firm-effect estimates and those from the baseline log-linear model, with an OLS slope close to one (Appendix Figure A13 and Table A17).

Finally, the baseline outcome uses concurrent annual expenditure, which is the natural margin for workplace influences in this setting: workers remain tied to the employer during sickness absence for up to two years, and in the Grossman framework contemporaneous shifts in the health-depreciation rate map into contemporaneous expenditure flows. Consistent with this, Table A16 shows that firm-effect dispersion attenuates with the horizon when future expenditure is used as the dependent variable, reinforcing that the baseline estimates primarily capture contemporaneous firm effects on healthcare expenditure.

VII Discussion and Conclusion

The findings show that firms differ materially in the contribution they make to otherwise similar workers' healthcare expenditure. In the Dutch setting, where basic health insurance is universal, individual, and portable across jobs, these firm effects are unlikely to be driven by employer-specific insurance coverage or cost sharing. A one-standard-deviation difference in firm effects corresponds to a 17.8% difference in expected healthcare expenditure for the same worker; the gap between the top and bottom quintiles of firm effects is comparable to the expenditure gradients by wealth and income; most of the dispersion lies among firms in the same detailed industry; and exposure to firms with higher estimated effects predicts higher subsequent disability and mortality. Together, these results show that firms make economically important contributions to workers' healthcare expenditure, that these contributions predict subsequent health, and that they are not a matter of what firms produce so much as of how they produce it.

The estimated firm effect should be interpreted as a composite measure of the workplace-related contribution to healthcare expenditure. It may reflect physical risks, psychosocial strain, work pace, task composition, safety practices, managerial quality, peer norms, and prevention or reintegration policies. Because job moves change several of these margins at once, and because detailed occupations, tasks, and workplace practices are not observed, the data do not allow a decomposition into separate mechanisms. The estimates instead capture the net effect of being employed at a particular firm on contemporaneous healthcare expenditure.

Several pieces of evidence point to workplace-related health risk as an important component of this net effect. Firms with higher healthcare-expenditure effects are concentrated in sectors with plausibly demanding physical or psychosocial environments, and firm effects are correlated with observable markers of demanding work environments, such as overtime. Workers who move to firms whose employees have higher average healthcare expenditure increase use of pain, anti-inflammatory, and muscle-relaxant medications, while comparable changes are not visible for cardiometabolic or cancer-related medications. Firm effects also predict disability and mortality, including among workers displaced by firm closures. These patterns are difficult to reconcile with a pure utilisation or access interpretation.

Alternative explanations appear less central. Estimated firm-effect dispersion is stable after controlling for pay, residential location, and local healthcare access. It remains sizeable when the sample is restricted to workers who face near-zero marginal prices for additional care and when the outcome is GP expenditure, which is exempt from the deductible. These results suggest that prices, access, screening, and broad utilisation spillovers are unlikely to be the main drivers. Narrower peer effects in care-seeking may still matter, especially for musculoskeletal complaints, and peer-driven health behaviour could also play a role. But such channels are unlikely to fully explain the sharp changes in contemporaneous healthcare expenditure, and the downstream increases in disability and mortality.

The wage–health comparison is consistent with compensating differentials between industries, with no indication of wage compensation within industries, where most of the dispersion lies. The firm-specific component of healthcare expenditure is therefore priced neither in the labour market nor, because the medical costs it generates are financed collectively, by the firm. In a system such as the Dutch one, this is a fiscal externality: firms do not internalise the healthcare expenditure associated with the risks they generate. This does not mean that firms face no health-related costs. Dutch employers already bear substantial responsibilities through sickness pay, occupational-health obligations, reintegration requirements, and experience rating in disability insurance. Basic healthcare expenditure, however, remains outside these instruments. The targeting exercise shows that the instruments could be extended: a regulator ranking employers on estimated firm effects would locate a substantial share of the excess expenditure that perfect information would, within as well as across industries.

Several caveats are important. The outcome is healthcare expenditure under the statutory basic package, not health itself and not the full social cost of work-related health risks. Care financed through supplementary insurance is not observed. Some part of the estimated dispersion in firm effects may also reflect differences in screening, referral, or care-seeking behaviour rather than underlying health, and the lower tail of the firm-effect distribution, which a few large employers dominate, should not be read symmetrically as evidence of health-improving workplaces.

Two directions for future research are especially important. The first is to identify and quantify the mechanisms through which firms affect health: physical

strain, work intensity, schedule control, managerial practices, psychosocial stress, or peer behaviour. The second is to study whether stronger cost internalisation can reduce the dispersion in firm effects documented here. Broader experience rating or more employment-linked financing of healthcare could strengthen prevention incentives, but it could also create distortions, including selection against less healthy workers. Understanding this trade-off is central for policy design.

Overall, the results show that the firm should be treated as an important unit in the study of health disparities. Because workplace environments contribute to otherwise similar workers' healthcare expenditure and are linked to later health outcomes, and because most of that contribution arises among firms doing the same work, policies focused only on individual behaviour, healthcare access, or industry regulation will miss an important source of inequality.

References

Abowd, J. M., F. Kramarz, and D. N. Margolis (1999). High wage workers and high wage firms. *Econometrica 67*(2), 251–333.

Achlioptas, D. (2003). Database-friendly random projections: Johnson-lindenstrauss with binary coins. *Journal of Computer and System Sciences 66*(4), 671–687.

Adema, Y., L. Bettendorf, E. Van Bezooijen, D. Freeman, and B. Wache (2025). Dynamics, productivity, and innovation in the dutch economy.

Ahammer, A., U. Glogowsky, M. Halla, and T. Hener (2023). The parenthood penalty in mental health: Evidence from austria and denmark.

Ahammer, A., A. Packham, and J. Smith (2026). Firms and worker healthcare utilization. Working paper.

Andrews, M. J., L. Gill, T. Schank, and R. Upward (2008). High wage workers and low wage firms: Negative assortative matching or limited mobility bias? *Journal of the Royal Statistical Society Series A: Statistics in Society 171*(3), 673–697.

Arnould, R. J. and L. M. Nichols (1983). Wage-risk premiums and workers' compensation: a refinement of estimates of compensating wage differential. *Journal of Political Economy 91*(2), 332–340.

Bhattacharya, A. (2014). Costs of occupational musculoskeletal disorders (msds) in the united states. *International Journal of Industrial Ergonomics 44*(3), 448–454.

Blackburn, B., T. Chan, E. Cherot, R. B. Freeman, X. Hu, E. Matt, and C. A. Rhodes (2023). Beyond burnout: from measuring to forecasting. Working Paper w30895, National Bureau of Economic Research.

Bonhomme, S., K. Holzheu, T. Lamadon, E. Manresa, M. Mogstad, and B. Setzler (2023). How much should we trust estimates of firm effects and worker sorting? *Journal of Labor Economics 41*(2), 291–322.

Bíró, A. and P. Elek (2025). Firm quality and health maintenance. *Journal of Health Economics 103*, 103045.

Card, D., A. R. Cardoso, J. Heining, and P. Kline (2018). Firms and labor market inequality: Evidence and some theory. *Journal of Labor Economics 36*(S1), S13–S70.

Card, D., A. R. Cardoso, and P. Kline (2016). Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women. *The Quarterly Journal of Economics 131*(2), 633–686.

Card, D., J. Heining, and P. Kline (2013). Workplace heterogeneity and the rise of west german wage inequality. *The Quarterly Journal of Economics 128*(3), 967–1015.

(CBS), S. N. (2025, November). Werkzame beroepsbevolking; arbeidsduur (statline table 85275ned). Data set. Retrieved 25 November 2025 from https://www.cbs.nl/nl-nl/cijfers/detail/85275NED.

Cesarini, D., E. Lindqvist, R. Östling, and B. Wallace (2016). Wealth, health, and child development: Evidence from administrative data on swedish lottery players. *The Quarterly Journal of Economics 131*(2), 687–738.

Chang, Y., X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, et al. (2024). A survey on evaluation of large language models. *ACM transactions on intelligent systems and technology 15*(3), 1–45.

Chen, J. and J. Roth (2024). Logs with zeros? some problems and solutions. *The Quarterly Journal of Economics 139*(2), 891–936.

Collaborators, G. . R. F. et al. (2018). Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks for 195 countries and territories, 1990-2017: a systematic analysis for the global burden of disease study 2017. *The Lancet 392*(10159), 1923–1994.

Currie, J. and E. Zwiers (2025). Medication of postpartum depression and maternal outcomes: Evidence from geographic variation in dutch prescribing. *Journal of Human Resources 60*(4), 1093–1125.

Cutler, D., A. Deaton, and A. Lleras-Muney (2006). The determinants of mortality. *Journal of Economic Perspectives 20*(3), 97–120.

Danesh, K., J. T. Kolstad, W. D. Parker, and J. Spinnewijn (2024). The chronic disease index: Analyzing health inequalities over the lifecycle. Technical report, National Bureau of Economic Research.

De Groot, N. and P. Koning (2016). Assessing the effects of disability insurance experience rating. the case of the netherlands. *Labour Economics 41*, 304–317.

Erreygers, G. and T. Van Ourti (2011). Measuring socioeconomic inequality in health, health care and health financing by means of rank-dependent indices: a recipe for good practice. *Journal of health economics 30*(4), 685–694.

Eurostat (2023). Unmet health care needs statistics. Statistics Explained, European Commission. Accessed 24 November 2025.

Finkelstein, A., M. Gentzkow, and H. Williams (2016). Sources of geographic variation in health care: Evidence from patient migration. *The Quarterly Journal of Economics 131*(4), 1681–1726.

Finkelstein, A., M. Gentzkow, and H. Williams (2021). Place-based drivers of mortality: Evidence from migration. *American Economic Review 111*(8), 2697–2735.

Galama, T. J. and H. Van Kippersluis (2019). A theory of socio-economic disparities in health over the life cycle. *The Economic Journal 129*(617), 338–374.

Gan, W. Q., H. W. Davies, and P. A. Demers (2011). Exposure to occupational noise and cardiovascular disease in the united states: the national health and nutrition examination survey 1999–2004. *Occupational and environmental medicine 68*(3), 183–190.

García-Gómez, P., A. C. Gielen, C. Riumallo Herl, and J. Rossi (2026). Health after birth: The persistent health penalty of becoming a parent. Draft manuscript.

Gilardi, F., M. Alizadeh, and M. Kubli (2023). Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences 120*(30), e2305016120.

Grossman, M. (1972a). *The Demand for Health: A Theoretical and Empirical Investigation*. Columbia University Press.

Grossman, M. (1972b). On the concept of health capital and the demand for health. *Journal of Political Economy 80*(2), 223–255.

Halla, M., J. Schmieder, and A. Weber (2020). Job displacement, family dynamics, and spousal labor supply. *American Economic Journal: Applied Economics 12*(4), 253–287.

Hamermesh, D. S. (1999). Changing inequality in markets for workplace amenities. *The Quarterly Journal of Economics 114*(4), 1085–1123.

Handel, B. R., J. T. Kolstad, T. Minten, and J. Spinnewijn (2020). The social determinants of choice quality: Evidence from health insurance in the netherlands.

Ho, K. and R. S. Lee (2019). Equilibrium provider networks: Bargaining and exclusion in health care markets. *American Economic Review 109*(2), 473–522.

Hummels, D., J. R. Munch, and C. Xiang (2025). No pain, no gain: Work demand, work effort, and worker health. *Review of Economics and Statistics 107*(4), 898–913.

Jochmans, K. and M. Weidner (2019). Fixed-effect regressions on network data. *Econometrica 87*(5), 1543–1560.

Johansson, N., S. C. de New, J. S. Kunz, D. Petrie, and M. Svensson (2023). Reductions in out-of-pocket prices and forward-looking moral hazard in health care demand. *Journal of Health Economics 87*, 102710.

Johnson, M. S. (2020). Regulation by shaming: Deterrence effects of publicizing violations of workplace safety and health laws. *American economic review 110*(6), 1866–1904.

Jolivet, G. and F. Postel-Vinay (2025). A structural analysis of mental health and labour market trajectories. *Review of Economic Studies 92*(3), 1920–1954.

Jones, D., D. Molitor, and J. Reif (2019). What do workplace wellness programs do? evidence from the illinois workplace wellness study. *The Quarterly Journal of Economics 134*(4), 1747–1791.

Kakwani, N., A. Wagstaff, and E. Van Doorslaer (1997). Socioeconomic inequalities in health: measurement, computation, and statistical inference. *Journal of econometrics 77*(1), 87–103.

Kline, P. (2024). Firm wage effects. In *Handbook of Labor Economics*, Volume 5, pp. 115–181. Elsevier.

Kline, P., E. K. Rose, and C. R. Walters (2022). Systemic discrimination among large u.s. employers. *The Quarterly Journal of Economics 137*(4), 1963–2036.

Kline, P., R. Saggio, and M. Sølvsten (2020). Leave-out estimation of variance components. *Econometrica 88*(5), 1859–1898.

Kline, P. M. (2025). Branching fixed effects: A proposal for communicating uncertainty. NBER Working Paper 34486, National Bureau of Economic Research.

Kline, P. M., E. K. Rose, and C. R. Walters (2024). A discrimination report card. Working Paper w32313, National Bureau of Economic Research. https://www.nber.org/papers/w32313.

Koning, P. (2009). Experience rating and the inflow into disability insurance. *De Economist 157*(3).

Kronenberg, K. and M. A. Carree (2010). Job and residential mobility in the netherlands: The influence of human capital, household composition and location. Technical Report MPRA Paper No. 25840, Munich Personal RePEc Archive (MPRA). Based on Statistics Netherlands (CBS) register microdata; full-time employees ($\geq$ 0.8 FTE), selected sectors, Sept. 2003–Sept. 2004. Reports counts used to compute that about 17.5% of job changes coincide with a residential move.

Lamadon, T., M. Mogstad, and B. Setzler (2022). Imperfect competition, compensating differentials, and rent sharing in the us labor market. *American Economic Review 112*(1), 169–212.

Lee, J. M. and L. O. Taylor (2019). Randomized safety inspections and risk exposure on the job: Quasi-experimental estimates of the value of a statistical life. *American Economic Journal: Economic Policy 11*(4), 350–374.

Lleras-Muney, A., H. Schwandt, and L. R. Wherry (2025). Poverty and health. *Annual Review of Economics 17*(1), 31–56.

Loef, B., I. Meulman, G.-C. M. Herber, G. J. Kommer, M. A. Koopmanschap, A. E. Kunst, J. J. Polder, A. Wong, and E. Uiters (2021). Socioeconomic differences in healthcare expenditure and utilization in the netherlands. *BMC Health Services Research 21*(1), 643.

Loomis, D., N. Guha, A. L. Hall, and K. Straif (2018). Identifying occupational carcinogens: an update from the iarc monographs. *Occupational and environmental medicine 75*(8), 593–603.

Marin, A. and G. Psacharopoulos (1982). The reward for risk in the labor market: evidence from the united kingdom and a reconciliation with other studies. *Journal of Political Economy 90*(4), 827–853.

Mas, A. and A. Pallais (2017). Valuing alternative work arrangements. *American Economic Review 107*(12), 3722–3759.

Miller, S., E. Rhodes, A. W. Bartik, D. E. Broockman, P. K. Krause, and E. Vivalt (2024). Does income affect health? evidence from a randomized controlled trial of a guaranteed income. Technical Report w32711, National Bureau of Economic Research.

Mincer, J. A. (1974). The human capital earnings function. In *Schooling, experience, and earnings*, pp. 83–96. NBER.

Moghani, V. (2026, August). When is the mover-design event study coefficient a place-effect share? Tinbergen Institute Discussion Paper 2026-056/III, Tinbergen Institute.

Mullahy, J. and E. C. Norton (2024). Why transform y? the pitfalls of transformed regressions with a mass at zero. *Oxford Bulletin of Economics and Statistics 86*(2), 417–447.

Nederlandse Arbeidsinspectie (2019). Arbo in bedrijf 2018: Naleving van arboverplichtingen en blootstelling aan risico's. Technical report, Nederlandse Arbeidsinspectie, Den Haag. Rapport en tabellenboek.

OECD and European Observatory on Health Systems and Policies (2019). Netherlands: Country health profile 2019. Technical report, OECD Publishing, Paris.

OECD and European Observatory on Health Systems and Policies (2021). Netherlands: Country health profile 2021. Technical report, OECD Publishing, Paris.

O'Donnell, O., E. Van Doorslaer, and T. Van Ourti (2015). Chapter 17 - health and inequality. In A. B. Atkinson and F. Bourguignon (Eds.), *Handbook of Income Distribution*, Volume 2, pp. 1419–1533. Elsevier.

Prudon, R. (2025). Is delayed mental health treatment detrimental to employment? *Review of Economics and Statistics*, 1–46.

Ravesteijn, B., H. v. Kippersluis, and E. v. Doorslaer (2018). The wear and tear on health: What is the role of occupation? *Health Economics 27*(2), e69–e86.

Rosen, S. (1974). Hedonic prices and implicit markets: product differentiation in pure competition. *Journal of political economy 82*(1), 34–55.

Rosen, S. (1986). Chapter 12 the theory of equalizing differences. In *Handbook of Labor Economics*, Volume 1, pp. 641–692. Elsevier.

Royer, H., M. Stehr, and J. Sydnor (2015). Incentives, commitments, and habit formation in exercise: evidence from a field experiment with workers at a fortune-500 company. *American Economic Journal: Applied Economics 7*(3), 51–84.

Schlotheuber, A. and A. R. Hosseinpoor (2022). Summary measures of health inequality: a review of existing measures and their application. *International journal of environmental research and public health 19*(6), 3697.

Shreekumar, A. and P. L. Vautrey (2022). Managing emotions: The effects of online mindfulness meditation on mental health and economic behavior. Technical report, MIT.

Simonsen, M. and L. Skipper (2025). Healthy at work? evidence from a social experimental evaluation of a firm-based wellness program. *American Economic Journal: Economic Policy 17*(3), 471–495.

Sockin, J. (2022). Show me the amenity: Are higher-paying firms better all around?

Sorkin, I. (2018). Ranking firms using revealed preference. *The quarterly journal of economics 133*(3), 1331–1393.

Tilipman, N. (2022). Employer incentives and distortions in health insurance design: Implications for welfare and costs. *American Economic Review 112*(3), 998–1037.

Törnberg, P. (2023). Chatgpt-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning.

Tran, K. B., J. J. Lang, K. Compton, R. Xu, A. R. Acheson, H. J. Henrikson, J. M. Kocarnik, L. Penberthy, A. Aali, Q. Abbas, et al. (2022). The global burden of cancer attributable to risk factors, 2010–19: a systematic analysis for the global burden of disease study 2019. *The Lancet 400*(10352), 563–591.

Van Kleef, R. C., R. C. Van Vliet, and W. P. Van de Ven (2013). Risk equalization in the netherlands: an empirical evaluation. *Expert Review of Pharmacoeconomics & Outcomes Research 13*(6), 829–839.

van Ours, J. (2019). Health economics of the workplace: Workplace accidents and effects of job loss and retirement. In *Oxford Encyclopedia of Health Economics*. Oxford University Press.

Vektis (2016). Verzekerden in beeld 2016. Technical report, Vektis.

Viscusi, W. K. and J. E. Aldy (2003). The value of a statistical life: A critical review of market estimates throughout the world. *Journal of Risk and Uncertainty 27*, 5–76.

Viscusi, W. K. and M. J. Moore (1987). Workers' compensation: Wage effects, benefit inadequacies, and the value of health losses. *The Review of Economics and Statistics*, 249–261.

Wagstaff, A. (2002). Inequality aversion, health inequalities and health achievement. *Journal of health economics 21*(4), 627–641.

Wagstaff, A. and E. v. Doorslaer (2004). Overall versus socioeconomic health inequality: a measurement framework and two empirical illustrations. *Health economics 13*(3), 297–301.

Wagstaff, A., P. Paci, and E. Van Doorslaer (1991). On the measurement of inequalities in health. *Social science & medicine 33*(5), 545–557.

Appendices

A1 Institutional Background Details

This appendix provides additional institutional detail for the background discussion in Section II.

A1.I Dutch Healthcare and Insurance System

The Netherlands has a system of universal basic health insurance provided by private insurers under tight public regulation. All residents must purchase the basic package. Insurers must accept all applicants and charge community-rated premiums, while a national risk-equalization scheme compensates insurers for differences in enrollee risk. Financing is supplemented by tax revenue.[42] Basic insurance is individual and portable: coverage does not change when workers change jobs. Employers may offer collective contracts, sometimes with premium discounts of up to about 10%, but these contracts do not alter the statutory content or cost-sharing rules of the basic package (Handel et al., 2020).

Cost sharing in the basic scheme is largely standardized through annual deductibles. In 2015, all adults faced a compulsory deductible of €375 and could choose an additional voluntary deductible of up to €500. The deductible applied to most specialist and hospital care and to many prescription drugs,[43] but not to GP care, maternity care, or several preventive services. More than 90% of adults, including almost all individuals with high expected healthcare expenditure, chose the minimum deductible (Handel et al., 2020). Switching between insurers was also infrequent: only 6.8% of individuals changed insurer in 2015 (Handel et al., 2020).

Access to care is organized around the general practitioner. Individuals are required to register with a GP, typically near their residence (Currie and Zwiers, 2025). GPs provide the first point of contact for non-emergency care and act as gatekeepers for almost all non-urgent specialist and hospital treatment. GP consultations are fully covered and exempt from the deductible. Pharmacies are

[42] In 2015, the annual premium for a standard policy with the minimum deductible was around €1,200; low-income individuals received a healthcare benefit that offset part of the premium and deductible (Handel et al., 2020).

[43] Prescription-drug cost sharing in the Dutch basic scheme is not identical across all medications, since some drugs can also be subject to drug-specific co-payments or reimbursement limits in addition to the general deductible.

widespread and usually located near GP practices. This organization makes basic access to primary care fairly uniform across workers and largely independent of the employer.

The content of the basic package is set by the government and is the same across insurers. It covers medically necessary GP care, specialist and hospital services, most prescription drugs, maternity and neonatal care, emergency transport, mental health services,[44] and selected medical aids. Routine adult dental care, eyeglasses and contact lenses, and many elective or alternative treatments are excluded. Physiotherapy is reimbursed only for chronic conditions and only after the first 20 sessions, so most short-term physiotherapy is either paid out of pocket or financed through supplementary insurance.[45]

For this paper, the key implication is that basic insurance in the Netherlands is not attached to the employment relationship. Cross-firm differences in observed basic-package expenditure therefore arise in a setting with limited employer-driven variation in insurance generosity, provider networks, or statutory cost sharing.

A1.II Occupational Health, Screening, and Sickness Provisions

Dutch employers face extensive obligations in occupational health and sickness management. Under the Working Conditions Act (*Arbowet*), firms must contract a certified occupational health service or expert, maintain a written risk inventory and evaluation (*RI&E*) covering physical, chemical, ergonomic, and psychosocial risks, and implement a corresponding plan of action. Employers must also provide employees with access to a registered company doctor who supervises sickness absence and reintegration, and they must continue paying wages for up to two years during illness. When indicated by the RI&E, firms must offer a job-specific occupational health examination (PAGO), and they may also provide broader preventive medical examinations (PMO). Participation is voluntary except in a small set of legally defined high-risk or safety-critical jobs.

These obligations are reinforced by experience rating in disability-related

[44] Although workplaces have expanded mental health and well-being services in recent years (Shreekumar and Vautrey, 2022), in the setting of this study most mental healthcare is still provided through the formal basic health insurance system (Prudon, 2025).

[45] Vektis reports that 15.9% of insured persons in 2015 had no supplementary insurance, implying that about 84% held some form of supplementary insurance (Vektis, 2016, p. 23).

insurance. Employers pay an experience-rated contribution to the Return-to-Work Fund (Whk), which finances benefits under the Work and Income Act (WIA) and the Sickness Benefits Act (ZW). The premium depends on the firm's own inflow into sickness and disability benefits.[46] Dutch firms therefore face both legal duties and financial incentives to limit workplace-related health problems.

Compliance, however, is incomplete. Around 80% of firms report having a contract with an occupational health provider and roughly three quarters have a formal sickness-absence policy, but only about half maintain a written RI&E and fewer than 40% combine it with a concrete plan of action. When four core requirements are considered jointly—RI&E, an occupational health contract, in-company emergency response, and a designated prevention officer—only about one third of firms comply with all four (Nederlandse Arbeidsinspectie, 2019). Compliance is higher among large and high-risk firms than among small employers.

Preventive use of occupational health services is even narrower than formal coverage. Although most firms have access to a company doctor through an external provider, only about 7% report that employees used an open consultation hour in the previous year. Screening is similarly limited: PAGO/PMO is explicitly included in contracts for roughly one quarter of firms, but only about 4% report that an examination actually took place during the previous 12 months (Nederlandse Arbeidsinspectie, 2019). These programmes are concentrated in large firms and in sectors with well-identified exposures such as noise, chemicals, and heavy physical work. For most workers, day-to-day healthcare still runs through the ordinary GP-based system. Occupational doctors mainly provide work-related advice and reintegration support rather than medical treatment.

In short, the Dutch institutional setting combines substantial employer responsibility for prevention and sickness management with limited employer control over the content of ordinary medical care. That combination is useful for separating workplace health effects from insurance design.

[46] Studies find that linking premiums to firm-specific disability inflow reduces DI entries among firms whose premiums actually adjust (Koning, 2009), and that removing experience rating raises DI inflow and lowers outflow for small employers (De Groot and Koning, 2016).

A1.III Workplace and Basic Healthcare Expenditure

Firms' contributions to basic healthcare expenditure can operate through two broad channels: differences in workers' ability to seek care and differences in the health consequences of the workplace itself.[47] The Dutch institutional setting is particularly informative because several features narrow the scope for the first channel. Out-of-pocket payments for basic care are capped by the annual deductible, more than 90% of adults choose the minimum deductible, and the marginal price of additional basic care is zero for high spenders once the deductible is exhausted. Existing evidence also finds limited income effects on healthcare expenditure (Miller et al., 2024; Cesarini et al., 2016). Time barriers also seem modest. Collective agreements and labour law provide paid time off for sickness and medical visits, employers must continue wage payments during illness for up to two years, and reintegration is formally supervised by occupational physicians. Consistent with this, self-reported unmet medical need in the Netherlands is among the lowest in the EU and shows little income gradient (e.g. OECD and European Observatory on Health Systems and Policies, 2019, 2021); lack of time is cited by less than 1% of respondents (Eurostat, 2023). Taken together, these features make the setting well suited to studying persistent firm effects on healthcare expenditure.

Workplace health conditions provide a natural set of mechanisms through which firms can have persistent effects on healthcare expenditure. One set of mechanisms runs through safety and physical working conditions: injury risks and ergonomic strain vary across firms and sectors and respond to regulation and enforcement (Hamermesh, 1999; Lee and Taylor, 2019; Johnson, 2020; Johansson et al., 2023). Despite substantial progress, occupational injuries and work-related musculoskeletal disorders remain major health hazards (Bhattacharya, 2014; van Ours, 2019). A second set concerns psychosocial stress: high workloads, long hours, job insecurity, and poor management practices are linked to stress, depression, burnout, and related somatic conditions (Hummels et al., 2025; Blackburn et al., 2023; Jolivet and Postel-Vinay, 2025). Third, workplaces differ in environmental exposures such as noise, dust, chemicals, heat, and air pollution, which are

[47] Appendix A4 formalises this point in a Grossman-style framework (Grossman, 1972b), where individuals choose health investments given preferences, income and time constraints, and a workplace-specific health-depreciation rate.

established risk factors for respiratory, cardiovascular, renal, and cancer outcomes (e.g. Collaborators et al., 2018; Loomis et al., 2018; Tran et al., 2022; Gan et al., 2011). Finally, firms may shape daily health behaviours through food provision, exercise incentives, and workplace norms around smoking, alcohol use, and physical activity (Royer et al., 2015; Jones et al., 2019; Simonsen and Skipper, 2025).

A2 Summary Statistics and Additional Results

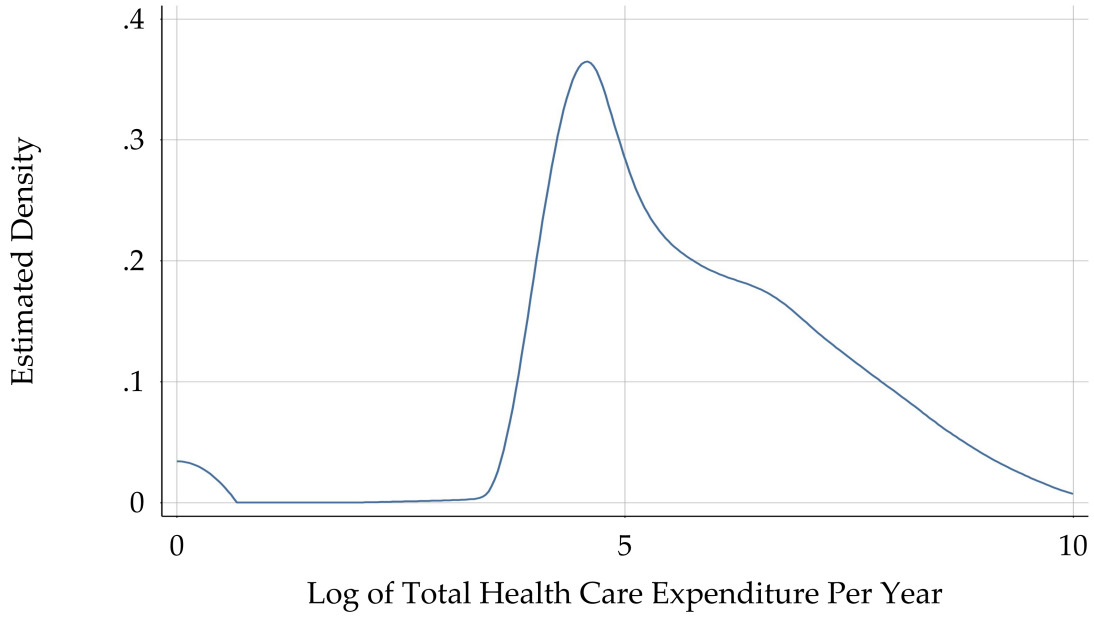


Figure A1: Log of total health expenditure per year for the connected set of employees (pooled male and female employees)

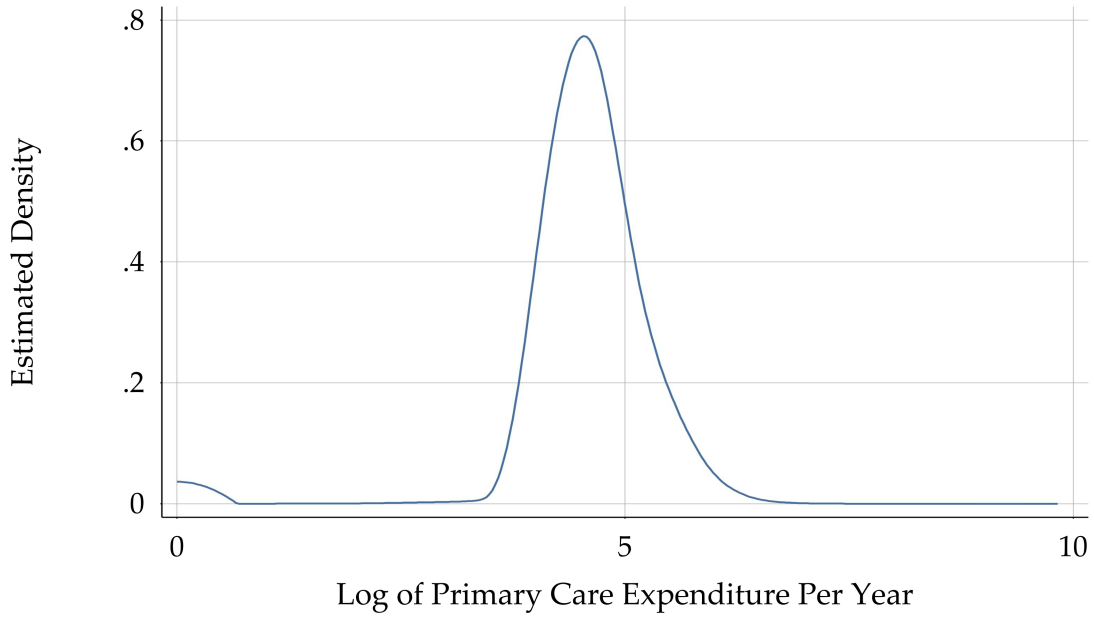


Figure A2: Log of primary health care expenditure per year for the connected set of employees (pooled male and female employees)

Statistic	**Plug-In Estimates (Biased)**
Log Health Exp. Var.: $var(h_i)$	3.01
Firm FE Var.: $var(\psi_j)$	0.33
Firm FE SD: σ_ψ	0.574
$100[e^{\sigma_\psi} - 1]$ (%)	77.5
Employee FE Var.: $var(\alpha_i)$	1.94
Firm and Employee FE's Cov.: $cov(\alpha_i, \psi_j)$	-0.04
No. Movers	1,232,499
No. Firms	132,899
No. of Observations	17,300,199

Note: This table displays the variance and covariance components from Equation (4), using the naive plug-in estimates of the parameters estimated with the OLS method from Equation (4). The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]\%$, using the reported SD.

Table A1: The variance decomposition results using the fixed effects estimated in the TWFE model (4): Male workers

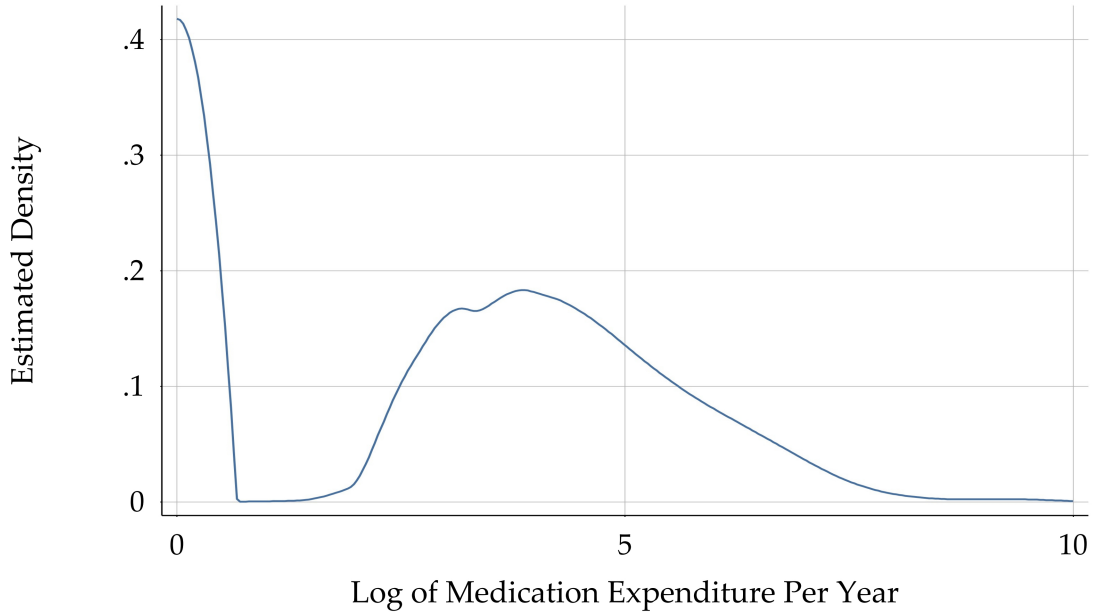


Figure A3: Log of medication costs per year for the connected set of employees (pooled male and female employees)

	Low group mean $\hat{h}_{it}$	High group mean $\hat{h}_{it}$	Difference (low − high)	Implied % difference in spending[a]
Household wealth 2009 (bottom vs top quintile)	0.147	-0.111	0.258	29.4
Total income 2009 (bottom vs top quintile)	0.153	-0.133	0.286	33.1
Education (primary vs master/doctor)	0.257	-0.140	0.398	48.8

[a] The implied percentage difference is $\left[\exp(\text{Difference}) - 1\right] \times 100$.

Notes: Outcome is log annual basic-package healthcare spending residualised on age and calendar-year dummies, $\hat{h}_{it}$. Each cell reports the mean of $\hat{h}_{it}$ for the indicated group, averaged over person–year observations for full-time male employees aged 25–65 in the main analysis sample (2009–2016). Household wealth and total personal income are measured in 2009 and divided into five quintiles; the corresponding rows compare the bottom and top quintiles of these distributions. Education is measured as highest completed level, and the row compares workers with only primary education ("Basisonderwijs") to those with a master's degree or doctorate.

Table A2: Gradients in age- and year-adjusted log healthcare spending by SES

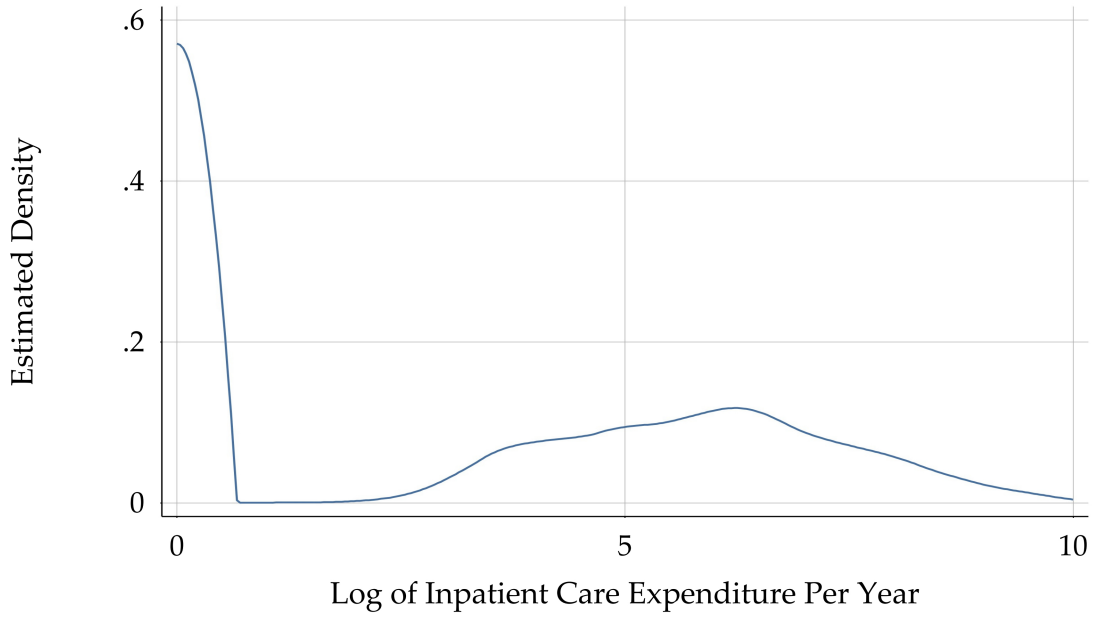


Figure A4: Log of inpatient care expenditure per year for the connected set of employees (pooled male and female employees)

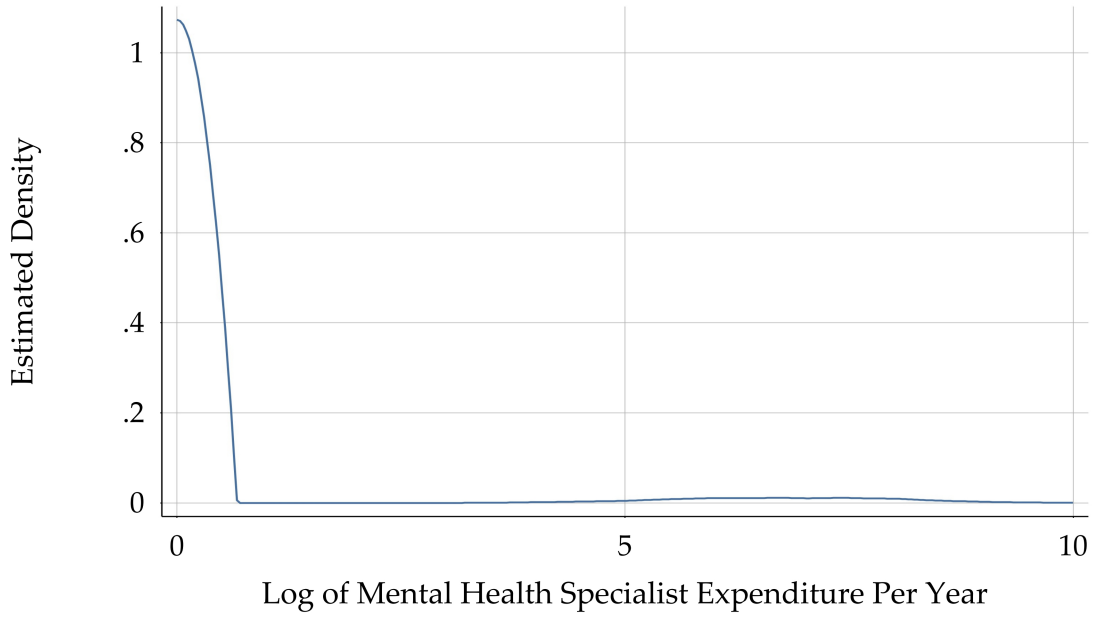
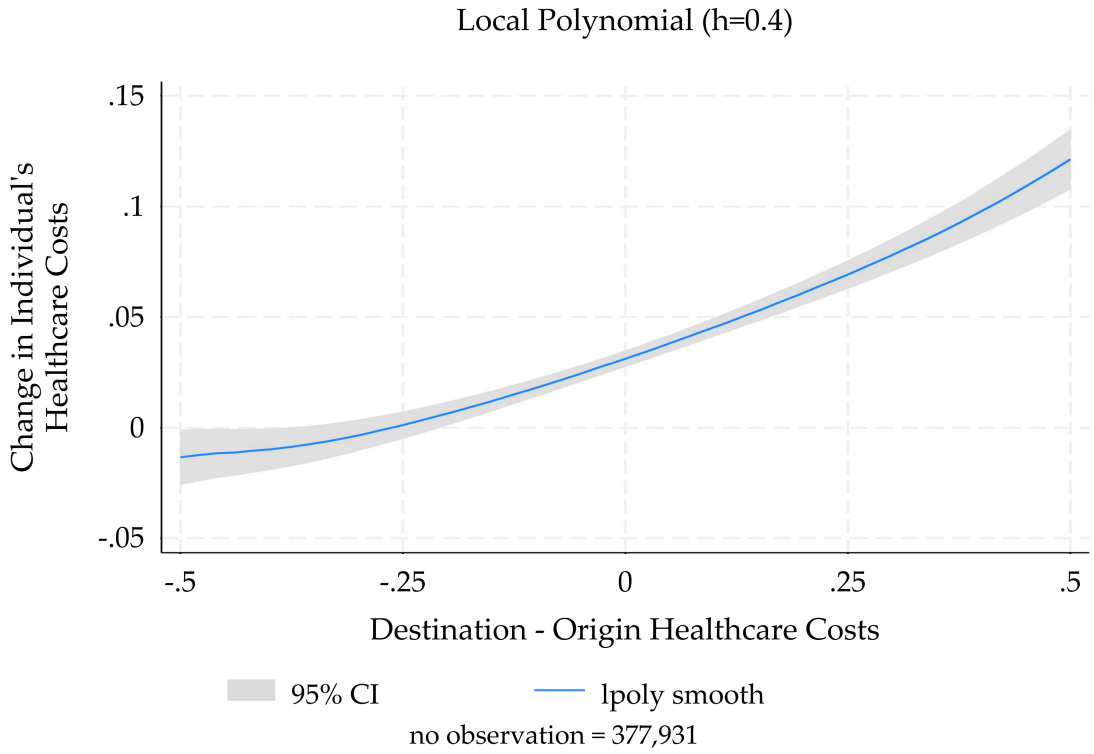
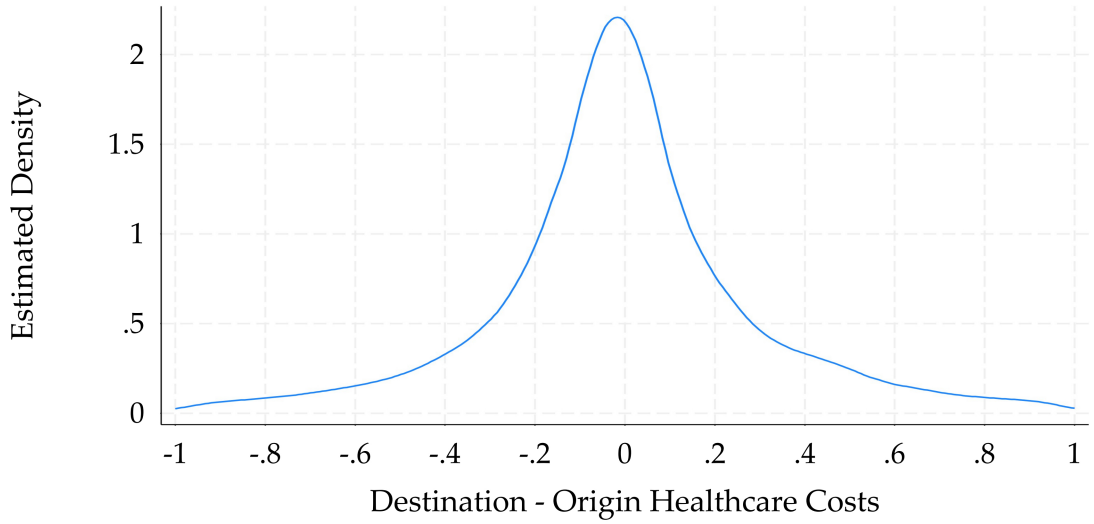


Figure A5: Log of mental health care expenditure (excluding primary care) per year for the connected set of employees (pooled male and female employees)



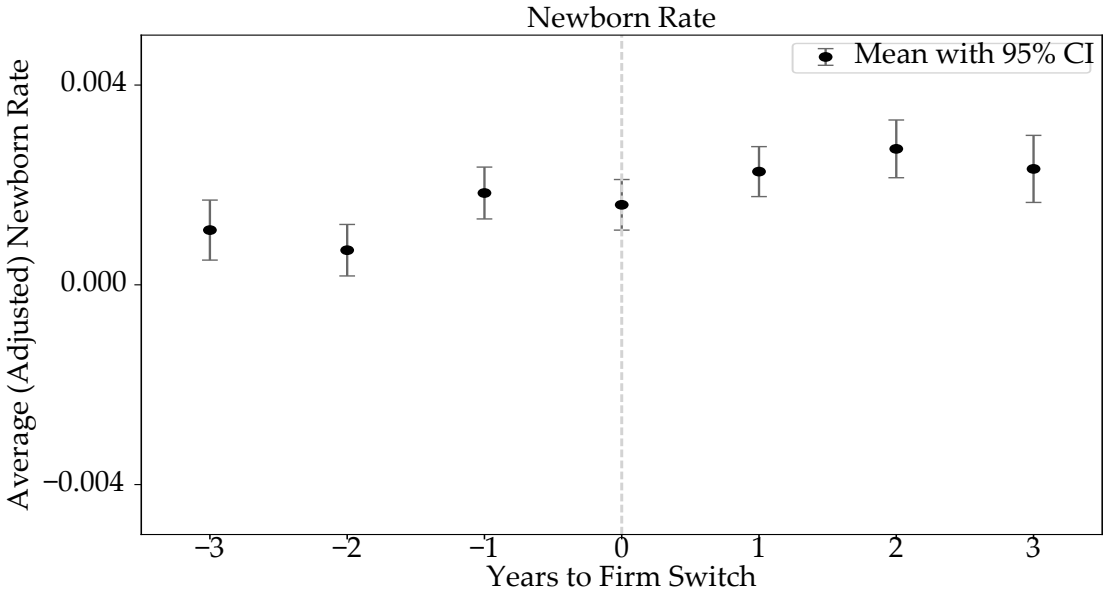
(a) Average Changes in Expenditure (binscatter)



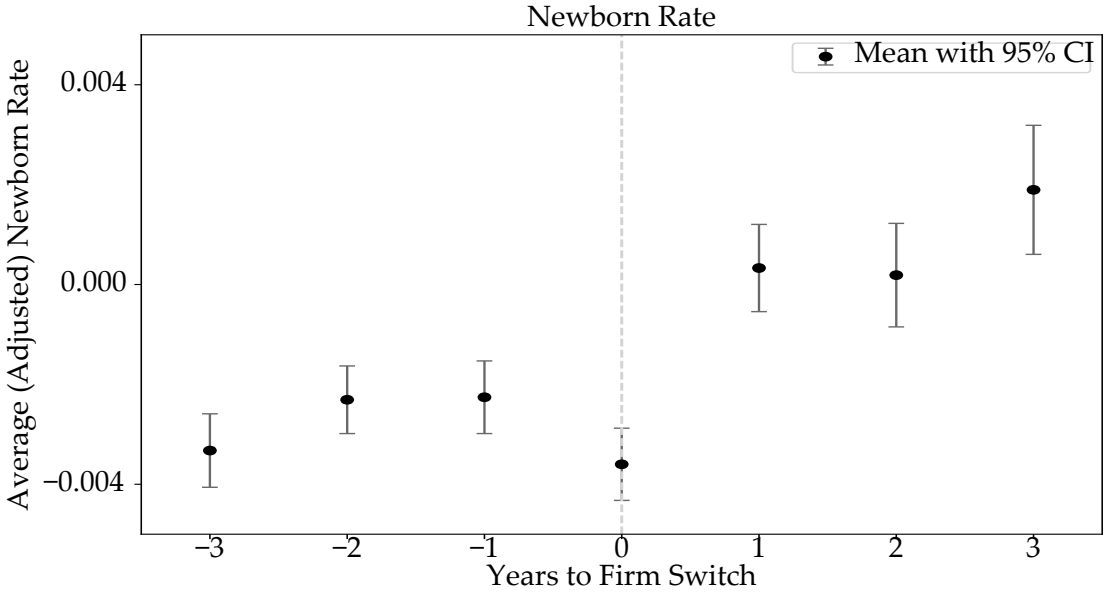
(b) Distribution of changes

Figure A6: Comparison of changes in workers' healthcare expenditure with changes in their colleagues' expenditure.

Note: Full-time male one-time movers, 2009–2016. Spending is age- and year-residualised $\log(c+1)$. Panel (a) relates pre/post changes to the destination-minus-origin leave-*i*-out mean coworker-spending gap; panel (b) shows its distribution.



(a) Men



(b) Women

Note: The mean annual new birth rate is approximately 4% for both samples.

Figure A7: Average annual probability of having a new child around the time of a job switch (time = 0) for movers. Estimates are adjusted for age and education using a flexible age by education functional form using a linear regression.

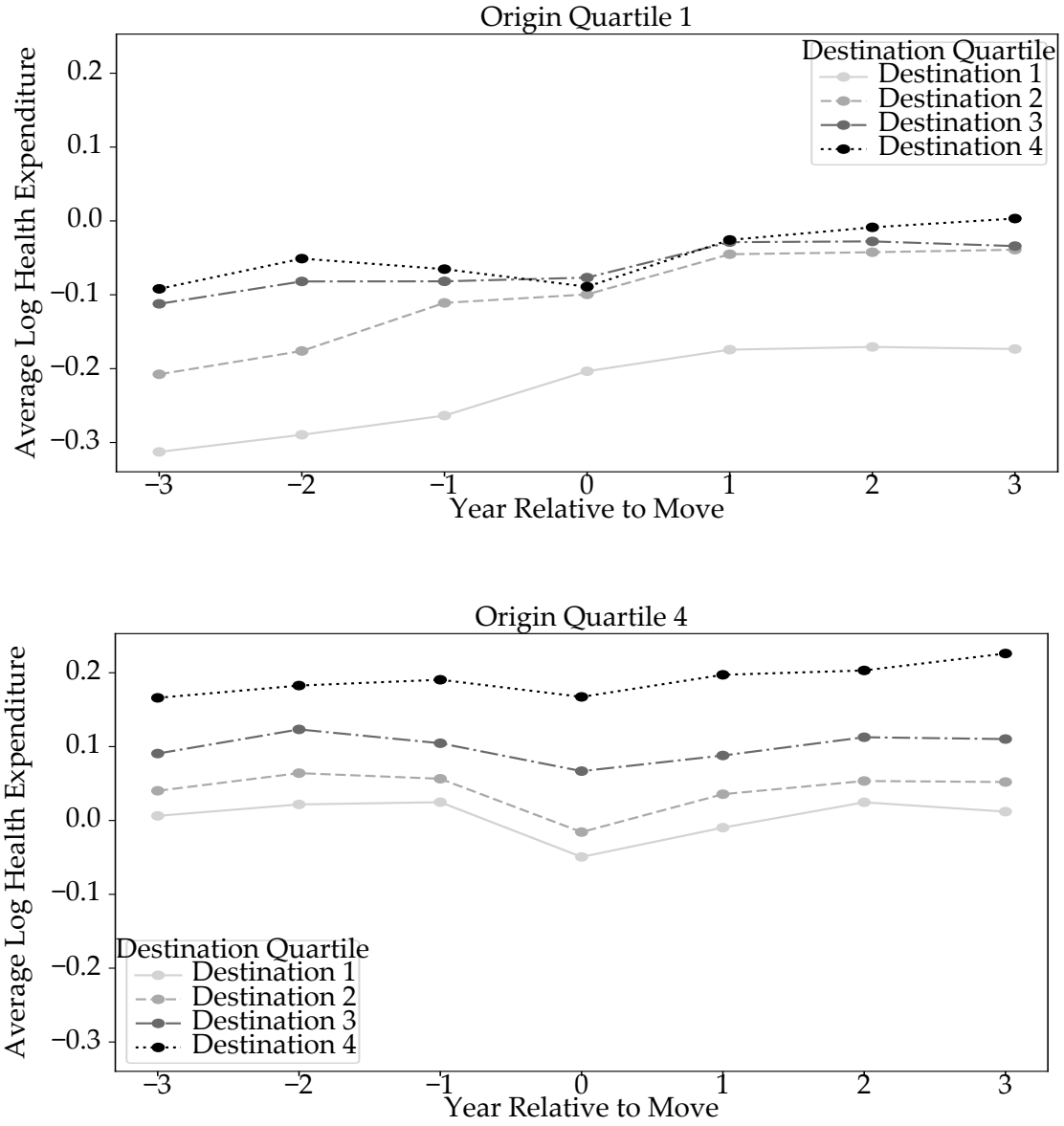
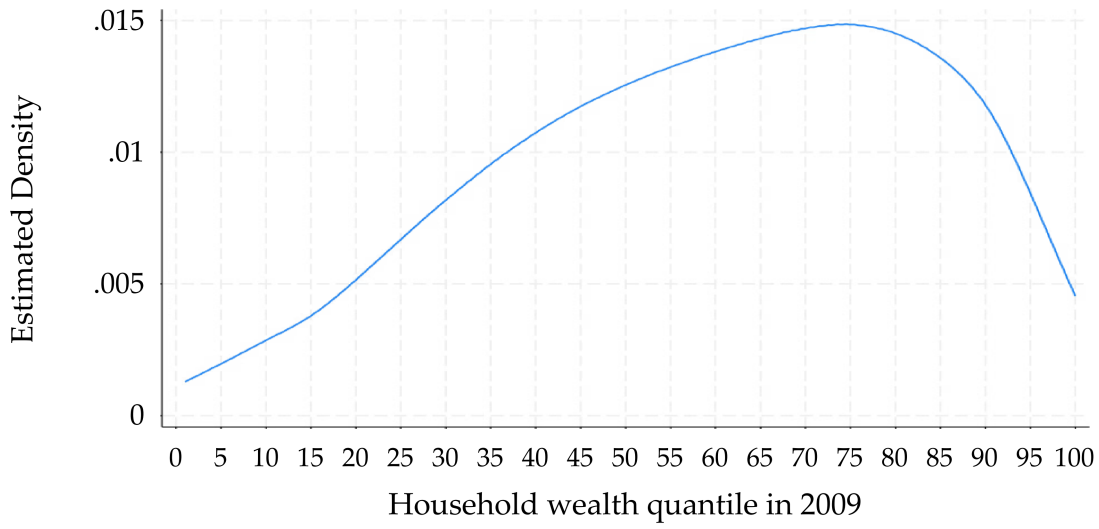
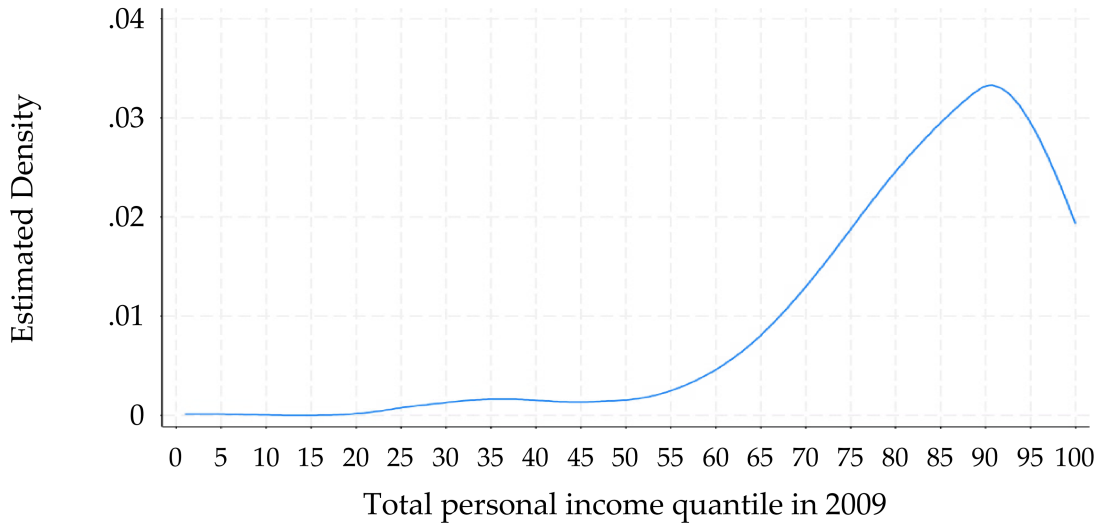


Figure A8: Mean age-adjusted log healthcare expenditures of movers by coworker expenditure quartile.

Note: Average age-adjusted logarithmic healthcare expenditures for employees who changed firms between 2009 and 2016, remaining at the origin firm for at least two years and at the destination firm for at least two years. Firms are grouped into quartiles by the average coworker (leave-i-out) health expenditure within the firm.



(a) Household wealth quantile in 2009



(b) Total personal income quantile in 2009

Figure A9: Location of the analysis sample in the national wealth and income distributions

Note: Each panel plots the kernel density of the position of the full-time male main sample in the national distribution of the corresponding SES measure, expressed as percentiles (0–100) of the Dutch population in 2009. For every individual in the main analysis sample, I use the Statistics Netherlands reported rank for 2009. The sample consists of full-time male employees aged 25–65 in the 2009–2016 analysis period.

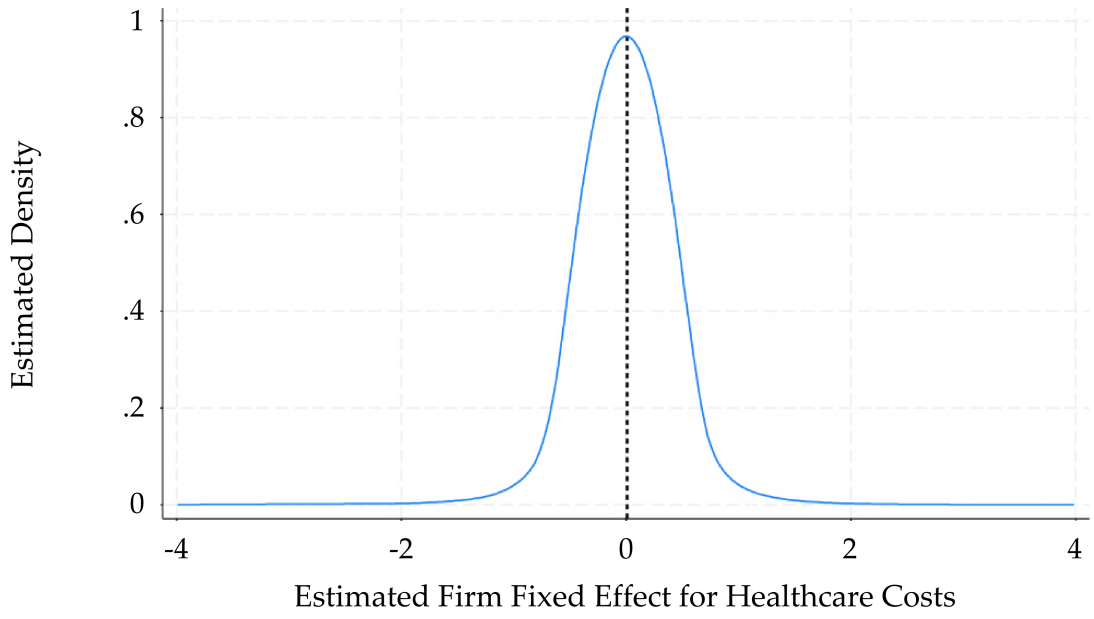


Figure A10: Distribution of estimated firm fixed effects

Note: Estimated density of the demeaned estimated firm fixed effects $\hat{\psi}_j$ from the AKM specification in Equation (4) for all (individual-year weighted) firms in the connected set used in the main analysis. The outcome is log annual basic-package healthcare spending, $h_{it} = \log(c_{it} + 1)$, and the model includes worker and firm fixed effects as well as age–education profiles and calendar-year dummies. The vertical dashed line shows the median.

Statistic	**KSS Estimates**
Log Wage Var.: var(w_i)	0.176
Firm FE Var.: var(ψ_j^w)	0.017
Firm FE SD: σ_ψ	0.129
$100[e^{\sigma_\psi} - 1]$ (%)	13.8
Employee FE Var.: var(α_i^w)	0.271
Firm–Employee Cov.: cov(α_i^w, ψ_j^w)	−0.001
Correlation $\rho(\alpha_i^w, \psi_j^w)$	−0.020
Explained share: workers & firms	1.621
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Log Wage: mean(w_i)	3.141

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log hourly wage. Hourly wages are constructed from gross annual contract earnings, contractual full-time equivalent (FTE), and contract length, assuming that a full-time job corresponds to 52 weeks of 40 hours per week, as in the data construction in Section IV. The sample consists of full-time male employees in the leave-one-out connected set of firms between 2009 and 2016. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j^w)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A3: Variance Decomposition of Log Hourly Wages

Panel A: KSS Variance Components	
Statistic	**KSS Estimates**
Log Health Exp. Var.: var(h_i)	2.617
Firm FE Var.: var(ψ_j)	0.027
Firm FE SD: σ_ψ	0.164
$100[e^{\sigma_\psi} - 1]$ (%)	17.8
Employee FE Var.: var(α_i)	1.515
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.018
Explained share: workers & firms	0.586
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Health Exp.: mean(h_i)	5.608
Panel B: Regression of Firm Effects with Pay Control on Baseline Firm Effects	
Dependent variable	$\hat{\psi}_j^{\text{HC,pay}}$
Coefficient on baseline firm FE (worker level) $\hat{\psi}_j$	1.000
No. of workers	16,976,071

Notes: Panel A reports components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year. The regression includes worker and firm fixed effects, age–by–education profiles, calendar-year dummies, and an additional control for workers' annual gross contract payment (yearly labour income). The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD. Panel B reports an OLS regression of the firm fixed effects from the healthcare model with the contract-payment control, $\hat{\psi}_j^{\text{HC,pay}}$, on the baseline healthcare firm fixed effects without that control, $\hat{\psi}_j$. The slope is close to one and the intercept is close to zero, indicating that adding the pay control leaves the pattern of estimated firm health effects almost unchanged.

Table A4: Variance Decomposition of Log Healthcare Expenditure (Controlling for Workers' Annual Contract Payments)

	High baseline costs *($\hat{\alpha}_i$ above median)*	**Low baseline costs** *($\hat{\alpha}_i$ below median)*
Panel A. KSS variance components		
Log Health Exp. Var.: var(h_i)	2.101	1.745
Firm FE Var.: var(ψ_j)	0.022	0.035
Firm FE SD: σ_ψ	0.147	0.186
$100[e^{\sigma_\psi} - 1]$ (%)	15.8	20.4
Employee FE Var.: var(α_i)	0.550	0.563
Firm–Employee Cov.: cov(α_i, ψ_j)	0.032	0.032
Correlation $\rho(\alpha_i, \psi_j)$	0.297	0.228
Explained share: workers & firms	0.303	0.379
No. Movers	551,201	606,960
No. Firms	80,230	89,384
No. of Observations	8,035,638	8,110,825
Mean log Health Exp.: mean(h_i)	6.447	4.775
Panel B. Descriptives and cross-sample link		
Mean annual basic-package costs (€)[a]	2,011	286.52
No. obs. used for mean costs[a]	8,488,038	8,488,033
Bias-corrected slope (worker level), firm FE (low baseline) on firm FE (high baseline)[b]	0.602	
No. worker–year obs. in slope regression[b]	7,635,508	

[a] Mean annual basic-package costs are computed directly from raw annual spending c_{it} (in euros) in the two samples, split by whether the estimated individual healthcare fixed effect $\hat{\alpha}_i$ from the baseline AKM lies above or below its median (median split defined over all connected-set worker–year observations).

[b] Slope from an OLS regression of the firm fixed effects estimated in the low-baseline-costs subsample on the firm fixed effects estimated in the high-baseline-costs subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the ratio of KSS-corrected to plug-in firm-effect variances following Kline et al. (2020). Because KSS is implemented on the leave-one-out connected set *within each subsample*, the two subsamples need not form an exact partition of the baseline connected set, and some firms (and their worker–year observations) appear in only one subsample and therefore drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A5: Variance Decomposition of Log Healthcare Expenditure by Baseline Individual Health-Cost Type

	High-wage workers *(above-median hourly wage)*	Low-wage workers *(below-median hourly wage)*
Panel A. KSS variance components		
Log Health Exp. Var.: var(h_i)	2.543	2.706
Firm FE Var.: var(ψ_j)	0.022	0.039
Firm FE SD: σ_ψ	0.150	0.197
$100[e^{\sigma_\psi} - 1]$ (%)	16.2	21.8
Employee FE Var.: var(α_i)	1.198	1.630
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.006	−0.005
Explained share: workers & firms	0.475	0.613
No. Movers	547,051	631,701
No. Firms	58,556	103,831
No. of Observations	8,061,912	8,278,053
Mean log Health Exp.: mean(h_i)	5.567	5.653
Panel B. Descriptives and cross-sample link		
Mean hourly wage (€)[a]	33.13	16.39
Mean annual contract payment (€)[a]	67,783	33,306
No. obs. used for wage descriptives[a]	8,488,041	8,488,030
Bias-corrected slope (worker level), firm FE (low wage) on firm FE (high wage)[b]	0.622	
No. worker–year obs. in slope regression[b]	6,644,986	

[a] Wage groups are defined using an estimated hourly wage (in euros) computed from annual contract payment and work intensity (as in Section IV). The median hourly wage in the connected-set sample is 20.9; "high wage" corresponds to observations above this threshold. Panel B means are computed directly from the raw wage variables in the connected-set person–year sample and are descriptive.

[b] Slope from an OLS regression of the firm fixed effects estimated in the low-wage subsample on the firm fixed effects estimated in the high-wage subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the ratio of KSS-corrected to plug-in firm-effect variances following Kline et al. (2020). Because KSS is implemented on the leave-one-out connected set *within each wage subsample*, the two subsamples need not form an exact partition of the baseline connected set, and some firms (and their worker–year observations) appear in only one subsample and therefore drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A6: Variance Decomposition of Log Healthcare Expenditure by Wage Group

	Young workers *(age ≤ 45)*	**Old workers** *(age > 45)*
Panel A. KSS variance components		
Log Health Exp. Var.: var(h_i)	2.390	2.680
Firm FE Var.: var(ψ_j)	0.029	0.014
Firm FE SD: σ_ψ	0.171	0.120
$100[e^{\sigma_\psi} - 1]$ (%)	18.6	12.7
Employee FE Var.: var(α_i)	1.369	1.649
Firm–Employee Cov.: cov(α_i, ψ_j)	0.005	−0.001
Explained share: workers & firms	0.590	0.620
No. Movers	742,335	409,353
No. Firms	108,472	49,383
No. of Observations	8,653,958	7,154,491
Mean log Health Exp.: mean(h_i)	5.317	5.944
Panel B. Cross-sample link		
Bias-corrected slope (worker level), firm FE (young) on firm FE (old)[a]	0.846	
No. worker–year obs. in slope regression[a]	7,236,057	

[a] Slope from an OLS regression of the firm fixed effects estimated in the young subsample on the firm fixed effects estimated in the old subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the reliability ratio for the right-hand-side firm effect, $Var_{KSS}(\psi)/Var_{PI}(\hat{\psi})$, following Kline et al. (2020) (equivalently, $\hat{\beta}_{BC} = \hat{\beta}_{OLS} \times Var_{PI}(\hat{\psi}_{old})/Var_{KSS}(\psi_{old})$). Because KSS is implemented on the leave-one-out connected set *within each age subsample*, the two subsamples need not form an exact partition of the baseline connected set; some firms appear in only one subsample and drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A7: Variance Decomposition of Log Healthcare Expenditure by Age Group

	White-collar workers *(education proxy)*	**Blue-collar workers** *(education proxy)*
Panel A. KSS variance components		
Log Health Exp. Var.: var(h_i)	2.142	2.286
Firm FE Var.: var(ψ_j)	0.020	0.032
Firm FE SD: σ_ψ	0.140	0.178
$100[e^{\sigma_\psi} - 1]$ (%)	15.0	19.5
Employee FE Var.: var(α_i)	1.071	1.143
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.005	−0.012
Explained share: workers & firms	0.504	0.503
No. Movers	319,969	158,616
No. Firms	49,485	40,897
No. of Observations	3,719,229	1,903,635
Mean log Health Exp.: mean(h_i)	5.443	5.638
Panel B. Cross-sample link		
Bias-corrected slope (worker level), firm FE (white collar) on firm FE (blue collar)[b]	1.063	
No. worker–year obs. in slope regression[b]	1,598,774	

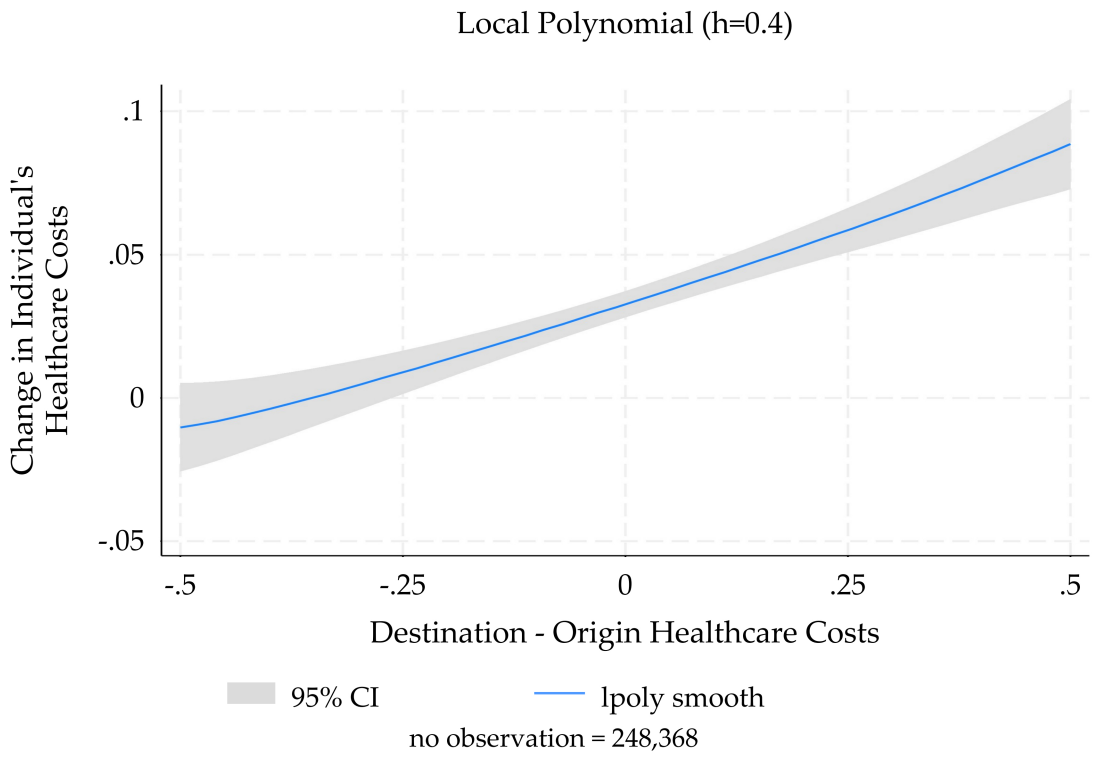
[a] Collar type is proxied by education titles; ambiguous titles are excluded (Appendix A8).
[b] Slope from an OLS regression of the firm fixed effects estimated in the white-collar subsample on the firm fixed effects estimated in the blue-collar subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the right-hand-side reliability ratio $Var_{KSS}(\psi)/Var_{PI}(\hat{\psi})$ following Kline et al. (2020), i.e. $\hat{\beta}_{BC} = \hat{\beta}_{OLS} \times Var_{PI}(\hat{\psi}_{blue})/Var_{KSS}(\psi_{blue})$.
Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A8: Variance Decomposition of Log Healthcare Expenditure by Collar Type (proxied by education)

Statistic	Primary care (GP)	Medication	Hospital (Inpatient + Specialist)	Mental health
Log Health Exp. Var.: var(h_i)	0.655	5.689	9.811	1.632
Firm FE Var.: var(ψ_j)	0.018	0.010	0.024	0.003
Firm FE SD: σ_ψ	0.133	0.098	0.154	0.050
$100[e^{\sigma_\psi} - 1]$ (%)	14.2	10.3	16.6	5.1
Employee FE Var.: var(α_i)	0.521	3.209	2.911	0.371
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.003	−0.001	−0.004	−0.002
Correlation $\rho(\alpha_i, \psi_j)$	−0.027	−0.006	−0.017	−0.061
Explained share: worker & firm	0.815	0.565	0.298	0.227
No. Movers	1,225,799	1,225,799	1,225,799	1,225,799
No. Firms	132,820	132,820	132,820	132,820
No. of Observations	16,976,071	16,976,071	16,976,071	16,976,071
Average Health Exp.: mean(h_i)	4.521	2.637	2.732	0.240

Notes: All columns report leave-one-match-out KSS estimates (Kline et al., 2020). Outcomes are the log of subcategory-specific healthcare expenditure: primary care (GP), medication, hospital (inpatient + specialist), and specialist mental health. The sample is the full connected set. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

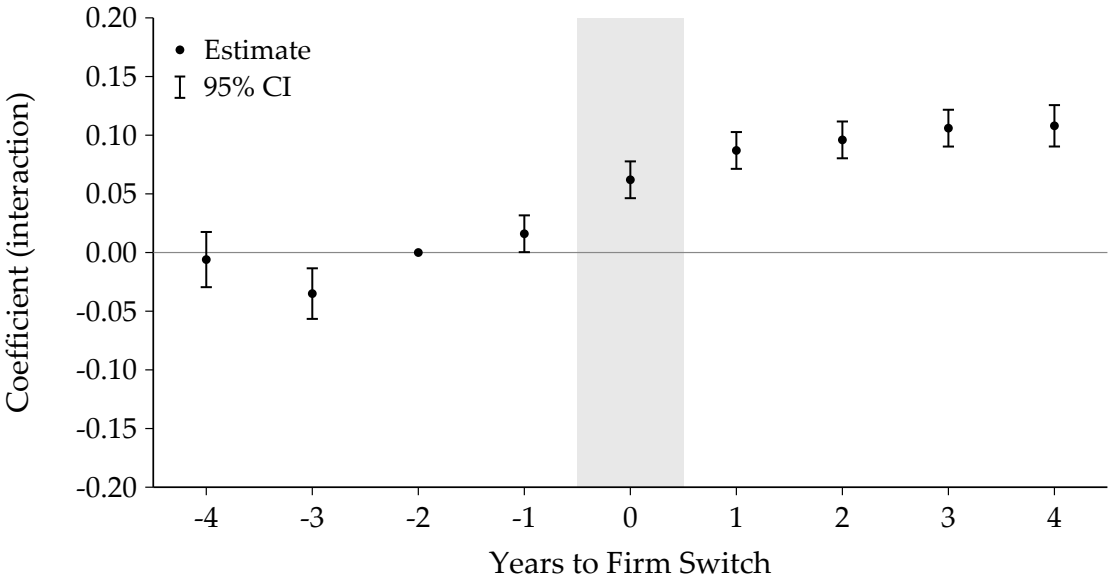
Table A9: Variance Decomposition of Log Healthcare Expenditure (KSS) by Cost Category



(a) Average change vs. destination exposure gap

Figure A11: Workers' healthcare spending around firm closures and reemployment

Notes. Sample of full-time male employees aged 25–65 whose employer exits during the panel and who are observed in consecutive years from 2009 and around the closure. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. The closure year is the last year in which the origin employer identifier is observed. In panel (a), the curve plots the average change in individual annual residualised spending from the pre-closure year to the post-reemployment year against the destination exposure gap, defined as the difference in leave-*i*-out mean residualised spending between the destination and origin employers. The line shows a non-parametric fit, where the intercept captures the common post-displacement increase and the slope measures pass-through associated with the destination exposure gap. In panel (b), the markers show coefficients from an event-study specification that interacts event-time indicators with the destination exposure gap, with event time measured relative to the closure year and $t = -2$ as the reference year.



(b) Event study around closure and reemployment

Figure A11 (continued): Workers' healthcare spending around firm closures and reemployment

Notes. Sample of full-time male employees aged 25–65 whose employer exits during the panel and who are observed in consecutive years from 2009 and around the closure. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. The closure year is the last year in which the origin employer identifier is observed. In panel (a), the curve plots the average change in individual annual residualised spending from the pre-closure year to the post-reemployment year against the destination exposure gap, defined as the difference in leave-*i*-out mean residualised spending between the destination and origin employers. The line shows a non-parametric fit, where the intercept captures the common post-displacement increase and the slope measures pass-through associated with the destination exposure gap. In panel (b), the markers show coefficients from an event-study specification that interacts event-time indicators with the destination exposure gap, with event time measured relative to the closure year and $t = -2$ as the reference year.

	Closure Sample *(Worker-year observations)* N = 1,550,889
Variable	Mean (SD)
Age	46.03 (9.43)
Dutch background (dummy)	0.85 (0.36)
First-generation migrant (dummy)	0.08 (0.27)
Second-generation migrant (dummy)	0.07 (0.25)
Education: Elementary (dummy)	0.02 (0.13)
Education: Lower sec./MBO1 (dummy)	0.05 (0.21)
Education: Upper sec./MBO (dummy)	0.19 (0.40)
Education: Bachelor (HBO/WO) (dummy)	0.18 (0.38)
Education: Master/Doctorate (dummy)	0.11 (0.31)
Education: Missing (dummy)	0.46 (0.50)
Total # children	1.54 (1.20)
New child (dummy)	0.04 (0.19)
Gross payment of contract (€)	55,101 (40,796)
Mover (dummy)	1.00 (0.00)
Full-time factor (FTE)	1.00 (0.01)
Estimated hourly wage (€)[a]	26.84 (19.78)
Business size (employees)	1,012 (815.15)
Total health care costs (€)	1,159 (4,574)
Mental health (specialist) (€)	75.13 (1,188)
Hospital costs (€)	700.58 (3,867)
GP costs (€)	110.45 (65.88)
Number of Observations	1,550,889

[a] Hourly wage computed from gross contract payment assuming a full FTE works 52 weeks × 40 hours per week.

Notes: Firm-closure sample (worker–year observations). Variables are defined as in Table 1. All observations are movers by construction (Mover dummy = 1). Business size is observed for 1,544,519 worker–year observations (all other variables are observed for 1,550,889).

Table A10: Summary Statistics: Firm-Closure Sample

	(1)	(2)	(3)
	Outcome measured pre-move		
Destination firm health-cost effect (sample-split)	0.0068	-0.3305***	-0.3879***
	(0.0092)	(0.0132)	(0.0201)
Mean of dep. var.	0.2054	0.2054	0.1960
N	209,235	209,235	206,945
R^2	0.084	0.089	0.146
Baseline controls & pre-move postal-code FEs	Yes	Yes	Yes
Origin firm-effect level (control)	No	Yes	No
Origin-firm fixed effects	No	No	Yes

Notes: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). The outcome is average pre-move basic-package healthcare spending. Covariates are the same as in Table 5. Column (2) additionally controls for the origin firm-effect level; column (3) includes origin-firm fixed effects. Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A11: Firm Closures: Pre-Move Healthcare Spending

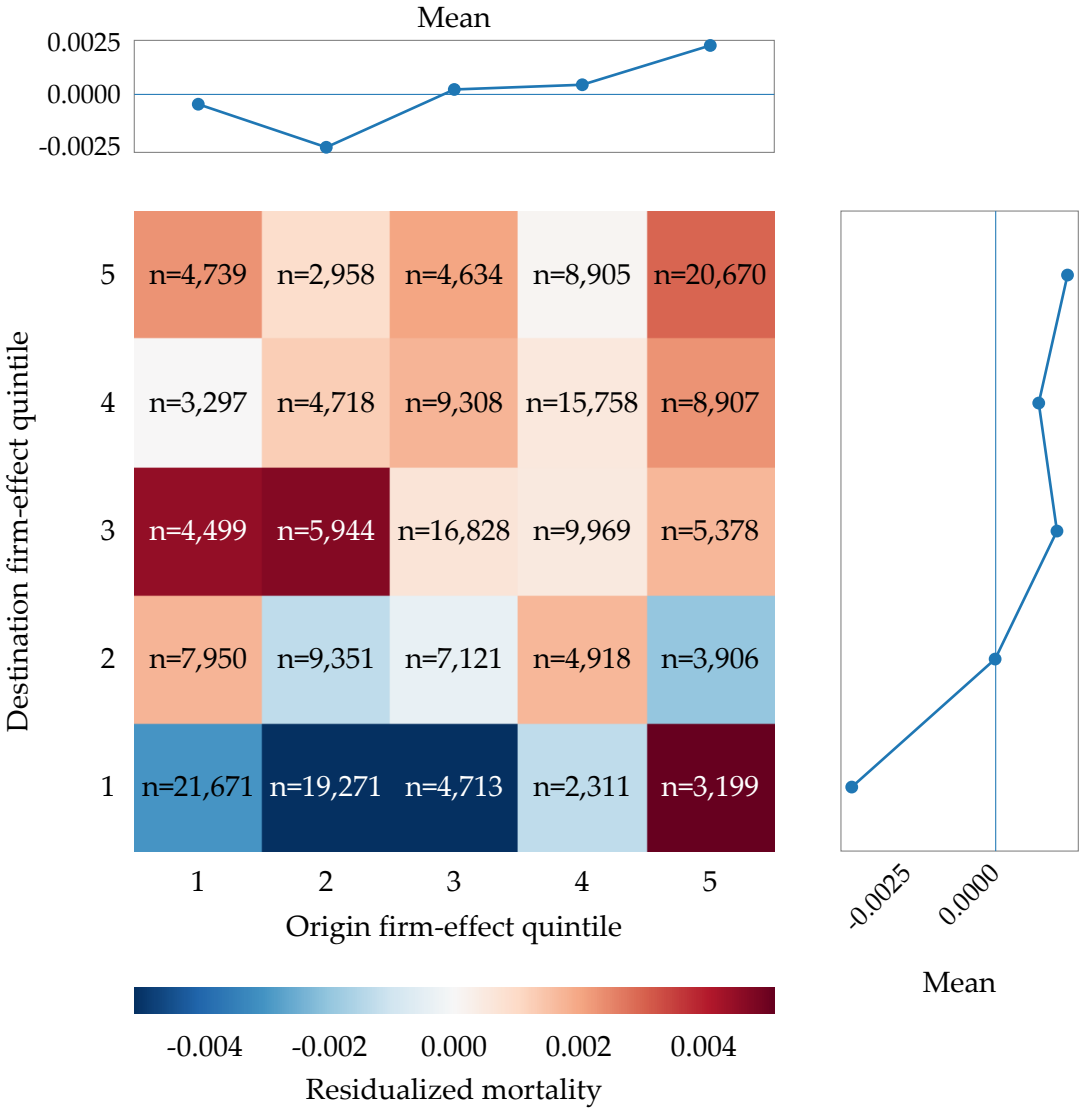
	All movers			Firm closures		
	(1)	(2)	(3)	(1)	(2)	(3)
Panel A: Selected accidents and injuries						
Destination firm effect	0.000230*	0.000478***	0.000279	0.000021	0.000171	-0.000170
	(0.000128)	(0.000171)	(0.000249)	(0.000128)	(0.000184)	(0.000282)
Mean of dep. var.		—			0.00034	
R^2	0.021	0.021	0.068	0.020	0.020	0.072
Panel B: Cancer						
Destination firm effect	0.002122***	0.003345***	0.002995***	0.001772***	0.002752***	0.002729**
	(0.000556)	(0.000740)	(0.001103)	(0.000570)	(0.000820)	(0.001267)
Mean of dep. var.		—			0.00687	
R^2	0.035	0.035	0.077	0.037	0.037	0.069
Panel C: Circulatory diseases and diabetes						
Destination firm effect	0.000323	0.000298	0.000075	0.000341	0.000774	0.000570
	(0.000352)	(0.000468)	(0.000699)	(0.000366)	(0.000527)	(0.000815)
Mean of dep. var.		—			0.00278	
R^2	0.021	0.021	0.059	0.024	0.024	0.057
Panel D: Selected respiratory diseases						
Destination firm effect	0.000092	0.000083	-0.000058	0.000086	0.000131	0.000091
	(0.000092)	(0.000122)	(0.000185)	(0.000084)	(0.000121)	(0.000189)
Mean of dep. var.		—			0.00015	
R^2	0.019	0.019	0.044	0.028	0.028	0.046
Panel E: Self-harm, substance-related causes, and selected sequelae						
Destination firm effect	0.000168	0.000058	-0.000088	0.000050	-0.000458	-0.000906*
	(0.000202)	(0.000269)	(0.000394)	(0.000217)	(0.000312)	(0.000477)
Mean of dep. var.		—			0.00096	
R^2	0.016	0.016	0.071	0.019	0.019	0.061
Panel F: Other causes						
Destination firm effect	0.000738**	0.001144***	-0.000272	0.000990***	0.001377***	0.001039
	(0.000317)	(0.000422)	(0.000630)	(0.000313)	(0.000451)	(0.000698)
Mean of dep. var.		—			0.00203	
R^2	0.016	0.016	0.055	0.019	0.019	0.052
N	289,755	289,755	282,051	209,235	209,235	206,945
Baseline controls & postcode FEs	Yes	Yes	Yes	Yes	Yes	Yes
Origin firm-effect level	No	Yes	No	No	Yes	No
Origin-firm fixed effects	No	No	Yes	No	No	Yes

Notes: Male movers, with all eligible movers in the first group and workers displaced by firm closures in the second. Both samples restrict destination firms to more than 75 employees. The regressor is the sample-split destination-firm health-cost effect. Within each group, column (2) adds the sample-split origin firm-effect level; column (3) includes origin-firm fixed effects. Baseline controls include demographics, pre-move healthcare expenditure, and pre-move postal-code fixed effects, as in Table 5. Outcome means for all movers are not available in the export; closure means refer to the descriptive sample (N = 211,539).

Causes use the first three characters of the underlying ICD-10 code: (A) V01–V99, W00–W59, X00–X39, X46–X59, Y85–Y86; (B) C00–C97; (C) I00–I99 and E10–E14; (D) J40–J47 and J60–J70; (E) X60–X84, Y87, F10–F19, X40–X45, Y10–Y15; (F) all remaining nonmissing causes. X40–X45 are assigned to (E), not (A). Group (A) covers selected accidental causes, without a workplace restriction. Group (D) covers chronic lower respiratory diseases and lung diseases due to external agents. Y87 in group (E) includes sequelae of self-harm, assault, and events of undetermined intent.

Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

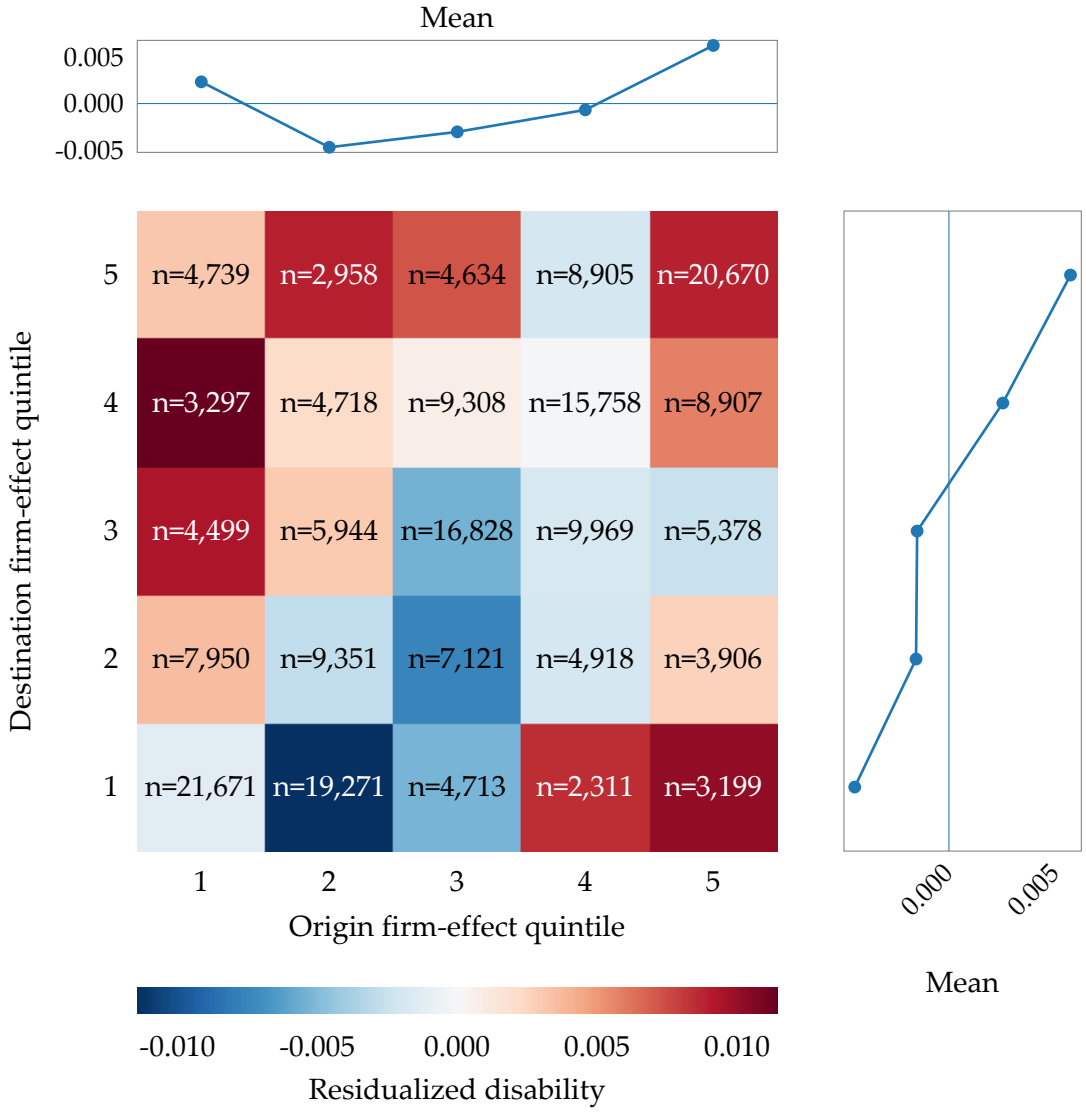
Table A12: Cause-Specific Mortality in 2019: All Movers and Firm Closures



(a) Mortality in 2019

Figure A12: Long-run outcomes by origin×destination firm-effect quintiles after firm closures

Note: Sample and residualization are the same as in Figure 5. The sample-split origin and destination firm health-cost effects are binned into quintiles (1 = lowest, 5 = highest). Cells show binned means of residualized outcomes, and annotations report the number of observations in each origin×destination cell.



(b) Disability in 2019

Figure A12 (continued): Long-run outcomes by origin×destination firm-effect quintiles after firm closures

Note: Sample and residualization are the same as in Figure 5. The sample-split origin and destination firm health-cost effects are binned into quintiles (1 = lowest, 5 = highest). Cells show binned means of residualized outcomes, and annotations report the number of observations in each origin×destination cell.

Statistic	KSS Estimates
Log Health Exp. Var.: var(h_i)	2.609
Firm FE Var.: var(ψ_j)	0.013
Firm FE SD: σ_ψ	0.114
$100[e^{\sigma_\psi} - 1]$ (%)	12.1
Employee FE Var.: var(α_i)	1.402
Firm–Employee Cov.: cov(α_i, ψ_j)	0.006
Correlation $\rho(\alpha_i, \psi_j)$	0.045
Explained share: workers & firms	0.547
No. Movers	1,113,768
No. Firms	117,247
No. of Observations	16,737,318
Average Health Exp.: mean(h_i)	5.611

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). Outcome is log healthcare expenditure per calendar year. The sample is restricted to full-time employees in firms with a non-missing sector code in the business registry, 2009–2016. Relative to the full connected sample in Table 3, the variance of firm effects is smaller in this restricted set, so firm effects account for a slightly smaller share of var(h_i). The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A13: Variance Decomposition of Log Healthcare Expenditure: Registry-Sector Sample

Statistic	PC4 FE (Residence)	PC4 FE + Workplace Access
Log Health Exp. Var.: var(h_i)	2.393	2.386
Firm FE Var.: var(ψ_j)	0.025	0.025
Firm FE SD: σ_ψ	0.157	0.157
$100[e^{\sigma_\psi} - 1]$ (%)	17.0	17.0
Employee FE Var.: var(α_i)	1.360	1.368
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.006	−0.007
Correlation $\rho(\alpha_i, \psi_j)$	−0.031	−0.037
Explained share: worker, firm & edu–age	0.574	0.578
No. Movers	1,212,637	1,173,299
No. Firms	131,992	129,198
No. of Observations	16,806,677	16,263,162
Average Health Exp.: mean(h_i)	5.652	5.653

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). Outcome is log healthcare expenditure per calendar year. The left column includes four-digit postcode (PC4) fixed effects for workers' residential location; the right column additionally controls for PC4-level workplace-area healthcare access (number of GPs, distance to nearest pharmacy, number of nearby hospitals). The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A14: Variance Decomposition of Log Healthcare Expenditure (KSS, with Residential PC4 FEs and Workplace Access Controls)

Statistic	Plug-in (Full Sample)	Plug-in (≥50-Move Firms)	KSS (≥50-Move Firms)
Log Health Exp. Var.: var(h_i)	2.617	2.675	2.675
Firm FE Var.: var(ψ_j)	0.113	0.049	0.030
Firm FE SD: σ_ψ	0.337	0.222	0.174
$100[e^{\sigma_\psi} - 1]$ (%)	40.1	24.9	19.0
Employee FE Var.: var(α_i)	1.839	1.790	1.520
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.077	−0.026	−0.008
Correlation $\rho(\alpha_i, \psi_j)$	−0.169	−0.086	−0.039
Explained share: worker, firm & edu–age	0.687	0.668	0.573
No. Movers	1,225,799	723,530	723,530
No. Firms	132,820	10,643	10,643
No. of Observations	16,976,071	11,295,814	11,295,814
Average Health Exp.: mean(h_i)	5.608	5.631	5.631

Notes: Plug-in columns report naive AKM quadratic components. The KSS column applies the leave-one-match-out bias correction of Kline et al. (2020). Outcome is log healthcare expenditure per calendar year. The ≥50-move sample restricts to firms with at least 50 observed worker moves. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

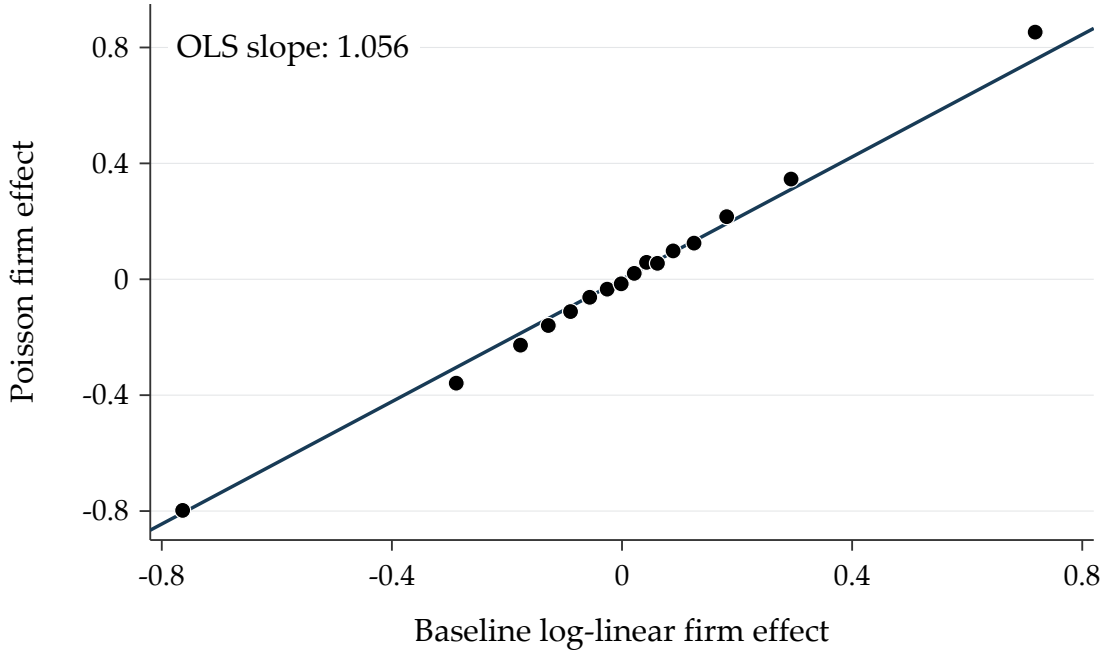
Table A15: Variance Decomposition of Log Healthcare Expenditure: Plug-in vs. KSS, Full Sample vs. ≥50-Move Firms

Statistic	1-year Ahead	2-year Ahead	3-year Ahead	4-year Ahead	5-year MA
Log Health Exp. Var.: var(h_i)	2.645	2.723	2.885	3.044	1.763
Firm FE Var.: var(ψ_j)	0.023	0.009	0.004	0.004	0.009
Firm FE SD: σ_ψ	0.151	0.094	0.067	0.066	0.097
$100[e^{\sigma_\psi} - 1]$ (%)	16.3	9.9	6.9	6.8	10.2
Employee FE Var.: var(α_i)	1.490	1.544	1.662	1.723	1.610
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.003	−0.001	−0.001	−0.003	0.000
Correlation $\rho(\alpha_i, \psi_j)$	−0.018	−0.005	−0.007	−0.039	−0.003
Explained share: worker, firm & edu–age	0.569	0.570	0.577	0.565	0.918
No. Movers	1,225,799	1,225,799	1,225,799	1,225,799	1,225,799
No. Firms	132,820	132,820	132,820	132,820	132,820
No. of Observations	16,976,071	16,976,071	16,976,071	16,976,071	16,976,071
Average Health Exp.: mean(h_i)	5.669	5.706	5.722	5.739	5.689

Notes: All columns report leave-one-match-out KSS estimates (Kline et al., 2020). Outcomes are the log of healthcare expenditure measured at different horizons relative to the baseline year: one- through four-year ahead, and a five-year moving average (current plus next four years). The sample is the full connected set. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A16: Variance Decomposition of Log Healthcare Expenditure (KSS) Across Horizons

Poisson specification. I compare firm effects estimated using a Poisson pseudo-maximum-likelihood two-way model with the baseline log-linear firm effects. Figure A13 displays the binned relationship, and Table A17 reports the corresponding OLS comparison of estimated firm effects.



Notes: Points report mean estimated firm effects in 16 equally sized bins. The horizontal axis shows baseline log-linear firm effects; the vertical axis shows Poisson firm effects. The line is the fitted OLS relationship from the underlying worker-year comparison reported in Table A17, rather than a regression on the 16 bin means.

Figure A13: Firm Effects from Poisson and Log-Linear Specifications

	Poisson firm effect
Log-linear firm effect	1.056***
	(0.000308)
Observations	16,976,071
R^2	0.410

Notes: OLS regression of estimated Poisson firm effects on estimated baseline log-linear firm effects, with an intercept. Observations are worker-years. Reported standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A17: OLS Comparison of Estimated Firm Effects

A3 Mover Evidence and Firm-Effect Dispersion

This appendix clarifies what the mover evidence measures. The event study asks how much of the origin–destination gap in firm-average healthcare expenditure passes through after a move. The main estimates instead recover firm effects and quantify both their dispersion across the workforce and their contribution to individual-level expenditure variation. These are different empirical objects; the formal results are developed in Moghani (2026).

Consider a two-period model with worker and firm fixed effects,

$$Y_{it} = \alpha_i + \psi_{j(i,t)} + \varepsilon_{it}, \tag{5}$$

where α_i is an individual effect and ψ_j is a firm effect. Let $\bar{\alpha}_j$ denote the average individual effect among workers at firm j, so that the firm's average outcome is

$$\bar{Y}_j = \psi_j + \bar{\alpha}_j.$$

For a mover from origin $o(i)$ to destination $d(i)$, define $\Delta\psi_i = \psi_{d(i)} - \psi_{o(i)}$ and $\Delta\bar{\alpha}_i = \bar{\alpha}_{d(i)} - \bar{\alpha}_{o(i)}$. With sufficiently large firms, the origin–destination gap in average outcomes converges to

$$\Delta\bar{Y}_i = \Delta\psi_i + \Delta\bar{\alpha}_i.$$

Under the usual exogenous-mobility condition, the mover coefficient therefore converges to

$$\theta_M = \frac{\mathrm{Cov}_M(\Delta\bar{Y}_i, \Delta\psi_i)}{\mathrm{Var}_M(\Delta\bar{Y}_i)}. \tag{6}$$

Equation (6) shows that the mover coefficient is a transition parameter. It depends on which firms movers leave and enter, how far those moves travel in the firm-effect distribution, and how changes in firm effects covary with changes in worker composition along those routes. Economies with the same cross-sectional dispersion of firm effects and the same sorting of workers across firms can therefore produce different mover coefficients when their mobility patterns differ.

A cross-sectional interpretation is available when moves reproduce the cross-sectional variation in firm effects and worker composition in the same proportions.

Moghani (2026) calls this condition *directional isotropy*. In compact form, for some scalar $\kappa > 0$,

$$\mathrm{Var}_M\begin{pmatrix}\Delta\psi_i \\ \Delta\bar{\alpha}_i\end{pmatrix} = \kappa\ \mathrm{Var}_M\begin{pmatrix}\psi_j \\ \bar{\alpha}_j\end{pmatrix}, \tag{7}$$

where cross-sectional moments weight firms by their incidence among movers. Under this condition,

$$\theta_M = \frac{\mathrm{Cov}_M(\psi_j, \bar{Y}_j)}{\mathrm{Var}_M(\bar{Y}_j)}.$$

Crucially, this equality concerns only the cross-sectional relationship between firm effects and firm-average outcomes. Both ψ_j and $\bar{Y}_j$ are firm-level objects, and the moments are computed across firms visited by movers. It does not recover the individual-level firm share, $\mathrm{Var}(\psi_{j(i,t)})/\mathrm{Var}(Y_{it})$. That denominator also contains variation among workers within firms and transitory expenditure shocks, and its workforce weights generally differ from the mover-incidence weights above. Directional isotropy aligns observed transitions with cross-sectional variation in firm effects and worker composition; it does not connect variation in firm averages to variation in individual healthcare expenditure.

Even at the firm-average level, the mover coefficient is not generically a variance share or a standard-deviation share. It lies midway between the share of cross-firm variance attributable to firm effects and the share of cross-firm variance that would disappear if firm effects were equalized. If firm effects are orthogonal to worker composition and mobility is directionally isotropic, it equals $\mathrm{Var}_M(\psi_j)/\mathrm{Var}_M(\bar{Y}_j)$. If firm composition is an affine, residual-free function of the firm effect with a non-negative slope, it equals $\mathrm{sd}_M(\psi_j)/\mathrm{sd}_M(\bar{Y}_j)$ for any non-degenerate mobility pattern. Both ratios concern variation across firms. Squaring the coefficient in the latter special case yields only the corresponding cross-firm variance ratio; it still does not identify the contribution of firm effects to individual-level healthcare-expenditure variation.

The empirical strategy follows directly from this distinction. I use mover event studies to assess pre-trends and document the timing and pass-through of healthcare expenditure after workers change firms. Firm-effect dispersion and its scale relative to individual-level expenditure variation are estimated directly from the limited-mobility-corrected two-way fixed-effects variance components.

A4 A Health-Capital Interpretation of the Log-Additive Specification

This appendix is an interpretive device, not a microfoundation of the estimator. It uses a Grossman-style health-capital framework for two limited purposes: to separate the two channels through which a firm can shift its workers' healthcare expenditure—the health consequences of the workplace and the cost of seeking care—and to show conditions under which individual and firm factors enter expenditure multiplicatively, so that logs are additive, as in the two-way fixed-effects specification of the main text. The derivation relies on a set of simplifying approximations that are stated where they are used (a linear investment technology, negligible interaction between the depreciation rate and the growth of the marginal cost of health, a local steady state, and age-dependent terms that are absorbed into the age profile); the result is a log-additive structure of the same form as the empirical specification, up to these approximations. The framework abstracts from health shocks such as accidents, from any consumption value of health, and from workers' choice of firm, and it therefore neither predicts the medication, disability, and mortality patterns documented in the paper nor is required for them.[48]

I extend the life-cycle health demand framework of Grossman (1972b,a). Individuals do not derive direct utility from health; instead, health affects utility indirectly through time lost to sickness and through the resources (time and money) devoted to healthcare. This structure is well suited to the Dutch context, where basic health insurance is universal and largely independent of the employer. Firms affect health-related spending primarily by changing the health environment (workplace risks, stress, exposures) and, to a lesser extent, the time needed to obtain care, rather than the price or generosity of basic coverage (see Section II).

The model introduces two firm-specific elements. First, a firm-specific health-depreciation component, capturing how the work environment accelerates or slows health deterioration through physical risks, psychosocial stress, and environmental exposures. Second, a firm-specific time cost of care-seeking, reflecting workplace flexibility, ease of taking time off for appointments, and the presence of occupational health services or screening. In the Dutch institutional setting, the first channel (health depreciation) is likely to be the dominant component of firms' effects on

[48] The framework describes how a given workplace affects health and spending on healthcare, not why workers choose different firms.

basic-package spending, while the second channel is unlikely to generate large or systematic differences (see Section II). I nevertheless keep the time-cost term in the model to make clear that, in principle, firms can affect both the health-depreciation environment and the convenience of care-seeking; in the empirical analysis, I interpret the estimated firm effects mainly through the depreciation channel.

Setup

Consider an employee working in firm j, with intertemporal utility

$$\sum_{t=0}^{T} \beta^t U(Z_t), \tag{8}$$

where T is the length of life,[49] β is the subjective discount factor, and Z_t is a composite of consumption goods and leisure:

$$Z_t = Z(C_t, L_t), \tag{9}$$

with C_t denoting market consumption and L_t leisure at age t.

Following Grossman (1972b), poor health reduces the available time endowment (normalised to 1). The employee faces the time constraint

$$T_t^w + T_t^h + L_t = 1 - T_t^{loss}, \tag{10}$$

$$T_t^{loss} = T^{loss}(H_t), \tag{11}$$

where H_t is the health stock at age t, T_t^{loss} the time lost due to sickness, and $T^{loss}(\cdot)$ a decreasing function of H_t. The variables T_t^w and T_t^h denote time spent working and time spent on health-care activities (care-seeking) at age t, respectively.

[49] In line with Grossman (1972b,a), life continuation is predicated on maintaining a minimum health level, denoted as $\bar{H}$. Therefore, an individual's lifespan can be affected by their health and health investment decisions.

The health stock evolves according to

$$H_{t+1} - H_t = I_t - \delta_{t,j} H_t, \tag{12}$$

$$I_t = I(\Theta_t), \tag{13}$$

$$T_t^h = \chi_j \Theta_t, \tag{14}$$

where I_t is health investment at age t, $\delta_{t,j}$ is an age- and firm-specific depreciation rate, Θ_t is a composite of medical care goods, and χ_j measures the time cost of obtaining a given amount of medical care. A higher χ_j means more time must be spent on care-seeking per unit of Θ_t; this reflects lower flexibility, more rigid schedules, or less convenient access to care. In a firm with lower χ_j, employees can obtain the same care with less time loss.

At age zero (which may be interpreted as entry into the labour market), the individual faces the full-income budget constraint

$$\sum_{t=0}^{T} \frac{f_t \Theta_t + p_t C_t}{(1+r)^t} = \sum_{t=0}^{T} \frac{w_{t,j} T_t^w}{(1+r)^t} + A_0, \tag{15}$$

where f_t and p_t are the market prices of medical care and consumption goods, respectively, $w_{t,j} T_t^w$ is labour income at age t, $w_{t,j}$ is the wage rate in firm j, A_0 is initial wealth, and r is the interest rate. In the Dutch setting, basic-package prices and coverage are set nationally and do not vary across firms; I therefore treat f_t as homogeneous across firms, so that differences in medical spending attributable to firms arise through differences in quantities Θ_t rather than from price differences.

Normalising the discount factor so that $\beta = \frac{1}{1+r}$, the utility maximisation problem is equivalent to

$$\begin{aligned} &\max_{C_t, L_t, \Theta_t, T_t^h, T_t^w} \sum_{t=0}^{T} \frac{U(Z_t)}{(1+r)^t} \\ &\text{s.t.} \quad \sum_{t=0}^{T} \frac{f_t \Theta_t + p_t C_t + w_{t,j}\left(T_t^h + L_t + T_t^{loss}\right)}{(1+r)^t} = \underbrace{\sum_{t=0}^{T} \frac{w_{t,j}}{(1+r)^t} + A_0}_{A}, \end{aligned} \tag{16}$$

where the term $w_{t,j}(T_t^h + L_t + T_t^{loss})$ collects forgone earnings from time spent in

care-seeking, leisure, and sickness. The right-hand side A represents full lifetime potential income (the value of all time at work plus initial wealth).

A4.I Optimal Health Investment Level

Focusing on health investments, consider the first-order condition with respect to $I_{\tau-1}$. In equilibrium,

$$\frac{\pi_{\tau-1}}{(1+r)^{\tau-1}} = -\sum_{t=\tau}^{T} w_{t,j} \frac{\frac{\partial H_t}{\partial I_{\tau-1}} \frac{dT^{loss}}{dH_t}(\cdot)}{(1+r)^t}, \tag{17}$$

where

$$\pi_\tau = f_\tau \frac{\partial \Theta_\tau}{\partial I_\tau} + w_{\tau,j} \frac{\partial T^h_\tau}{\partial I_\tau}$$

is the marginal cost of health investment (the combined monetary and time cost at age τ). Equation (17) states that the present value of the cost of an additional unit of health investment at age $\tau - 1$, $\pi_{\tau-1}$, equals the present value of the future gains from reduced time lost to sickness (and therefore higher effective labour supply and earnings).

Following the standard recursive manipulation in Grossman (1972b,a), this condition can be rewritten as

$$(1+r)\pi_{\tau-1} = -w_{\tau,j} \frac{dT^{loss}}{dH_t}(\cdot) + (1-\delta_{\tau,j})\pi_\tau, \tag{18}$$

or, equivalently,

$$\pi_{\tau-1}\left(r + \delta_{\tau,j} - \tilde{\pi}_\tau\right) = -w_{\tau,j} \frac{dT^{loss}}{dH_t}(\cdot), \tag{19}$$

where $\tilde{\pi}_\tau = \frac{\pi_\tau - \pi_{\tau-1}}{\pi_{\tau-1}}$ is the gross growth rate of the marginal cost of health.[50]

[50] For (19) to hold in this simple form, I follow Grossman (1972b,a) and assume that cross-terms of the form $\delta_{\tau,j}\tilde{\pi}_\tau$ are negligible. This keeps the focus on the main dependence on $\delta_{\tau,j}$ and $w_{\tau,j}$ without materially affecting the qualitative implications.

A4.II Assumptions and a Closed-Form Demand Function

To obtain a closed-form expression for medical care demand, I introduce a set of standard simplifying assumptions.

(i) Linear investment in health goods. Assume health investments have constant returns to scale with respect to health goods:

$$I_\tau = I_0 \Theta_\tau, \tag{20}$$

with $I_0 > 0$ a constant. Introducing diminishing returns would be more realistic but would prevent a simple closed-form solution. Under (20),

$$\pi_\tau = \frac{1}{I_0}\left(f_\tau + \chi_j w_{\tau,j}\right). \tag{21}$$

In the Dutch context, the monetary price component f_τ under the basic package is largely uniform across firms and, for high spenders, the marginal out-of-pocket price is often zero once the deductible is exhausted. For our purposes, any non-negligible but *firm-invariant* f_τ only affects the intercept of log spending and does not contribute to firm-level variation. To simplify the algebra and isolate firm-specific variation in care-seeking costs, I therefore treat the time-cost term $\chi_j w_{\tau,j}$ as the relevant driver of variation in π_τ across firms and write, up to a proportionality constant,

$$\pi_\tau \propto \chi_j w_{\tau,j}. \tag{22}$$

This implies that the growth rate of π_τ is approximately the growth rate of wages:

$$\tilde{\pi}_\tau \approx \tilde{w}_\tau. \tag{23}$$

(ii) Sick time as a function of health. Following the literature (e.g. Galama and Van Kippersluis, 2019), assume

$$T_\tau^{loss} = \phi H_\tau^{-\rho},$$

with $\phi > 0$ and $\rho > 0$. Substituting into (19) yields

$$\delta_{\tau,j} \frac{1}{I_0} \chi_j \left(1 + \frac{r - \tilde{w}_\tau}{\delta_{\tau,j}}\right) = \rho \phi H_\tau^{-(\rho+1)}. \tag{24}$$

(iii) Individual and firm components in depreciation and wages. Extend the functional forms commonly used in the life-cycle literature and assume

$$\delta_{\tau,j} = \delta_0 \delta_j e^{\kappa(\tau)},$$

where δ_0 and δ_j are individual- and firm-specific factors determining the level of health depreciation, and $\kappa(\tau)$ is an increasing function of age. By normalisation, the population averages of δ_0 and δ_j equal one, so that $e^{\kappa(\tau)}$ is the average age profile of depreciation.

Similarly, following Abowd et al. (1999), assume wages factor as

$$w_{\tau,j} = w_0 w_j e^{l(\tau)},$$

where w_0 is a common level term, w_j a firm-specific wage component, and $l(\tau)$ an age profile. This implies $\tilde{w}_\tau \approx l'(\tau)$.

Taking logs and rearranging, we obtain

$$\begin{aligned}&(1 + \rho) \log(H_\tau) \approx \\ &\log(\rho) + \log(\phi) + \log(I_0) - \log(\delta_0) - \log(\delta_j) - \log(\chi_j) - \kappa(\tau) - \log\left(1 + \frac{r - l'(\tau)}{\delta_{\tau,j}}\right).\end{aligned} \tag{25}$$

Assuming either $(r - l'(\tau)) \ll \delta_{\tau,j}$ or, more generally, that the last term in (25) can be well approximated by a function of age only,[51] we can write

$$\log(H_\tau) \approx \varepsilon \left[\log(\rho) + \log(\phi) + \log(I_0) - \log(\delta_0) - \log(\delta_j) - \log(\chi_j) - \check{\kappa}(\tau)\right], \tag{26}$$

where $\varepsilon = \frac{1}{\rho+1}$ and $\check{\kappa}(\tau)$ is an age function that absorbs $\kappa(\tau)$ and the term in brackets

[51] Formally, one can either neglect the last term in (25) or absorb it into an age-specific term, provided that the dependence on δ_0 and δ_j is weak. This is a standard approximation in health-capital models and preserves the separability between age and individual/firm factors.

in (25).

Using (12) and (20), and approximating health around a local steady state so that $I_\tau \approx \delta_{\tau,j} H_\tau$, we obtain an expression for medical care goods:

$$\begin{aligned}\log(\Theta_\tau) \approx\ & \underbrace{\varepsilon \log(\rho) + \varepsilon \log(\phi)}_{C} + \underbrace{(\varepsilon - 1)\log(I_0) + (1-\varepsilon)\log(\delta_0)}_{\alpha_0} \\ & + \underbrace{(1-\varepsilon)\log(\delta_j) - \varepsilon \log(\chi_j)}_{\alpha_j} + \underbrace{\kappa(\tau) - \varepsilon \check{\kappa}(\tau)}_{\eta(\tau)} + \log\left(1 + \frac{\varepsilon \check{\kappa}'(\tau)}{\delta_{\tau,j}}\right).\end{aligned} \tag{27}$$

As before, assuming either $\varepsilon\check{\kappa}'(\tau) \ll \delta_{\tau,j}$ or that the last term in (27) can be approximated by an age function,[52] we obtain the compact expression

$$\log(\Theta_\tau) \approx C + \alpha_0 + \alpha_j + \check{\eta}(\tau), \tag{28}$$

where $\check{\eta}(\tau)$ is an age function and

$$\alpha_j = (1-\varepsilon)\log(\delta_j) - \varepsilon \log(\chi_j).$$

In the Dutch basic-insurance system, the per-unit price of covered care is set nationally and does not vary across firms. Total basic-package spending at age t can therefore be written as $c_t \propto f_t \Theta_t$, with f_t common across firms. Taking logs and using (28), we obtain

$$\log c_t \approx \underbrace{\alpha_0 + C + \log f_t}_{\text{individual term}} + \underbrace{\alpha_j}_{\text{firm term}} + \underbrace{\check{\eta}(t)}_{\text{age term}}.$$

For a given worker i with expenditure c_{it} at age t, relabelling the individual component as α_i, the firm component as ψ_j, and the age term as part of $x_{it}'\beta$ gives, up to the approximations above, a log-additive structure of the same form as the two-way fixed-effects specification in the main text:

$$\log c_{it} \approx \alpha_i + \psi_j + x_{it}'\beta + r_{it},$$

[52] Here, I assume either the last term in equation (27) is negligible, or it can be absorbed into $\eta(\tau)$, leaving the dependence on δ_0, δ_j, and χ_j unchanged.

with r_{it} collecting idiosyncratic shocks. The empirical specification uses $h_{it} = \log(c_{it} + 1)$, an empirical convention to accommodate the small share of zero-expenditure years (Section IV).

Two observations follow. First, the firm term α_j combines a scaled firm-specific depreciation factor δ_j and the firm's care-seeking cost χ_j. In the Dutch institutional context, where basic prices and coverage are uniform across firms and time-related barriers to care are limited, the depreciation channel is the more natural reading of the estimated firm effects ψ_j; the empirical evidence on medication, disability, and mortality in the main text, rather than the framework, is what supports that reading.

Second, because individual and firm factors enter expenditure multiplicatively (through products of δ_0, δ_j, and χ_j), taking logarithms yields an additively separable individual term, firm term, and age profile. This is the property the main text uses: it motivates the log specification and the interpretation of ψ_j as a common proportional shift in expenditure across workers within a firm, an interpretation that the saturated-model, residual, and subgroup checks in Sections V and VI test directly.

A5 Leave-Out Variance Correction for the AKM Model

This appendix summarises how I implement the leave-out variance correction of Kline et al. (2020) (KSS) for the AKM specification in Equation (4). The goal is to obtain unbiased estimates of variance and covariance components of worker and firm effects in the presence of limited mobility.

A5.I Quadratic-form representation and plug-in bias

Write the AKM model in stacked form as

$$h = Z\nu + \varepsilon,$$

where h collects log healthcare expenditures h_{it}, Z is the design matrix containing worker and firm dummies and observed covariates, and ν stacks all parameters, $\nu = (\alpha_1, \ldots, \psi_1, \ldots, \beta')'$. Let z_o' denote the o-th row of Z, and $S_{zz} = \sum_{o=1}^{n} z_o z_o'$ the Gram matrix.

Many objects of interest in the AKM decomposition can be written as quadratic forms

$$\theta = \nu' A \nu,$$

for some known symmetric matrix A; for example, $Var(\psi_j)$ and $Cov(\alpha_i, \psi_j)$ correspond to particular choices of A.[53] The natural plug-in estimator is

$$\hat{\theta}_{PI} = \hat{\nu}' A \hat{\nu},$$

where $\hat{\nu}$ is the OLS estimator from Equation (4). As shown by Kline et al. (2020), $\hat{\theta}_{PI}$ is generally biased when worker mobility is limited: even if each element of $\hat{\nu}$ is unbiased, the quadratic form combines true dispersion and the dispersion of the estimation error.

Under weak conditions, KSS show that the bias of the plug-in estimator can be written as

$$\text{Bias}(\hat{\theta}_{PI}) = \mathbb{E}[\hat{\theta}_{PI}] - \theta = \sum_{o=1}^{n} z_o' S_{zz}^{-1} A S_{zz}^{-1} z_o \, \sigma_o^2,$$

[53] See Kline et al. (2020) for explicit A matrices for standard AKM variance components.

where $\sigma_o^2 = \mathbb{E}[\varepsilon_o^2]$ is the observation-specific error variance. Limited mobility bias is thus a weighted sum of these variances, with weights given by the leverage-like terms $z_o' S_{zz}^{-1} A S_{zz}^{-1} z_o$.

A5.II Leave-out error-variance estimation

To obtain an unbiased estimator of θ, KSS propose estimating each σ_o^2 using a leave-one-out construction. Let $\hat{\nu}_{-o}$ denote the OLS estimator from Equation (4) computed on the sample that excludes observation o. The leave-out prediction for h_o is $z_o' \hat{\nu}_{-o}$ and the associated error-variance estimator is

$$\hat{\sigma}_o^2 = h_o\Big(h_o - z_o' \hat{\nu}_{-o}\Big). \tag{29}$$

By construction, h_o is independent of $\hat{\nu}_{-o}$ (because h_o is not used in its estimation), and $z_o' \hat{\nu}_{-o}$ is an unbiased predictor of h_o conditional on Z. It follows that $\mathbb{E}[\hat{\sigma}_o^2 \mid Z] = \sigma_o^2$, so (29) is an unbiased estimator of the observation-specific error variance; see Kline et al. (2020) for a formal derivation.

Given $\hat{\sigma}_o^2$, an unbiased estimator of θ is then

$$\hat{\theta}_{KSS} = \hat{\nu}' A \hat{\nu} \; - \; \sum_{o=1}^{n} z_o' S_{zz}^{-1} A S_{zz}^{-1} z_o \, \hat{\sigma}_o^2, \tag{30}$$

which satisfies $\mathbb{E}[\hat{\theta}_{KSS}] = \theta$ under the conditions in Kline et al. (2020). Intuitively, $\hat{\theta}_{KSS}$ subtracts an estimate of the plug-in bias from the naive quadratic form.

In large panels, recomputing $\hat{\nu}_{-o}$ separately for every o is infeasible. KSS show that for linear models the leave-one-out prediction error can be written as

$$h_o - z_o' \hat{\nu}_{-o} = \frac{h_o - z_o' \hat{\nu}}{1 - \ell_o},$$

where $\ell_o = z_o' S_{zz}^{-1} z_o$ is the leverage of observation o. This identity implies that the leave-out prediction $z_o' \hat{\nu}_{-o}$, and therefore $\hat{\sigma}_o^2$, can be computed from the full-sample OLS quantities without repeated re-estimation.

A5.III Implementation for the AKM decomposition

In practice, I implement the KSS correction as follows. First, I partial out the observed covariates x_{it} from the outcome by estimating Equation (4) and forming the residualised dependent variable $h_{it} - x'_{it}\hat{\beta}$. The dimensionality problem affects the worker and firm effects, not the coefficients on x_{it}, so $\hat{\beta}$ is estimated precisely.

Second, I apply the leave-out formulas of Kline et al. (2020) to the worker–firm match level, treating an entire match (rather than a single person–year observation) as the leave-out unit to allow for serial correlation of errors within matches, as recommended by KSS. Using the quadratic-form representation, I construct $\hat{\theta}_{KSS}$ in (30) for the variance of firm effects, $Var(\psi_j)$, and for the covariance between worker and firm effects, $Cov(\alpha_i, \psi_j)$.

Finally, because the AKM design matrix is very large, I use the random-projection approximation proposed by Kline et al. (2020), based on the database-friendly projections of Achlioptas (2003), to evaluate the quadratic forms in (30) efficiently. This approximation delivers estimates that are numerically very close to the exact leave-out quantities at a fraction of the computational cost; see Kline et al. (2020) for details and accuracy guarantees.

A6 Higher Moments of the Firm-Effect Distribution

The standard deviation of firm effects measures the scale of expenditure differences across firms for the same worker. Quantile comparisons give that scale a more familiar form: how far apart low-, middle-, and high-effect firms are, on the same footing as the expenditure gradients by wealth, income, and education. The leave-out variance of Kline et al. (2020) identifies the second moment but not the shape of the distribution, and the shape matters most in the tails. This appendix estimates the third and fourth moments from independent worker splits and uses them to bound the quintile comparisons.

Four-way split. I start from the fixed-design framework of Kline et al. (2020): disturbances are conditionally mean zero and independent across workers, with arbitrary heteroskedasticity and arbitrary dependence within a worker's history. Workers are assigned to four disjoint splits by equal-probability draws, independently of the disturbances, with each worker's complete history kept in one split, and firm effects are maintained to be the same across splits. Conditional on the realised designs and the assignment, denoted $\mathcal{D}$, the estimation error of split s is a linear transformation of that split's own disturbances, $u^{(s)} = A_s \varepsilon_s$, so $\mathbb{E}[u^{(s)} \mid \mathcal{D}] = 0$ and the four error vectors are mutually independent. As in Kline (2025), products of estimates from distinct splits are therefore free of sampling error in expectation.

Let $\mathcal{J}$ be the set of firms identified in all four splits and let w_j be the worker-year weights on this set, normalised to sum to one. I centre each split with the same weights,

$$\widetilde{\psi}_j^{(s)} = \widehat{\psi}_j^{(s)} - \sum_{k \in \mathcal{J}} w_k \widehat{\psi}_k^{(s)}, \qquad \theta_j = \psi_j - \sum_{k \in \mathcal{J}} w_k \psi_k,$$

so that the arbitrary normalisation of each split's AKM estimates does not enter the moments. For any set S of r distinct splits,

$$\mathbb{E}\left[\prod_{s \in S} \widetilde{\psi}_j^{(s)} \,\middle|\, \mathcal{D} \right] = \theta_j^r, \qquad r = 2, 3, 4, \tag{31}$$

because every other term in the expansion of the product contains at least one split error on its own. Weighting these products by w_j gives the central moments. Four

splits provide six pairwise products for μ_2, four triple products for μ_3, and one four-way product for μ_4; the second and third moments are averages over their components.

The common support contains 47,144 firms and 14,950,514 worker-years, 88.1% of the baseline worker-years. Table A18 reports the moments. The standard deviation implied by μ_2 is 0.163 log points, against the baseline KSS estimate of 0.164, so the cross-split second moment reproduces the leave-out variance on a support that excludes a third of firms but keeps nine tenths of workers. The third moment is negative and the fourth is large relative to μ_2^2: the distribution has a pronounced left tail. Three readings are consistent with this. A small number of large employers may offer unusually low-hazard work environments; some employers may route part of their workers' care through channels outside the basic package, so that their measured expenditure understates care; or records may be incomplete for particular employers. The main text treats the left tail as a feature to be robust to rather than as evidence of health-improving firms, and the quintile comparisons below are the objects it uses.

	Full common support
Mean (normalization)	0
Second central moment, μ_2	0.026508
Third central moment, μ_3	−0.045691
Fourth central moment, μ_4	0.125747
Standard deviation	0.163

Notes: Firm effects are in log-healthcare-expenditure units. The moments use firm-effect estimates from four worker splits on their common support and worker-year weights. The second moment averages six pairwise cross-split estimates, whose standard deviation is 0.000541 and whose range is [0.025697, 0.027134]. Mean zero fixes the location of relative firm effects. Entries are point estimates.

Table A18: Cross-Split Moments of Firm Effects

Moments relative to industry means. For the industry-adjusted targeting benchmark in Section VI.III, I restrict attention to firms with an SBI code and at least 100 worker-year observations and compute the moments after subtracting each firm's SBI industry mean, separately in each split. Table A19 compares the moments before and after adjustment on the same 19,650 firms. Removing industry

means lowers the second moment from 0.0180 to 0.0135, so about three quarters of the dispersion among these firms lies within detailed industries; this is the noise-corrected counterpart of the within-industry share reported in the main text. The adjusted moments retain the left tail.

	Unadjusted	SBI-mean adjusted
Mean (normalization)	0	0
Second central moment, μ_2	0.018029	0.013549
Third central moment, μ_3	−0.023799	−0.020331
Fourth central moment, μ_4	0.054967	0.047356
Firms	19,650	19,650

Notes: Both columns use the same SBI-coded firms with more than 100 worker-year observations on the four-way common support and worker-year weights. The second column subtracts each firm's SBI industry mean separately in each split. Moments use products of estimates from distinct splits, as in equation (31). Mean zero fixes the location of relative firm effects. Entries are point estimates; distribution fitting uses the unrounded moments.

Table A19: Cross-Split Moments Before and After SBI-Mean Adjustment

Distributional comparisons. Four moments constrain a distribution without determining it, so I fit three families that match the mean and the second to fourth moments exactly and read the quintile comparisons as a range rather than a point. The families are chosen to differ in how they place the left tail. A two-normal mixture with a small lower-mean component (fixed at 0.5% or 1% of the weight) represents the tail as a separate group of firms and keeps the remaining distribution continuous; this is the specification I prefer, because it matches the moments with the least distortion of the bulk. The generalised lambda distribution in its FMKL parameterisation matches the same moments by placing a small amount of mass very far out and concentrating the rest near the centre, which compresses the quintile means and yields the conservative end of the range. A normal with the same variance ignores the asymmetry altogether and treats the right tail as the mirror image of the left; it is reported as the symmetric benchmark and marks the upper end.

For a fitted quantile function $Q(u)$, the bottom- and top-quintile means are $5\int_0^{0.2} Q(u)\,du$ and $5\int_{0.8}^{1} Q(u)\,du$, and the median is $Q(0.5)$. Under the log-linear specification, a top–bottom gap Δ corresponds to a percentage reduction in

expenditure of $100[1 - \exp(-\Delta)]$. Table A20 reports the fits on the full common support. The asymmetric fits imply top–bottom quintile gaps of 0.09 to 0.29 log points, or expenditure reductions of 9–25%, against 0.46 (37%) under the normal; the two mixtures, which I take as the reference, give 0.25–0.29. In every asymmetric fit the bottom quintile lies farther below the median than the top quintile lies above it. Most of the gap is the lower tail, but substantial dispersion remains from median to the upper tail.

	Top–bottom quintile	Median–bottom quintile	Top quintile –median
Full four-way common support			
Normal benchmark	0.456	0.228	0.228
Generalised lambda (FMKL), bounded	0.091	0.071	0.020
Two-normal mixture: 0.5%	0.290	0.165	0.125
Two-normal mixture: 1%	0.245	0.146	0.099

Notes: Entries are log-point differences in firm effects. Top and bottom refer to means within worker-year-weighted quintiles; the median is the point quantile. The asymmetric fits match the moments in Table A18 using unrounded values. The normal matches only the mean and variance; using the baseline KSS variance instead gives a gap of 0.460. Mixture percentages are the fixed weight of the lower-mean component, which allows the mixtures to represent the left tail of the full-support moments. All fits use the full four-way common support.

Table A20: Quintile Comparisons across Firm-Effect Distributions

A7 Split-Sample Validation of Firm-Effect Dispersion

Estimated firm effects contain sampling error, and their precision varies with firm size and mobility. As an independent check on the KSS-corrected firm-effect variance, I estimate firm effects in two disjoint worker samples and compare the resulting estimates for firms observed in both samples. Following the independent-sample logic of Kline (2025), the covariance between the two estimates isolates their common firm component rather than split-specific estimation noise.

I randomly partition workers into two mutually exclusive sets, $S = 1$ and $S = 2$, and estimate the two-way fixed-effects model separately in each subsample, including all worker-years for workers in that subsample. I restrict attention to firms that belong to the intersection of the two split-sample connected sets. To ensure comparability across splits, I impose the same normalization in each estimation by recentering firm effects so that $\sum_i \hat{\psi}^{(s)}_{j(i,t)} = 0$ for $s \in \{1, 2\}$. This produces two firm-effect estimates for each firm in the common set,

$$\hat{\psi}^{(1)}_j \;=\; \psi_j + u^{(1)}_j, \qquad \hat{\psi}^{(2)}_j \;=\; \psi_j + u^{(2)}_j,$$

where $u^{(1)}_j$ and $u^{(2)}_j$ are estimation errors induced by sampling variation in the two disjoint worker sets. Under random partitioning and the maintained two-way fixed-effects assumptions, $u^{(1)}_j \perp u^{(2)}_j \mid \psi_j$ and $\mathbb{E}[u^{(s)}_j \mid \psi_j] = 0$ for $s \in \{1, 2\}$ (Kline, 2025).

The cross-split covariance therefore identifies the variance of latent firm effects on the common support:

$$\mathrm{Cov}\left(\hat{\psi}^{(1)}_j, \hat{\psi}^{(2)}_j\right) \;=\; \mathrm{Var}(\psi_j).$$

I use this identity to assess the full-sample KSS estimate. The cross-split covariance is 0.026, close to the KSS-corrected firm-effect variance in the main sample. The KSS estimates obtained separately in the two worker splits are also close to the full-sample estimate (Table A21).

Statistic	Split 1 (KSS)	Split 2 (KSS)
Panel A. AKM variance components estimated within split (KSS-corrected)		
Log Health Exp. Var.: var(h_i)	2.632	2.622
Firm FE Var.: var(ψ_j)	0.030	0.029
Firm FE SD: σ_ψ	0.172	0.171
$100[e^{\sigma_\psi} - 1]$ (%)	18.8	18.6
Employee FE Var.: var(α_i)	1.598	1.486
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.007	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.033	−0.020
Explained share: workers & firms	0.613	0.575
No. Movers	576,056	576,410
No. Firms	84,852	85,160
No. of Observations	8,023,188	8,021,807
Average Health Exp. : mean(h_i)	5.614	5.613
Panel B. Cross-split covariance of firm effects (common firms)		
$\text{cov}\left(\hat{\psi}_j^{(1)}, \hat{\psi}_j^{(2)}\right)$	0.026	

Notes: Columns report leave-one-match-out KSS-corrected components (Kline et al., 2020) from the two-way fixed-effects model estimated separately in two random worker splits (all worker-years for workers in a split are included). The outcome is log annual healthcare expenditure; the sample covers full-time employees from 2009–2016. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD. Panel B reports the covariance between firm effects estimated in the two splits for firms observed in both. Under independent, mean-zero estimation error across splits, this covariance identifies the variance of the latent firm effects on the common support.

Table A21: Variance Decomposition of Log Healthcare Expenditure: Sample Splits

Held-out targeting. Let x_j denote a firm effect relative to the overall mean or its deviation from its SBI industry mean. For a share q of eligible firms, let $T_A(q)$ contain those with the highest split-A estimates. Using fixed size weights w_j, I evaluate this set in split B:

$$\widehat{m}_{A\to B}(q) = \frac{\sum_{j\in T_A(q)} w_j \widehat{x}_j^B}{\sum_{j\in T_A(q)} w_j}. \tag{32}$$

Following the independent-split logic above, conditionally mean-zero error vectors that are independent across splits imply that, conditional on the designs and the split-A ranking, this estimate has expectation equal to the weighted mean of the selected firms' latent effects. The same argument applies after industry adjustment when a common fixed demeaning operator is applied separately to each split. The

ranking remains noisy, but evaluation does not reuse the noise that selected the firms.

The reference is the overall or own-industry mean. The eligible targeting subset need not itself average zero relative to that reference, even when all eligible firms are selected. I report $100[1 - \exp(-\widehat{m}_{A\to B}(q))]$, the percentage reduction corresponding to moving the selected mean effect to the reference.

Oracle benchmark. For a fitted latent distribution with quantile function Q_F, selecting its upper tail gives

$$m_F^*(q) = \frac{1}{q}\int_{1-q}^{1} Q_F(u)\,du, \qquad 0 < q \le 1. \tag{33}$$

I apply the same percentage transformation to this mean. The moments describe worker-year-weighted distributions. To interpret q as a share of firms, I assume $\mathbb{E}[N_j \mid x_j] = \mathbb{E}[N_j]$, separately for raw and SBI-adjusted latent effects. Under this assumption, size weighting leaves the marginal distribution of x_j unchanged.

The overall fits use 47,144 firms on the four-way common support, and the SBI-adjusted fits use 19,650 SBI-coded firms with more than 100 worker-year observations. Feasible targeting uses eligible firms in the two-way split. These differences make the comparison illustrative. The generalised-lambda calibration's concentrated centre produces low upper-tail means beyond the most selective cutoffs; it lies below the feasible curve over much of the plotted range. The SBI-adjusted 1% mixture also lies slightly below it at wider cutoffs. The fitted curves therefore do not provide empirical bounds on feasible targeting or identify its efficiency loss.

A8 LLM-based collar classification from education codes

Because the administrative records do not contain occupational titles, I proxy blue- versus white-collar status using educational specialization. The data include Dutch SOI education codes and, for each code, an education/program title in Dutch (`OPLEIDINGSNAAM`). I treat this education title as a concise summary of the type of work the individual is trained for, and I map it into a broad collar category.

I implement this mapping using an automated large-language-model (LLM) classification routine. For each education title, I query the OpenAI Chat Completions API (model `gpt-4o-mini`) with a fixed prompt that asks the model to assign the title to one of three categories: (1) *blue collar*, for programs that typically lead to manual, physical, trade, or craft occupations; (2) *white collar*, for programs that typically lead to office-based, professional, managerial, or administrative occupations; and (3) *borderline*, for ambiguous or mixed programs (for example, titles that plausibly combine technical and supervisory content or are too generic to map cleanly). The prompt restricts the model to return only the numeric label (1–3), with no accompanying text. I run the model with temperature zero and a short output constraint, so the classification is deterministic and machine-readable.

In the analysis, I collapse these labels into the collar proxy used in the subsample exercises. White-collar workers are those whose education titles are classified as category (2), and blue-collar workers are those classified as category (1). The borderline category (3) captures mixed or insufficiently specific education titles; they are excluded from collar subsample analysis.

This approach is motivated by the fact that the task is a coarse, low-dimensional coding problem: the goal is not to infer a specific occupation, but to distinguish whether a program is predominantly geared toward manual/trade work versus office/professional work. A growing literature finds that LLMs achieve human-comparable accuracy in such straightforward classification settings. Gilardi et al. (2023) show that GPT-4 outperforms human crowdworkers in text annotation. Törnberg (2023) documents expert-level reliability in political text coding. Chang et al. (2024) report similar evidence from large-scale evaluations in social science contexts. Since mapping education titles into broad collar categories is simpler than the nuanced political or attitudinal coding studied in these papers, the resulting

proxy is expected to be reliable for the purpose of stratifying the AKM estimates by worker type.

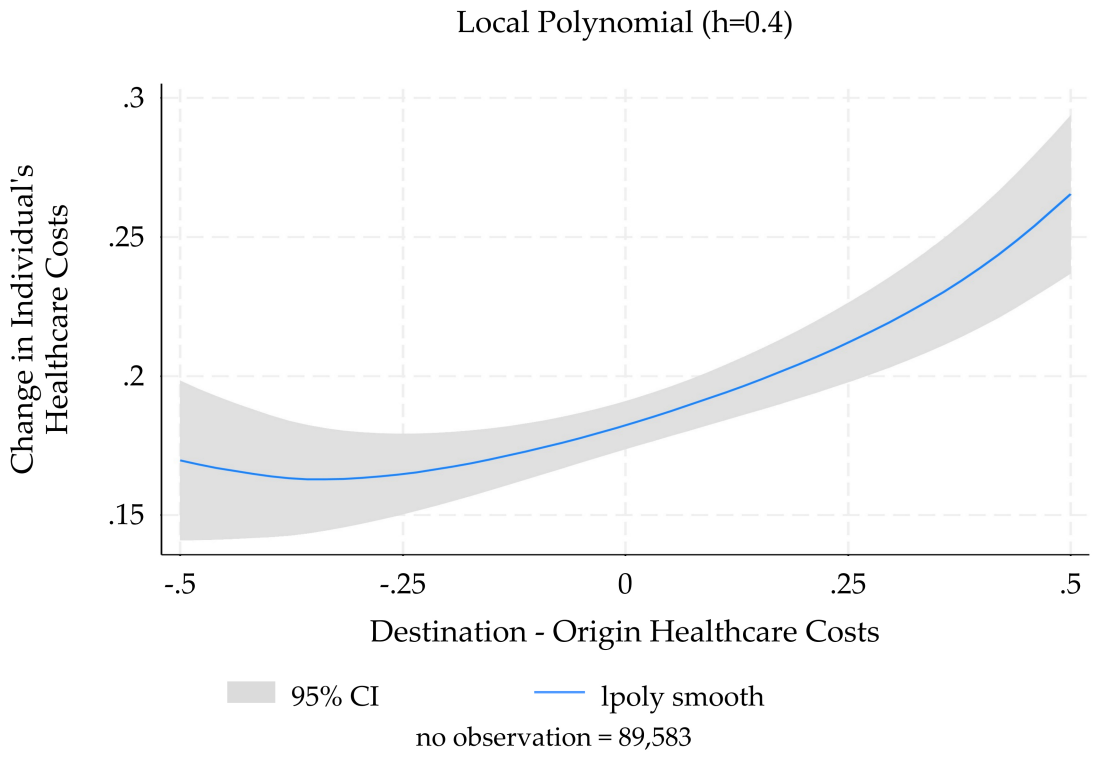
A9 Female Sample Analysis

Statistic	KSS Estimates
Panel A. KSS variance components	
Log Health Exp. Var.: var(h_i)	2.617
Firm FE Var.: var(ψ_j)	0.020
Firm FE SD: σ_ψ	0.143
$100[e^{\sigma_\psi} - 1]$ (%)	15.4
Employee FE Var.: var(α_i)	3.158
Firm–Employee Cov.: cov(α_i, ψ_j)	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.017
Explained share: workers & firms	1.211
No. Movers	366,643
No. Firms	44,578
No. of Observations	4,784,634
Average log Health Exp.: mean(h_i)	6.114
Panel B. Cross-sample link	
Bias-corrected slope, firm FE (male) on firm FE (female)[a]	1.149
No. worker–year obs. in slope regression[a]	4,607,681

[a] Slope from an OLS regression of the firm fixed effects estimated in the male subsample on the firm fixed effects estimated in the female subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the reliability ratio for the right-hand-side firm effect, $Var_{KSS}(\psi)/Var_{PI}(\hat{\psi})$, following Kline et al. (2020) (equivalently, $\hat{\beta}_{BC} = \hat{\beta}_{OLS} \times Var_{PI}(\hat{\psi}_{\text{female}})/Var_{KSS}(\psi_{\text{female}})$). In this application, $\hat{\beta}_{OLS} = 0.151$, $Var_{PI}(\hat{\psi}_{\text{female}}) = 0.155$, and $Var_{KSS}(\psi_{\text{female}}) = 0.020$, yielding $\hat{\beta}_{BC} = 1.149$. The male and female models use separate leave-one-out connected sets; the slope regression uses firms with effects estimated in both.

Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using 50 random projections. The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. The firm-effect SD is $\sigma_\psi = \sqrt{\text{var}(\psi_j)}$; a one-SD increase has the log-linear percentage equivalent $100[\exp(\sigma_\psi) - 1]$%, using the reported SD.

Table A22: Variance Decomposition of Log Healthcare Expenditure (Females)



(a) Average change vs. exposure gap

Figure A14: Workers' healthcare spending around moves between firms

Note: Sample of full-time female employees aged 25–65 who switch employer exactly once between 2009 and 2016 and belong to the connected set of firms. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. In panel (a), the curve plots the average change in individual annual residualised spending between the years before and the years after the move against the "exposure gap" D_i, defined as the difference in leave-i-out mean residualised spending between destination and origin firms; the line shows the (non-parametric) fitted relationship. In panel (b), the markers show coefficients from the mover event-study specification in equation (3), where the outcome is residualised log spending and the regressors are event-time indicators interacted with D_i, with event time measured relative to $t = -2$ as the reference year. Because workers are assigned to the firm where they spend most months, the year immediately before the move already contains some exposure to the destination firm.



(b) Event-study of mover responses

Figure A14 (continued): Workers' healthcare spending around moves between firms

Note: Sample of full-time female employees aged 25–65 who switch employer exactly once between 2009 and 2016 and belong to the connected set of firms. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. In panel (a), the curve plots the average change in individual annual residualised spending between the years before and the years after the move against the "exposure gap" D_i, defined as the difference in leave-i-out mean residualised spending between destination and origin firms; the line shows the (non-parametric) fitted relationship. In panel (b), the markers show coefficients from the mover event-study specification in equation (3), where the outcome is residualised log spending and the regressors are event-time indicators interacted with D_i, with event time measured relative to $t = -2$ as the reference year. Because workers are assigned to the firm where they spend most months, the year immediately before the move already contains some exposure to the destination firm.